

Human Genetics: Principles and Applications

ALEXEY G. NIKITIN



Human Genetics

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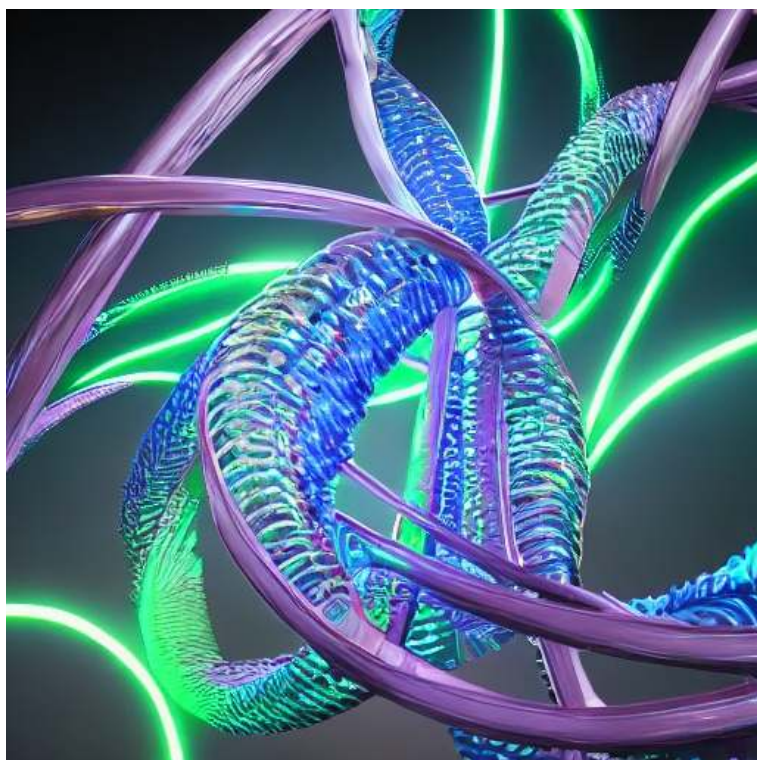
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Human Genetics: Principles and Applications

Alexey G. Nikitin, Ph.D.



Introduction

γνῶθι σεαυτόν

The phrase above is translated from Greek as “Know Thyself”. It was prominently featured as the main of the three maxims on the front of the Temple of Apollo at Delphi since the 5th century BCE (Before Common Era).



*Temple of
Apollo
(Delphi).
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Ancient Greek philosophers thought that no man could ever comprehend the human spirit and thought thoroughly, so it would have been almost inconceivable to know oneself fully. Therefore, the saying “Know Thyself” may refer to a less ambitious ideal, such as knowing one’s own habits, morals, temperament, ability to control anger, and other aspects of human behavior that we struggle with on a daily basis.

Knowledge helps solve problems. Knowing our genetic makeup will ultimately lead to solving health problems and make our lives qualitatively better as well as allowing us to better understand our origins and our connection with the rest of the living world.

Chapter 1. Overview

The subject of Genetics

Genetics is the study of inheritance and variation. Transmission Genetics studies mechanisms of gene transmission from generation to generation. Molecular Genetics studies the molecular basis of inheritance (structure and function of genetic material at the molecular level). Biotechnology (Genetic Engineering) is applied molecular genetics. Cytogenetics studies chromosomal structure and function. Population Genetics studies genetic variation in groups of organisms.

Human Genetics is the study of inheritance and variation in human beings. Medical Genetics is the study of medical aspects of human genetics.

The principles of inheritance in humans do not fundamentally differ from that in other living organisms. An understanding of human genetic inheritance is a vital asset in the diagnosis, prediction, and treatment of medical conditions that have a genetic foundation.

DNA and human cells

The human body is made up of 100 trillion cells. Each cell (except red blood cells) has at least one nucleus, which houses the chromosomes.

There is 1.8 m of DNA in each of our cells packed into a structure only 0.0001 cm across.

Most human somatic cells (cells of the body) contain 46

chromosomes: pairs of chromosomes 1-22 (autosomes), and a pair of sex chromosomes.

Females normally have two X chromosomes; males have an X and a Y.

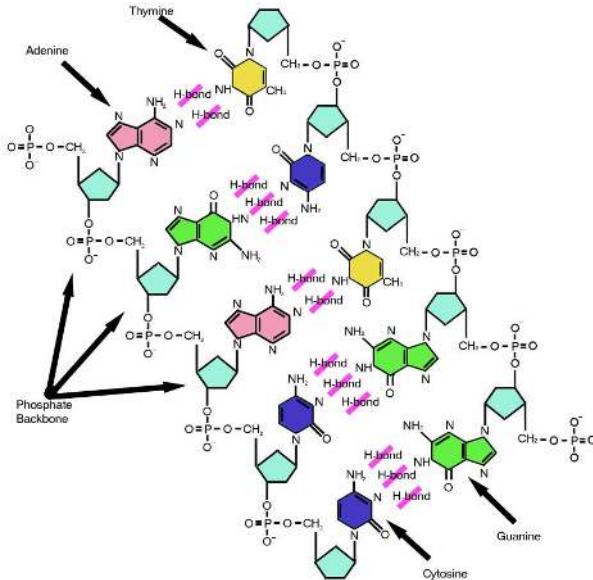
Human reproductive cells (sperm cells and oocytes) contain one set of 22 autosomes and one of each sex chromosomes.

The ABC of GATC: The chemical organization of nucleic acids

Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) comprise the nucleic acid family of bioorganic molecules. Besides nucleic acids, bioorganic molecular groups include proteins, carbohydrates and lipids.

DNA and RNA are biopolymers, meaning that they are formed from connecting together simple building blocks called monomers. In the case of DNA and RNA, the monomers are called nucleotides.

Nucleotides themselves consist of a pentose monosaccharide called ribose (in RNA) and deoxyribose (in DNA), a phosphate functional group and a compound called nitrogenous base. Deoxyribose is derived from ribose by the loss of an oxygen atom in the hydroxyl group connected to the second (2') carbon of the pentose molecule. There are five types of nitrogenous bases found in nucleic acids. Three of the five (adenine, guanine, cytosine) are found in both DNA and RNA, while thymine is only found in DNA and uracil is only found in RNA.



The chemical composition and bonding in double-stranded DNA molecule. Created by Raul654 on September 15, 2004. CC BY-SA 3.0, via Wikimedia Commons.

The Human Genome

The Human Genome Project (HGP) is an endeavor that could be compared in complexity and significance to a trip to Mars. If you were to read the entire genome of a human being, like it were a book, one base at a time, it would take you 140 years to complete! By 1985, when serious talks about the sequencing of the human genome began to develop, Jim Watson estimated that it would take 1,000 years to complete (with the existing technology at the time)! In October of 1990, when the HGP was officially launched, it was intended to be a 20-year project, where most of the time would be devoted to improving of technology, rather than actual sequencing. In 1997, the estimate was that it would take another 50 years to actually complete the sequencing. Despite the prognosis, the first “rough draft” of the human genome was completed

in 2000, at which point it was expected that the mapping of the genome would be completed in five years. The essentially completed human genome assembly was released in April 2003, 2 years earlier than planned.

Why is the knowledge of the human genome important?

The knowledge we have gained from the deciphering of the human genome has significantly impacted modern medicine. Doctors can now tailor medications to your specific genetic makeup. Pharmacogenetics, the field of medicine concerned with how your genes affect your response to drugs, can now find ways to reduce side effects and improve drug efficiency, by taking into consideration how your genes may react to medications you are prescribed. Other benefits include being able to anticipate the probability of developing cancer based on an individual genetic makeup as well as being able to predict the specific genetic conditions children of perspective parents may have. There are countless other benefits, which we will be discussing throughout the course.

The Human Genome At A Glance

Total number of chromosomes: 46

Haploid genome: 23 chromosomes

Molecular size: 3.3 billion (3,323,950,079) nucleotides (or base pairs, bp)

Largest chromosome #1 = ~263 million base pairs (bp)

Smallest chromosome Y = ~59 million bp

Known protein-coding genes: 19,940 (GENCODE reference

genome version 29, 2018, <https://www.gencodegenes.org/human/stats.html>)

RNA genes (regulatory, tRNA, rRNA etc.): 23,643

Pseudogenes: 14,729

Gene transcripts (detected mRNA): 206,694

Protein-coding transcripts: 83,129

Mobile genetic elements: 45% of the genome

Average size of a human gene: ~3,000 bp

Only about 5% of the human genome contains genes (coding sequences)

Less than 2% of the genome codes for proteins

Humans share most of the same protein families with worms, flies, and plants

Function of much of the genome is unknown

At least 20% of the non protein-coding portion of the genome is dedicated to regulating the transcriptional activity of the encoding 2%.

Repeated sequences that do not code for proteins make up at least 50% of the human genome. Recent studies suggest that that 66%–69% of the human genome is repetitive or repeat-derived (de Koning et al., 2011)

45% of these repeated sequences are able to move around within the genome (mobile genetic elements, or transposable elements)

Below is a link to a map of human chromosomes and associated genetic diseases:

<http://www.ncbi.nlm.nih.gov/books/NBK22266/>

Mobile genetic elements

Mobile genetic elements (MGEs), also referred to as transposable elements (TEs) or transposons, are nucleotide sequences normally housed within the genome that retain the ability to change their location. Transposons are classified into

retroelements (Class I MGEs) and DNA transposons (class II MGEs). The latter are sometimes referred to as “jumping genes” or transposable elements, since the movement of transposons requires the presence of a functional enzyme called transposase. Retroelements can be transcribed from DNA to RNA and the RNA produced is then reverse transcribed to DNA. Reverse transcription is accomplished with the help of an enzyme called reverse transcriptase (RNA dependent DNA polymerase), sometimes encoded by retroelements themselves. This copied DNA is then inserted at a new position into the genome. Transposons have a tendency to re-insert into the non-coding regions of the genome, but they can also insert in the vicinity of functional genes, thereby interrupting vital functions. DNA transposons comprise 8% of nuclear genome. MGEs are found on all chromosomes, but mostly on the X and chromosome 19. More on transposons in Chapter 10 (Mutations).

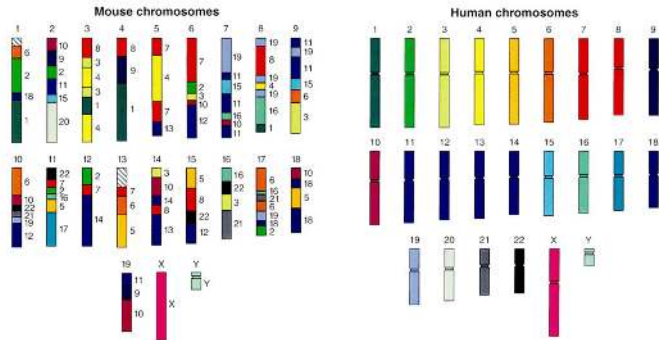
Comparative Genetics and its Importance

Basic assumption of genetic research: similar mechanisms control genetic processes in different organisms.

By studying the genetic mechanisms in simpler genetic models, we will be able to uncover basic principles of genetic inheritance.

Using the knowledge of genetics gained in the study of model organisms, we will be able to understand the mechanisms of genetic inheritance in human beings.

Mouse and Human Genetic Similarities



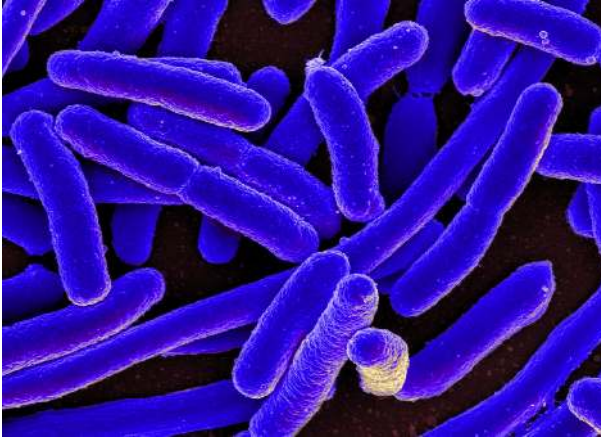
YCA 98-075R2

Courtesy Lisa Stubbs
Oak Ridge National Laboratory

This figure demonstrates the genetic similarity (homology) of the superficially dissimilar mouse and human species. The similarity is such that human chromosomes can be cut (schematically at least) into about 150 pieces (only about 100 are large enough to appear here), then reassembled into a reasonable approximation of the mouse genome. The colors and corresponding numbers on the mouse chromosomes indicate the human chromosomes containing homologous segments.

Favorite model organisms of geneticists

E. coli (4,377 genes, one chromosome)



Scanning electron micrograph of *Escherichia coli*.
<https://www.niaid.nih.gov/diseases-conditions/shiga-toxin-producing-e-coli>

Neurospora sp. (~10,000 genes, 14 chromosomes)

C. elegans (~20,000 genes, 12 chromosomes)

Drosophila sp. (~14,000 genes, 8 chromosomes)



Drosophila melanogaster.
Mr.checker,
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Mus musculus (3 billion bp, 23,786 genes, 40 chromosomes)



Mus musculus.
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A Brief Overview of the Scientific Method

Genetics, like any other field of science, deals with observable and measurable phenomena.

Scientific method is a thought process that scientists use to produce a comprehensive and objective (i.e., unbiased, not dependent on an individual opinion) explanation of the processes that govern natural world. Scientific knowledge doesn't have a value system; it generates and tests hypotheses based upon observations.

"Pluralitas non est ponenda sine neccesitate", plurality is not to be posited without necessity (Parsimony Rule/ Ockham's Razor).

"Of the two competing explanations, both of which are consistent with the observed facts, we regard it as right and obligatory to prefer the simpler" (Barker, 1961, p. 273).

Steps of scientific inquiry

- Observation and description of a phenomenon.
- Formulation of a tentative explanation for the observed phenomenon (formulation of a hypothesis).
- Making predictions based on the hypothesis.
- Testing the predictions by conducting a series of experiments.
- If the results of the tests support the predictions, the hypothesis becomes a theory.
- The theory becomes generally accepted when different research groups perform independent tests of the theory and obtain comparable results.

Science look-alikes

Beware of what's called "science by press conference," when

scientists announce their results to the media, before they've been reviewed by their peers and published in scientific journals. An example of such activity is the announcement in November of 2001 of the first successful attempt to clone a human embryo made by Dr. West, the director of the Advanced Cell Technology (ACT) company in MA, which was made at a press-conference, before other scientists in the field got a chance to take a look at it. Upon closer examination by peers, it appeared that the achievement wasn't so great, the clone didn't survive past the third cell division, but the media turned this premature scientific announcement into a sensation and the procedure drew sharp criticism, followed by health policy changes, from the US Government. Thus, to the public (and the government) it appeared as though the scientists succeeded in cloning a human being.

Chapter 2. The Rules of Genetic Inheritance

The Relationship Between Genotype and Phenotype

This chapter provides an overview of the principles of classical (Mendelian) genetics. The relationship between the physical appearance (phenotype) of a living being and the genetic determinants (factors, as they used to be called) was first described by a Czech monk Gregor Mendel in the 1860s.

- Dominant trait/allele
 - Recessive trait/allele
 - Homozygous/Heterozygous
 - Phenotype/Genotype
- Monohybrid cross (follows one character)/Dihybrid cross (two characters are being followed)

Classical (Mendelian) transmission genetics assumes a binary relationship and discrete phenotypes based on the predictable expression of contributing alleles. In this relationship, each allele is either dominant or recessive and it determines one of the alternative phenotypes (white/red, smooth/wrinkled etc.). A homozygous genotype produces the phenotype determined by the corresponding allele, while a heterozygous genotype produces the phenotype determined by the dominant allele.

Traditionally, the most common phenotype observed in natural populations, and hence its underlying genotype, was referred to as the wild type. However, when it comes to humans, the designation of a “natural population” and, consequently, the wild type, is problematic. Another way to think about the wild type is that to have it determine the original state of the phenotype (and the corresponding genotype).

To use this designation objectively, we need to have a good idea of the origin and distribution of the original state type, which is aided by the knowledge of the ultimate origin of a species.

When it comes to humans, there is ample scientific evidence that the cradle of the *Homo sapiens* species is in Africa.

Thus, the most common phenotypes observed in African and African-origin populations would be the most likely candidates for the original state types.



In humans, brown eye color is considered to be the original state type color of the eyes. About three-fourths of eye color variation can be explained by

genetic changes in and around the OCA2 gene located [on chromosome 15](#). While there is a wide variability in eye color, colors other than brown only exist among individuals of European descent. African and Asian populations are typically brown-eyed. In 2008 a team of researchers studying the OCA2 gene published results demonstrating that the allele associated with blue eyes occurred only within the last 6,000 – 10,000 years within the European population (Eiberg et al., 2008). Studies of DNA extracted from ancient human remains confirmed the timing of the blue color gene. Such as, the remains of a 7,000 year-old Mesolithic hunter gatherer from Spain revealed that he had blue eyes, along with dark hair and skin pigmentation. also suggesting that the light skin pigmentation arose in Europe after the blue eye color mutation came about (Olalde et al., 2014).



[https://de
epai.org](https://de.epai.org)

Phenotypes & genotypes: the relationship is rarely simple

In some cases, genetics may be difficult to interpret because of the complexity of how a genotype manifests itself as a phenotype.

Factors that affect the Mendelian phenotype/genotype relationship:

Dominance relationships:

- Overdominance (otherwise known as heterosis, hybrid

vigor or heterozygote advantage)

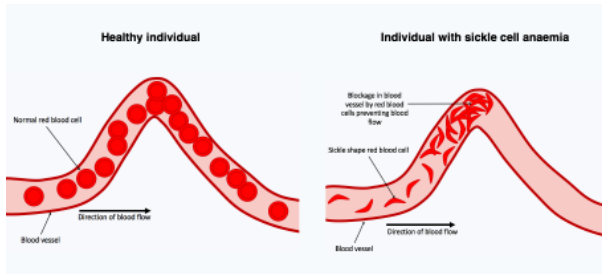
- Incomplete dominance
- Codominance
- Multiple alleles
- Epistasis / gene interaction
- Penetrance
- Expressivity
- Mosaicism
- Imprinting

Levels Of Dominance

Overdominance

Overdominance is a condition in genetics where the phenotype of the heterozygote lies outside the phenotypical range of both homozygote parents. Overdominance can also be described as heterozygote advantage, wherein heterozygous individuals have a higher fitness than homozygous individuals. An example in humans is sickle cell anemia.

Sickle cell anemia is a recessive disorder characterized by abnormal (sickle-like) shape of the red blood cells. This change in shape leads to a reduced capacity of erythrocytes to carry oxygen; which leads to a variety of health complications, including a lower life expectancy. The condition is a result of a mutation in the hemoglobin gene. In people heterozygous for the sickle cell allele, the health problems are minor, since the normal allele is able to produce normal hemoglobin.



*Sickle Cell Anaemia red blood cells in blood vessels.
Credit: Kep17, CC BY-SA 4.0, via Wikimedia Commons.*

However, this allele also yields some resistance to malaria. Therefore, in regions where malaria exerts or has exerted a strong selective pressure, the sickle cell allele has been selected for its conferred partial resistance to the disease. Since the gene is incompletely recessive, carriers can produce a few sickled red blood cells, not enough to cause symptoms, but enough to give resistance to malaria.

The sickle cell allele seems to have originated independently in at least five times in different geographic locations, such as Atlantic West Africa (the Senegal **haplotype**), Central West Africa (Benin haplotype), Cameroon (Cameroon haplotype), as well as India (the Adavasi haplotype) and eastern Saudi Arabia.

Incomplete dominance

In classical genetics, one allele is usually considered as completely dominant, one as completely recessive. Some mutations (amorphs or nulls) are completely recessive, but in many cases they have some residual activity (hypomorph or leaky) – incomplete dominance.

In situations of incomplete dominance, the heterozygote (Aa) has an intermediate phenotype between both homozygotes. One of the most well-studied examples of incomplete

dominance in humans occurs in the genes for curly hair. Inheriting a gene for curly hair from one parent and a gene for straight hair from the other parent will give a hair texture that is a blend of the two, wavy hair.



Wavy hair.
<https://deepai.org>

Codominance (additivity)

Codominance occurs when both alleles are separately expressed in the phenotype. In codominance, the heterozygote exhibit the phenotypes of both homozygotes.

A1A2 expresses both the A1 and A2 phenotypes (human MN, ABO blood groups).

Multiple alleles of a single gene

In a population of individuals, a given gene may have several alleles (for example, one wild type and the rest mutant). Some traits are determined by more than one variation of either allele.

Example: There are 60+ different mutations in the CFTR gene that could lead to cystic fibrosis of various degrees of severity (expressivity), depending on a particular mutation.

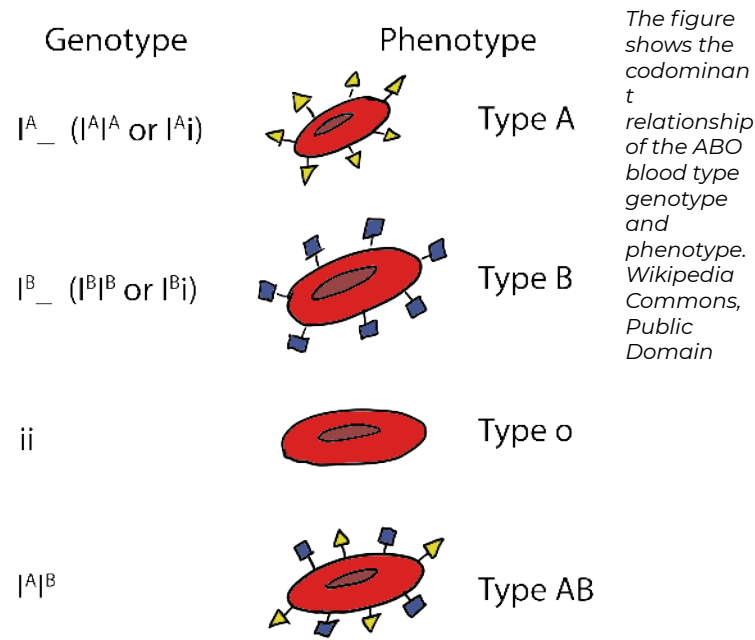
Although a gene may be represented by multiple alleles in a

population, a single individual will still carry only two of these alleles, one on each homologous chromosome.

Example of multiple alleles and co-dominance: ABO blood groups

Antigens are surface markers on the cell membrane that can stimulate an immune response. Antigens can be proteins, carbohydrates, or a combination of the two types of macromolecules (known as glycoproteins).

The red blood cell antigens of the ABO system are carbohydrates. ABO antigens are produced by a series of enzymatic reactions (additions of monosaccharide units) performed by glycosyl transferases. The A and B antigens are synthesized from a precursor oligosaccharide (H antigen). The H antigen itself does not stimulate an immune response.



The ABO blood type is determined by three alleles that specify enzymes that turn the precursor molecule into a functioning antigen:

- IA (produces A antigen, antibodies against B in plasma);
- IB (B antigen, anti-A antibodies);
- iO (lacks either A or B antigens (no modification to the precursor molecule), carries both anti-A and anti-B antibodies).

IA and IB are co-dominant to each other and both are dominant to iO. Thus, the ABO blood group system is also an example of multiple alleles.

Epistasis, interactions between genes

Epistasis is a genetic state where the effects of a given gene on a biological trait are masked or enhanced by one or more other genes.

Epistasis refers to the observation that some phenotypes only arise when specific alleles of two (or more) genes interact with each other to cause a specific phenotype. Genetic interaction can proceed at many different levels. In most familiar cases, it is the genes' products that interact with one another.

In humans, the Bombay phenotype is an example of epistasis. In the Bombay phenotype, the absence of a synthesis of a precursor antigen (H) can prevent the A or B antigens from being produced.

The H antigen is a precursor to the A and B antigens. If only recessive alleles for the H antigen are inherited (hh), the H antigen will not be produced. Subsequently, the A and B antigens also will not be produced. The result is an O phenotype by default since a lack of A and B antigens is the O type. In this case, an O blood type child can be born to parents with no iO allele.

Other examples of epistatic effects on traits include height and obesity.

Penetrance

Penetrance indicates the fraction of individuals of a particular genotype who show the expected phenotype.

An individual may have an abnormal genotype but show no phenotype.

Example: a human disease. If only 75% of AA individuals have the disease, the penetrance is 75%.

Incomplete penetrance in humans: polydactyly (extra digits), an autosomal dominant condition.

Polydactyly doesn't always express in all who carry the gene, but is passed on to the next generation even in those individuals who don't express it (skips a generation). Polydactyly also shows epistasis.

Another example of penetrance: PKU (environmentally affected).

Other genes may "modify" the expected phenotype.

Expressivity

Expressivity refers to a variation in a phenotype in individuals carrying a particular genotype.

Example of varying expressivity in humans: Neurofibromatosis (NF).

NF is an autosomal dominant disease of the nervous system. Mild NF symptoms include various pigmented spots on the skin. Severe NF causes additional abnormalities (bone deformities etc.).

There are two types of NF. NF1 is due to a mutation in the neurofibromin gene located on chromosome 17 (17q11.2). This

mutation can be caused by an insertion of a mobile genetic element (Alu). The frequency of NF1 in general population is 1 in 3,500.

NF2 is caused by a mutation in the gene that specifies neurofibromin 2, a tumor suppressor protein, located on chromosome 22 (22q12).

Example: Marfan Syndrome

Another example of variable expressivity is Marfan syndrome. Marfan syndrome is an autosomal dominant disease caused by a mutation in collagen formation. It affects about 1/5,000 live births. Symptoms of Marfan syndrome include skeletal, optical, and cardiovascular abnormalities. Skeletal abnormalities include arachnodactyly (long fingers and toes), extreme lengthening of the long bones, scoliosis, rib and sternum abnormalities, among others. Optical abnormalities almost always include ectopia lentis, a dislocation of the lens into the anterior chamber of the eye. Cardiovascular abnormalities may be numerous and include possible dissecting aneurysms, which are largely responsible for the shorter life span of Marfan syndrome patients as a group. Each patient may express all of the symptoms, or only a few. That is variable expressivity. Each patient with the mutant allele for Marfan syndrome expresses at least one of the symptoms, but the physician may have to look closely. Almost all are taller than average, but a lot of non-Marfan individuals are tall. Almost all have long fingers, but so do a lot of non-Marfan persons. Some may be very mildly affected and lead normal lives while others, more severely affected, have a shorter life expectancy. The disease is maintained in the population through recurrent mutations and the matings of less severely affected individuals with normal individuals. The extent of severity of affected does not affect the severity of expression in the next generation, that is, the offspring of mildly affected

individuals range from mildly affected to severely affected, with equal probability.

Famous people suspected to have had Marfan syndrome: Abraham Lincoln (1809-1865), Nicolò Paganini (1782-1840) (a famous Italian violinist, violist, guitarist, and composer).

Genetic mosaicism

In genetic medicine, a mosaic or mosaicism denotes the presence of two populations of cells with different genotypes in one individual, who has developed from a single fertilized egg.

Different types of mosaicism exist, such as gonadal mosaicism (restricted to the gametes) or tissue mosaicism; one of them is chimerism, where two or more genotypes arise from the fusion of more than one fertilized zygote in the early stages of embryonic development. In the more common mosaics, different genotypes arise from only a single fertilized egg cell, due to errors in early mitoses. In rare cases, intersex conditions can be caused by mosaicism where some cells in the body have XX and others XY chromosomes.

Cellular mosaicism in females most commonly results from a random inactivation of an X chromosome in females in early development, where all cells in an organism have the same genotype, but a different copy of the X chromosome is expressed in different cells. The coloring pattern of calico cats is the result of X inactivation. In an animal heterozygous for X-linked (associated with the X chromosome) alleles for fur color, different parts of the cat's skin contain genes that produce differently colored fur, based on which alternative allele is expressed from one or the other X chromosome in the group of skin cell beneath a particular color patch.

Dosage compensation and the

X-chromosome inactivation

Dosage compensation is a genetic regulatory mechanism, which operates to equalize the expression of characteristics determined by genes on the X chromosome so that they are equally expressed in the human XY male and the XX female.

Inactive X chromosomes can be seen in interphase nuclei as darkly staining heterochromatic masses. These masses are called Barr bodies after the cytologist who discovered them.

The mechanism of dosage compensation in humans is a random inactivation of an X chromosome in the embryo (day 16 post-fertilization in humans). At the 500-1000 cell stage, each embryonic cell randomly decides to inactivate either the paternally-derived X or the maternally-derived X. This “decision” is inherited by all of the descendants of the original cell. There is an X-linked mutation that causes anhidrotic ectodermal dysplasia (absence of sweat glands). Females heterozygous for the mutation display large patches of skin lacking sweat glands. This is because the decision to inactivate the X is made randomly at early stages of development, leading to mitotic clones that eventually become patches of normal and mutant skin.

Genomic Imprinting

The pattern of inheritance when expression of a gene depends on whether it is inherited from the mother or the father is called genomic imprinting. The mechanism of imprinting is uncertain, but, it may involve DNA methylation.

DNA methylation is a process by which methyl groups are added to the DNA molecule. Methylation can change the activity of a DNA segment without changing the sequence. When located in a gene promoter, DNA methylation typically acts

to repress gene transcription. DNA methylation is essential for normal development and is associated with a number of key processes including genomic imprinting, X-chromosome inactivation, repression of transposable elements, aging and carcinogenesis.

The Prader-Willi syndrome (PWS) in humans is caused by the absence of part of the long arm (q11-13) of the paternally derived chromosome 15. In 70-80% of PWS cases, the region is missing due to a deletion. In normal individuals, certain genes from this region are not expressed from the maternal chromosome, but are expressed from the paternal chromosome. When these paternally derived genes are absent or disrupted, the PWS phenotype results. When this same segment is missing from the maternally derived chromosome 15, a completely different disease, Angelman syndrome, arises.

Additional challenges to the genotype-phenotype relationship

Polygenic Inheritance

Classic phenotypes correspond to one of two sorts: green v. yellow, wrinkled v. smooth. Such phenotypic distribution is usually attributed to traits under a single gene control. Traits specified by more than one gene express continuous variation. Continuous variation, also known as quantitative variation, is observed when a spectrum of phenotypes exist rather than two or three discrete, qualitative differences.

When variation is quantitative, phenotypes often follow a normal distribution, in which the extreme phenotypes are relatively rare, and intermediate phenotypes are the most common.

In polygenic inheritance, several genes contribute to the

phenotype in an additive manner (i.e., enhancing or diminishing the trait's expression, like height or the intensity of skin pigmentation).

If polygenic inheritance involves just a few loci with complete dominance, there may be several distinct phenotypes among the offspring.

Pleiotropy: one gene affecting many traits

The condition where a single gene has an effect on more than one phenotype is called pleiotropy. Examples of pleiotropy in humans include juvenile-onset diabetes mellitus and PKU.

PKU is an autosomal recessive disorder that affects 1 in 10,000 newborns in the US (1 in 50 is a carrier). PKU results from the absence of phenylalanine hydroxylase (PAH) that converts phenylalanine to tyrosine. L-tyrosine is used to produce norepinephrine, epinephrine, dopamine. It is also a precursor of melanin. L-tyrosine aids in the functions of the adrenal, thyroid, and pituitary glands. Thyroxine (a thyroid hormone, made from L-tyrosine) is involved in regulation of metabolic rate, mental health, and growth rate.

Symptoms of PKU include mental retardation, melanin deficiency (lighter skin, blue eyes). The most effective treatment is dietary restriction of phenylalanine from birth to adolescence.

The history of PKU has been traced through Viking conquerors and French settlers to present day conditions; it has been suggested that there may be a heterozygous advantage in PKU that operates through protection against the toxic effects of ochratoxin A (Wolf, 1986). Ochratoxin A, an N-acyl derivative of phenylalanine, acts by competing with phenylalanine for phenylalanyl-tRNA synthetase, thus bringing protein synthesis to a halt. Doses of phenylalanine sufficient to raise the blood

phenylalanine concentration reduce or reverse the toxicity of ochratoxin A.

This mycotoxin is produced by several species of *Aspergillus* and *Penicillium* that infest stored grains and other foods. The mild, wet climate of Ireland and West Scotland tends to encourage the growth of molds. Furthermore, these areas have suffered repeated famines during which moldy food was eaten. Heterozygous women appear to have a lower spontaneous abortion rate (Eisensmith and Woo, 1992).

Another example of pleiotropy (in association with epistasis): red haired people may have a different perception of pain

The gene called MC1R causes 80% of all cases of red hair. (Other cases of red hair are due to mutations outside of the MC1R gene.) It is responsible for dark pigment deposition in hair. If an individual carries two recessive copies of the MC1R gene, the pigment is not deposited and the hair color appears red. Apparently, people who are homozygous for the inactive MC1R gene, have different pain sensitivity than the rest of us. Some red-haired people are more sensitive to pain, while others respond quite the opposite. The exact nature of this differential response to pain remains unknown, and the research on the genetic pathways of MC1R continues.

The Red Hair Gene

How often do you meet a person with red hair? It depends on where you are. In England and Scotland,

for example, where the trait shows up at its highest frequency, the chance is about 1/30. Overall in the world, people with red hair comprise about 2% of the population.

The gene responsible for hair color is called [MC1R](#). It is located on chromosome 16. The MC1R gene provides instructions for making a protein called the melanocortin 1 receptor. This receptor plays an important role in normal pigmentation. The receptor is primarily located on the surface of melanocytes, which are specialized cells that produce a pigment called melanin. Melanin is the substance that gives skin, hair, and eyes their color. Melanin is also found in the light-sensitive tissue at the back of the eye (the retina), where it plays a role in normal vision. The MC1R gene comes in several versions, called alleles.

The melanocortin 1 receptor controls which type of melanin is produced by melanocytes. When the receptor is activated, it triggers a series of chemical reactions inside melanocytes that stimulate these cells to make eumelanin (the brown and black melanin). If the receptor is not activated or is blocked, melanocytes make pheomelanin, which is responsible for lighter-colored pigmentation.



<https://deepai.org/>

Common alleles of the MC1R gene are associated with normal differences in skin and hair color. Certain genetic variations are most common in people with red hair, fair skin, freckles, and an increased sensitivity to sun exposure.

These MC1R alleles lead to the production of the melanocortin 1 receptor with a reduced ability to stimulate eumelanin production, causing melanocytes to make mostly pheomelanin. One must inherit two copies of a particular red hair-associated allelic variant of the MC1R gene to actually have red hair. A person carrying a red hair-producing allele on one of the two chromosomes #16 (you inherit one from your mother and another from your father) and an allele producing a fully functional version of MC1R will have darker hair color. Those alleles that require just one copy to be present in the genome to phenotypically express the trait they specify are called dominant. The alternative alleles, like the red hair-causing MC1R allele, are called recessive.

Heritability, environmental effects and phenotypic plasticity

Heritability of a trait is defined as ratio of genetic variance to total phenotypic variance (genetic and environment): $H^2 = V(G)/V(P)$. The parameter H^2 is the broad-sense heritability and reflects all possible genetic contributions to a population's phenotypic variance. Included are effects due to allelic variation

(additive variance), dominance variation or which act epistatically (multigenic interactions), as well as maternal and paternal effects, where individuals are directly affected by their parents' phenotype (such as with milk production in mammals). Pathological innumeracy, known as dyscalculia, is often associated with neurological lesions.

The genetic component of the phenotype's variation of a particular trait (heritability) can be measured by comparing the trait's expression in "affected" families with that of the general population. The genetic component has to show up more frequently in related individuals compared to the general population: twins > siblings > adopted children = general population.

How much of the phenotypic expression of a trait is inherited and how much of it is due to the environment? A meta-study of heritability correlations in twins published in 2015 by *Nature Genetics* (Polderman et al., 2015) revealed what geneticists have been suspecting all along: there's an almost even split of the influence between environment and genetics on the expression of our phenotypic traits. The finding is based on a review of 2748 studies involving 14 million twin pairs from across 39 countries. The researchers were able to determine the contribution of genetics and the environment on the trait by measuring how similar various traits are between identical twins and non-identical twins. "If the trait is genetic then you would expect identical twins will be more similar than the non-identical twin. The more similar an identical twin to a non-identical twin then we can infer the trait is largely due to the genetic factor," says Dr Beben Benyamin, one of the authors of the article from the Queensland Brain Institute in Australia. For all traits, the average heritability is about 0.49, or 49%, while the environment accounts for the remaining 51% of traits' variation. However the heritability of different traits varies widely. For example, schizophrenia is 70% heritable, while the heritability of longevity in humans is well below 10% (Ruby et al., 2018).

Height, one of the best-studied human traits with respect to heritability, is 86% heritable. At the same time, knowing the heritability of a trait does not provide information about which genes or environmental influences are involved, or how important they are in determining the trait.

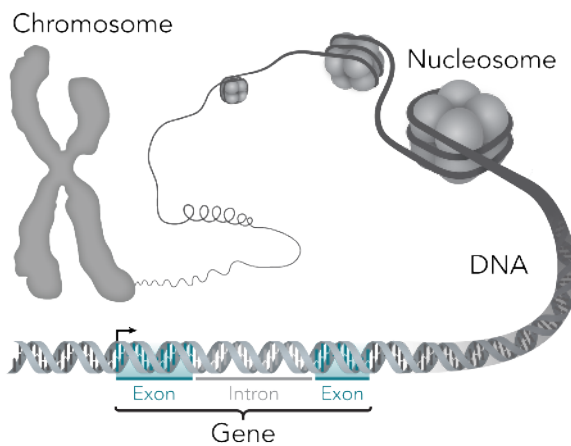
Phenotypic plasticity and norm of reaction

Phenotype will depend upon genetic and environmental factors, and the interaction between them. Phenotypic plasticity is the ability of an organism to change its phenotype in response to changes in the environment. The range of phenotypic possibilities for a single genotype, as influenced by the environment, is called the norm of reaction. The term was introduced to genetics by W. L. Jahannsen, a Danish biologist who also contributed to the advancement of our understanding of the basic principles of selection through his studies on true-bred lines.

Chapter 3.

Chromosomal Composition and Nomenclature

Chromosomes And Chromatin



A schematic depiction of a coding region in a segment of eukaryotic DNA. Thomas Spletstoeser (www.scistyle.com), CC BY-SA 4.0, via Wikimedia.

Just before cell division (and after DNA synthesis), the chromatin condenses into individual chromosomes. The dividing chromosomes appear as two chromatids. The chromatids appear to be made of coiled loops of 20-30 nm-thick chromatin fibers.

The structural unit of chromatin is the nucleosome.

Each nucleosome consists of eight histone molecules (two

each of histones H2A, H2B, H3, and H4) associated with 146 nucleotide pairs of DNA and a stretch of linker DNA about 50 nucleotide pairs in length. The diameter of the nucleosome “bead,” or core particle, is about 10 nm. Histone H1 (not shown) is thought to bind to the linker DNA and facilitate the packing of nucleosomes into 30-nm fibers.

Chromosome nomenclature

In human and other genomes, chromosomes are broken down into arms, regions, and bands.

Chromosome arms are defined with respect to the kinetochore (centromere). A centromere is a region of DNA found along the length of a chromosome where two identical sister chromatids come in contact. It is involved in cell division as the point of mitotic spindle attachment. Centromeres function in sister chromatid adhesion and pairing of homologous chromosomes during meiosis. During mitotic and meiotic divisions, a protein structure called kinetochore is formed on top of the centromeres. The kinetochores are the sites where the spindle fibers attach during the cell division to pull the chromosomes apart.

The centromere usually contains specific tandem repetitive DNA sequences, which are often called “satellite DNA”. At the end of chromosomes, DNA sequences called telomeres are composed of the same type repetitive sequences as well. The centromeres are, together with telomeres and origins of replication, one of the essential parts of any eukaryotic chromosome.

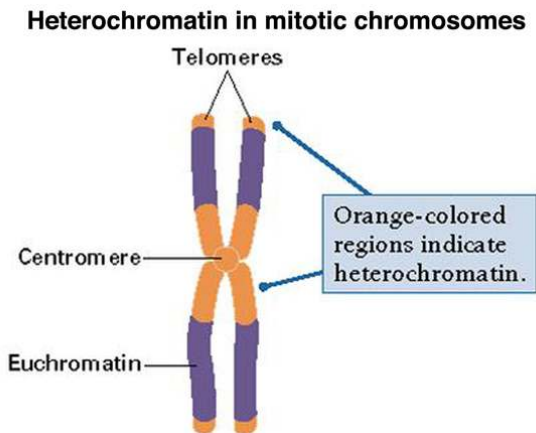
Although the terms “kinetochore” and “centromere” are used interchangeably in chromosome nomenclature, it is probably more correct to refer to kinetochore, since we don’t really see the centromere, which is simply a DNA sequence. Rather, we

see the proteinaceous kinetochore, and the distinct heterochromatic staining that occurs in the vicinity of the centromere.

The long arm of chromosomes is represented with the letter q and short arm is designated as p (p = "petite").

Heterochromatin and euchromatin

Heterochromatin is a tightly packed form of DNA, whereas euchromatin is a lightly packed form. Heterochromatin is found in genetically inactive regions, such as the centromeric and telomeric regions of chromosomes. Heterochromatin stains darker (more intensely) than euchromatin. Euchromatic regions usually display a higher amount of genetic activity, and the lighter packing allows access to the information encoded into the DNA of euchromatic regions.



Heterochromatin and euchromatin in eukaryotic chromosomes. Credit: <http://www.discoveryandinnovation.com/Biol202/notes/lecture18.html>

Types of chromosomes

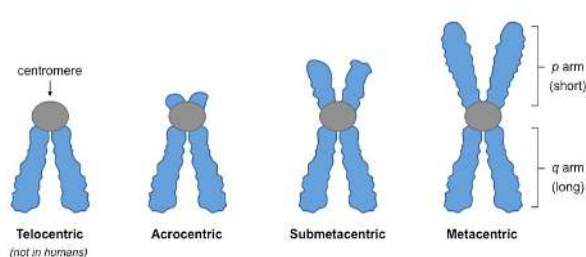
There are three types of chromosomes in humans, based upon the position of the centromere.

Metacentric: In this type of chromosome the arms are of equal length. Human chromosomes 1, 3, 16, 19, 20 are metacentric.

Submetacentric: In this type of chromosome one arm is slightly longer than the other side. Human chromosomes 2, 4, 5, 6-12, 17, 18, X are submetacentric.

Acrocentric: In this type of chromosome one arm is significantly longer than the other.

A small round or rod-shaped structure called satellite, attached by a very thin thread is variably present on the short arm of acrocentric chromosomes. There are five acrocentric chromosomes with satellites in the human genome: 13-15, 21, and 22. Chromosome Y is considered acrocentric as well but has no satellites.



Types of eukaryotic chromosomes according to the lengths of chromosomal arms. Telocentric chromosomes are not found in humans. Credit: <https://ib.bioninja.com.au/standard-level/topic-3-genetics/32-chromosomes/chromosome-types.html>

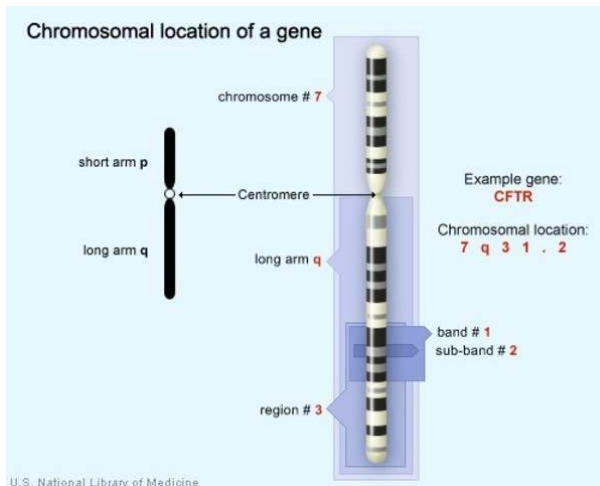
Banding pattern of human chromosomes

Chromosomes in metaphase can be identified using certain staining techniques (banding). Most conventional cytogenetic analyses depend on the karyotyping of banded metaphase chromosomes.

A band is defined as that part of a chromosome which is clearly distinguishable from its adjacent segments by appearing darker or brighter when stained with a dye. In G-bands (produced by Giemsa dye), the dark regions tend to be heterochromatic, whereas the light regions tend to be euchromatic. Hatched pattern represents regions that vary in staining intensity.

The chromosomal nomenclature starts with the

chromosome itself. For example, 15q represents the long arm of chromosome 15. The first digit after the arm indicates the region within the arm. Numbers increase going away from the kinetochore on each arm: 15p1, 15p2, 15p3, or 15q1, 15q2, 15q3, and so on. The next digit indicates specific band within that region. Such as, 15q13 refers to chromosome 15, long arm, region 1, band 3. Digits after decimal points indicate additional bands resolved more recently after the original nomenclature was established in 1971.

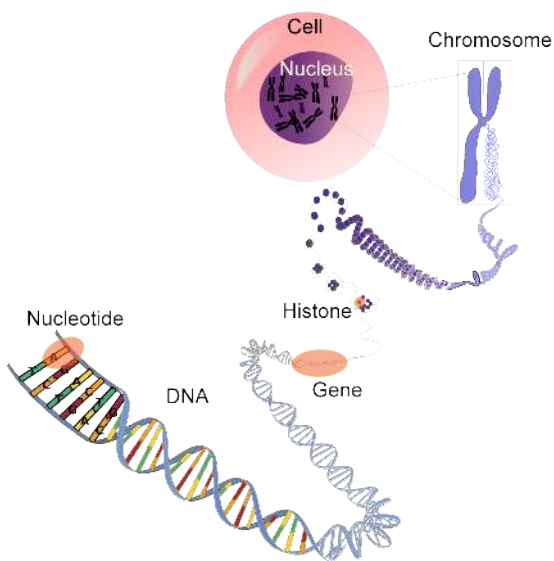


An example of the chromosomal location using the cystic fibrosis gene, CFTR. Credit: U.S. National Library of Medicine

Chapter 4. DNA Replication

DNA and Chromosomes

Apart from reproductive cells (gametes), each cell of the human body contains 23 pairs of chromosomes, each a packet of compressed and entwined DNA. Every strand of the DNA is a huge natural polymer of repeating nucleotide units, each of which comprises a phosphate group, a sugar (deoxyribose), and a base (either adenine, thymine, cytosine, or guanine). Every strand thus embodies a code of four characters (A's, T's, C's, and G's), the recipe for the machinery of human life.



*DNA organization in human cells.
Source: Wikimedia Commons, public domain.*

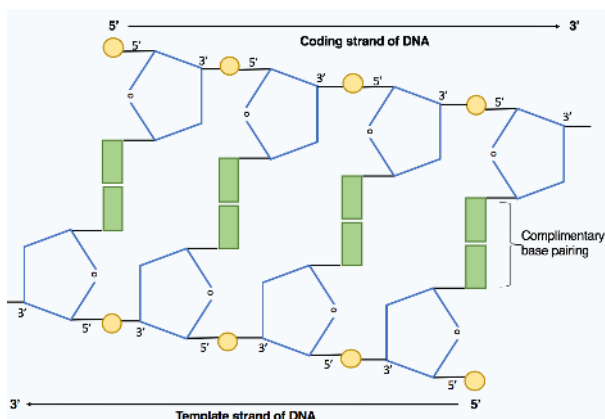
In its normal state, DNA takes the form of a highly regular double-stranded helix, the strands of which are linked by hydrogen bonds between adenine and thymine (A:T) and between cytosine and guanine (C:G). Each such linkage is said to constitute

a base pair; some three billion base pairs constitute the human genome. It

is the specificity of these base-pair linkages that underlies the mechanism of DNA replication illustrated in this chapter. Each strand of the double helix serves as a template for the synthesis of a new strand, the nucleotide sequence of which is strictly determined. Replication thus produces twin daughter helices, each an exact replica of its sole parent.

DNA directionality

DNA strands have directionality, and the different ends of a single strand are called the “3' (three-prime) end” and the “5' (five-prime) end”. These terms refer to the carbon atom in deoxyribose to which the next phosphate in the chain attaches. In addition to being complementary, the two strands of DNA are antiparallel: They are orientated in opposite directions. This directionality has consequences in DNA synthesis, because DNA polymerase can synthesize DNA in only one direction by adding nucleotides to the 3' end of a DNA strand.



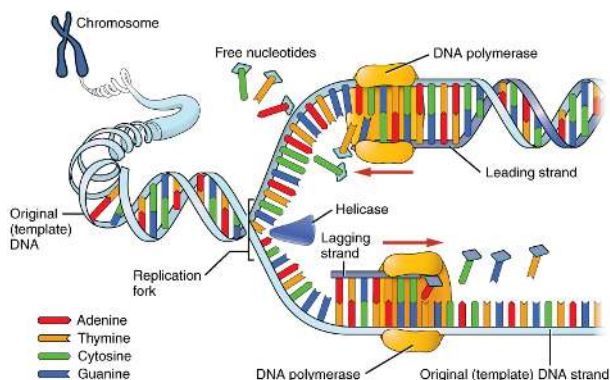
DNA directionality. Credit: Kep17, https://commons.wikimedia.org/wiki/File:Directionality_of_DNA_molecule.png, BY 4.0 (<http://creativecommons.org/licenses/by/4.0>)

DNA replication

DNA replication begins at a fixed origin of replication (ori) and usually involves both DNA strands at the same time. The replication is initiated by addition of an RNA primer by the enzyme primase.

In the newly forming nucleic acid strand, nucleotides are added one at a time to the 3'-OH (the 3rd carbon of deoxyribose) end of the growing chain (in 5' to 3' direction) by the enzyme DNA polymerase.

The fixed position of the ori sites and the 5'-3' direction of DNA synthesis makes it possible for only one out of two new DNA strands to be synthesized continuously. This strand is called the leading strand. The other strand (the lagging strand) is synthesized in pieces, called the Okazaki fragments.



DNA
transcription.
<https://images.app.goo.gl/CyyeQ2Q6fLkxyqp7>

Because DNA replication does not begin at either end of the DNA strand, but starts in the center, and considering that all DNA polymerases that have been discovered synthesize a new DNA strand in the 5' to 3' direction, the leading and the lagging strands on the DNA molecule are replicated differently.

On the leading strand, DNA polymerase can make a complementary DNA strand without any difficulty because it goes from 5' to 3'. However, there is a problem going in the other direction on the lagging strand. To counter this, short sequences of RNA acting as primers attach to the lagging strand a short distance ahead of where the initiation site was. The DNA polymerase can start replication at that point and go to the end of the initiation site. This causes the formation of Okazaki fragments. More RNA primers attach further on the DNA strand and DNA polymerase comes along and continues to make a new DNA strand.



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view them online here: <https://pressbooks.gvsu.edu/humangenetics/?p=72#oembed-1>

Replication of the ends of DNA molecules

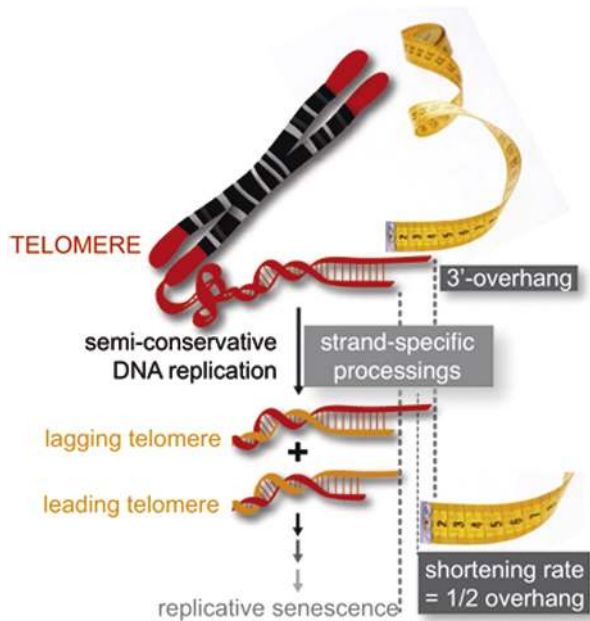
Eventually, the last RNA primer attaches at the end of the lagging strand, and DNA polymerase, RNA nuclease (to remove RNA nucleotides of the primer), and DNA ligase come along to convert the RNA (of the primers) to DNA and to seal the gaps in between the Okazaki fragments. But, in order to change RNA to DNA, there must be another DNA strand in front of the RNA primer. This happens at all the sites of the lagging strand, but it does not happen at the 5' end of either newly formed DNA strands where the last (for the lagging strand) or first (for the leading strand) RNA primer is attached. Ultimately, that RNA is destroyed by enzymes that degrade any RNA left on the DNA.

Thus, a section of the telomere is lost during each cycle of replication at the 5' end of both the leading and the lagging strands. As a result, repeated rounds of replication produce shorter and shorter DNA molecules in somatic cells.

Telomeres

To prevent damage to DNA ends during repeated rounds of replication, eukaryotic chromosomes are capped with repeated DNA sequences called telomeres. In all vertebrates, the sequence is TTAGGG. The repetitive sequences that compose telomeres do not carry any essential genetic

information and, thus, their loss during replication does not result in functional problems for the chromosome. But, with enough rounds of replication, the repetitive ends of chromosomes do “wear out.” When telomeres become too short, a cell can seize any subsequent cell division and enter into a state of “replicative senescence.”



Telomere shortening and replicative senescence
 . Credit: Soudet,, Jolivet & Teixeira, 2014,
<https://doi.org/10.1016/j.molcel.2014.02.030>

In reproductive cells, embryonic stem cells and cancer cells, telomere sequences are extended by the enzyme telomerase. Telomerase belongs to a class of replicative enzymes called **reverse transcriptases**, which use RNA molecule as a template to synthesize a complementary DNA strand. Telomerase adds telomeric sequence repeats (TTAGGG) to the terminal 3' ends of the template DNA strands and DNA strands and DNA polymerase then extends the telomeric 5' ends of the newly

synthesized strands using the RNA oligonucleotide carried by telomerase as a primer. The existence of a compensatory shortening of telomere (telomerase) mechanism was first predicted by a Russian biologist Alexey Olovnikov in 1973, who also suggested the telomere hypothesis of aging and the telomere's connections to cancer. Telomerase was discovered by Carol W. Greider and Elizabeth Blackburn in 1984 in the ciliate *Tetrahymena*. Together with Jack W. Szostak, Greider and Blackburn were awarded the 2009 Nobel Prize in Physiology or Medicine for their discovery.

While the original discoveries of telomerase and its role have concluded that there is little, if any, telomerase activity in somatic cells, this understanding has been challenged in recent years. A NASA study of twin astronauts (the preliminary findings of which were announced in January 2018) revealed that telomeres do get significantly extended during a prolonged space flight. Remarkably, it appears that the majority of those telomeres shorten within a couple of days of the return to Earth.

A recent study determined that the level of telomerase activity is dependent on concentrations of deoxyguanosine diphosphate (dGDP). Guanine happens to be the first nucleotide added by telomerase to the terminal 3' end of the template DNA strand. A substitution in the RNA template of telomerase leading to the switch in first nucleotide added from G to A leads to a shift in telomerase repeat addition activity stimulation from being dGDP-dependent to being dATP-dependent, leading to a dramatic increase in telomerase activity in somatic cells (Chen et al., 2018).

Role of telomeres and telomerase

Cancer cells are able to escape cellular senescence by

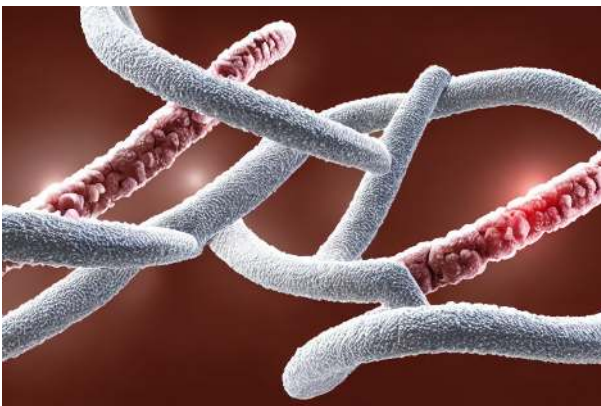
rebuilding their telomeres after each replication round. This is possible because cancer cells can reactivate their telomerase, which is normally inactive in adult somatic cells. Several mechanisms of telomerase activation in cancer cells have been proposed. One of these mechanisms is a mutation in the promoter region of the [hTERT](#) gene on chromosome 5 that encodes telomerase in humans. When mutated, the hTERT promoter is able to initiate the transcription of this somatically silent gene, leading to the production of functional telomerase.

Telomeres are also involved in tumor initiation and progression. Longer telomeres seem to offset the progression of carcinomas.

Telomeres are involved in aging. Shortening of telomeres shuts down mitosis and leads to progressive “wearing out” of actively dividing tissues. Telomere length could be inherited, contributing to the heritability of longevity. At the same time, longer telomeres encourage malignant proliferation.

Telomeres also participate in the regulation of meiosis. Abnormal telomere extension seems to disrupt meiotic division.

Telomeres are involved in the maintenance of bone marrow function. Telomerase function appears to help maintain and even accelerate osteogenesis in human bone marrow cells.



<https://deepai.org>

Chapter 5. Gene Structure

What are Genes

Genes are basic units of inheritance. Genes hold the information to build and maintain an organism's cells and pass genetic traits to offspring. All organisms have many genes corresponding to various different biological traits, some of which are immediately visible, such as eye color or number of limbs, and some of which are not, such as blood type or increased risk for specific diseases, or the thousands of basic biochemical processes that comprise life.

The genetic code

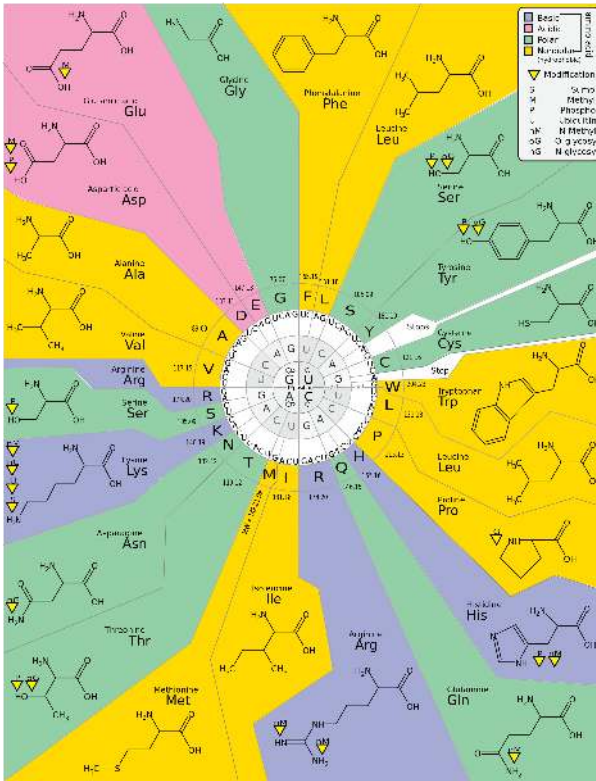
The genetic code is the set of rules by which information encoded in genetic material (DNA or mRNA sequences) is translated into proteins (amino acid sequences) by living cells.

The code defines how sequences of three nucleotides, called codons, specify which of the 20 naturally occurring amino acids will be added next during protein synthesis. This was understood by the biochemists since the beginning of the analysis of DNA as the blueprint of life. The challenge was to understand how the DNA molecule, that consists only of four "letters" (A, T, G, and C) can encode 20 amino acids. The simplest assumption is that one DNA "letter" specifies one amino acid. This, however, leaves 16 amino acids without a code. If we increase the complexity one step forward, and

suppose that two DNA “letters” code for one amino acid. $4^2=16$; we are still four “words” short of 20. Increasing the complexity one step further, we can imagine that a combination of three “letters” specifies one amino acid. $4^3 = 64$; now we have too many “words”, but this is the most parsimonious combination.

The theoretical genetic “vocabulary” was verified experimentally in the early 1960s. It has been determined that of the 64 nucleotide “triplets” (codons), 61 code for amino acids; three out of 64 serve as termination sequences (stop codons) during protein synthesis. Many amino acids are coded by more than one codon (the code is redundant).

The vast majority of genes are encoded using the same code, but there are also many variant codes. For example, protein synthesis in human mitochondria relies on a genetic code that differs from the standard genetic code.



The Genetic Code.
 Credit:
 dunk,
<https://www.flickr.com/photos/dullhunk/8676707450/>
 / (CC BY 2.0)

Not every nucleotide in a DNA sequence part of the genetic code. All organisms' DNA contains regulatory sequences, intergenic segments, chromosomal structural areas, and other non-coding DNA that can contribute greatly to phenotype. Those elements operate under sets of rules that are distinct from the codon-to-amino acid paradigm underlying the genetic code.

Codons appear on the coding (sense) DNA strand as they are read in the 5' to 3' direction. Each protein-coding gene is transcribed into a molecule of the related polymer RNA, using

the other DNA strand in the double helix (called the template stand).

Therefore, the sequence of nucleotides in the RNA copy of the DNA code appears exactly the same as the code of the DNA, except that the nucleotide thymidine (T) is found in place of uridine (U). Since the discovery of the genetic code took place through the analysis of RNA copies of the genetic message, the traditional genetic code table consisted of RNA codons. However, with the rise of computational biology and genomics, proteins have become increasingly studied at a genomic level. As a result, the practice of representing the genetic code as a DNA codon table has become more popular.

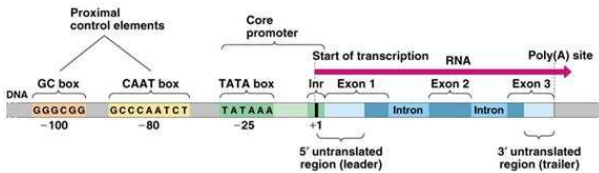
Structure of a eukaryotic gene

A gene is a sequence of nucleotide triplets of a DNA molecule bound by a start codon (ATG) and a stop codon (TGA, TAA, or TAG) that specifies a cellular product.

Most of the time, the cellular product “coded” in DNA is a protein, in which case the nucleotide triplets in a gene will specify a specific sequence of amino acids on a polypeptide chain. A gene’s final product can, sometimes, be an RNA molecule (rRNA, tRNA).

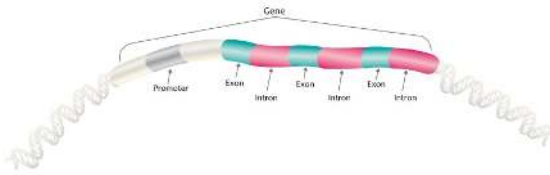
In humans, like in other eukaryotes, the region of the DNA coding for a protein is usually not continuous. This region is composed of alternating stretches of exons (the actual nucleotide sequences that carry a message delivered to the site of polypeptide assembly) and introns (non-coding “spacer” sequences that take part in regulation of genetic expression at the mRNA level). During transcription, both exons and introns are transcribed onto the RNA, in their linear order. Thereafter, a process called splicing takes place, in which, the intron sequences are excised and discarded from the RNA sequence.

The remaining RNA segments, the ones corresponding to the exons are ligated to form the mature RNA strand.



Structure of a eukaryotic gene and the accompanying control elements (discussed in detail in Chapter 6). [image015.jpg](https://www.copyright.gov/title17/92chap1.html#107) The use of the image adheres to the Fair Use application of 17 U.S. Code § 107, <https://www.copyright.gov/title17/92chap1.html#107>.

A typical human multi-exon gene starts with the promoter region, which is followed by a transcribed but non-coding region called 5' untranslated region (5' UTR), which contains the start codon (ATG). Following the 5'UTR, there is an alternating series of introns and internal exons, followed by the terminating exon, which contains the stop codon (either TGA, TAA, or TAG). It is followed by another non-coding region called the 3' UTR. Ending the gene, there is a stretch of the adenine nucleotide repeating several times (also called the polyadenylation (polyA) signal). The exon-intron boundaries (i.e., the splice sites) are signaled by specific short (2 nucleotide-long) sequences.



Components of a eukaryotic gene.

Credit: Genomics Education Programme, CC BY 2.0, via Wikimedia Commons.

On average, a vertebrate gene is around 30,000 nucleotides (kilobases, or Kb) long, out of which the coding region is only about 1 Kb long. The average coding region consists of six exons, each about 150 nucleotides (base pairs, since DNA is double stranded, abbreviated as bp) long. Huge deviations from the average are observed. For example, the gene called dystrophin is 2.4 million (Mb) bp long. Blood coagulation-factor VIII has 26 exons whose size varies from 69 bp to 3106 bp, with the total coding region reaching length around 186 Kb and the introns lengths adding up to 32.4 Kb. Intron number 22 produces two transcripts unrelated to this gene, one for each strand. An average 5' UTR is 750 bp long, but it can be longer and span several exons (for example, in the MAGE family). On average, the 3' UTR is about 450 bp long, but examples exist where its length is 5 Kb (e.g., the gene for Kallman's syndrome).

Chapter 6.

Mechanisms of Gene Expression

The flow of genetic information

Otherwise known as the Central Dogma of Molecular Biology (Crick, 1970), the flow of genetic information in living cells proceeds from DNA to RNA to protein. Since the final folding of polypeptides is the result of interaction of the R groups of amino acids that polypeptides are made of, the last element in the three-step genetic information flow is more correctly referred to as being a polypeptide that makes up a protein. Genetics contains no information about the protein function, only the protein structure.

The transfer of genetic information consists of three stages. It begins with the DNA replication, which was discussed earlier. The other two stages convert DNA information into a sequence of RNA nucleotides (transcription), which is subsequently converted into a sequence of amino acids to produce a polypeptide molecule (translation). Transcription and translation are the required steps to express genetic information as a functional product or a process that leads to the outward appearance (i.e., the phenotype) of most organismal traits. This **gene expression** begins with transcription of DNA into an RNA molecule, which is a tightly regulated process.

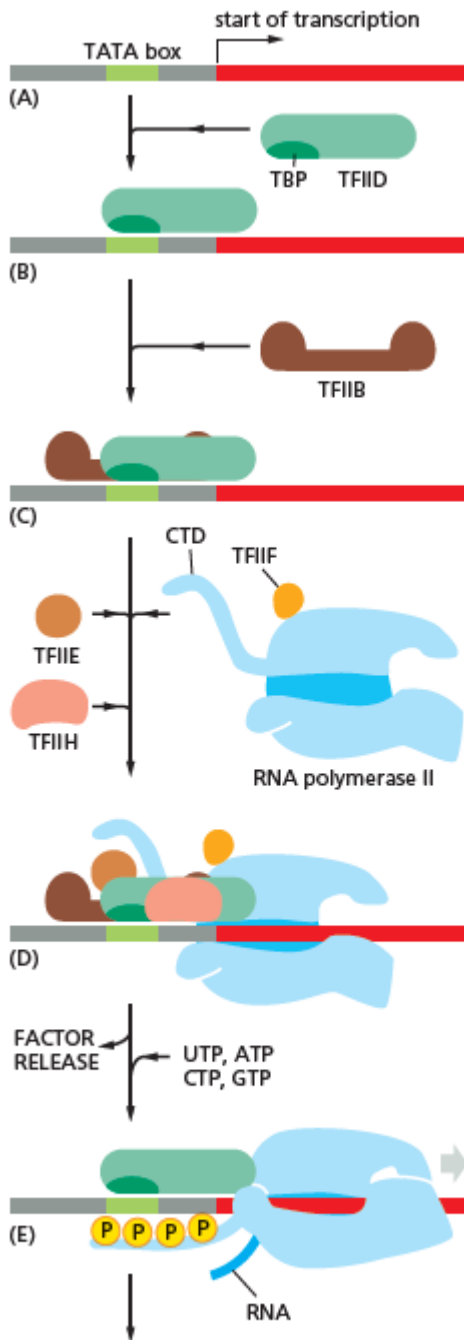
Transcriptional regulatory elements

In eukaryotes, RNA polymerase, and therefore the initiation of transcription, needs to be able to find a gene to be transcribed among the billions of nucleotides in the genome. To do that, it relies on specific activity of various regulatory elements. With respect to their location relative to the gene, regulatory elements can be proximal and distal. They can also be cis- and trans-acting. (Cis: on the same DNA molecule. Trans – on a different molecule.) The main groups of regulatory sequences are promoters and enhancers.

Promoter elements can be found immediately next to the Transcription Initiation Site (TSS) or at a distance from it. The former elements together with TSS compose the core promoter. Proximal promoter elements are regions located up to a few hundred base pairs from the core promoter, and typically contain multiple binding sites for other regulatory elements.

The core promoter is the region at the start of a gene that serves as the docking site for the transcriptional machinery. Promoters are regions of DNA that promote transcription and, in eukaryotes, are found at -30, -75, and -90 base pairs upstream from TSS. Core promoters are sequences within the promoter that are essential for transcription initiation. RNA polymerase is able to bind to core promoters in the presence of various specific transcription factors.

The most common component of core promoter in eukaryotes is a short DNA sequence known as a TATA box, found 25-30 base pairs upstream from TSS. The TATA box is the binding site for a transcription factor known as the TATA-binding protein (TBP), which is itself a subunit of another transcription factor, called Transcription Factor II D (TFIID).



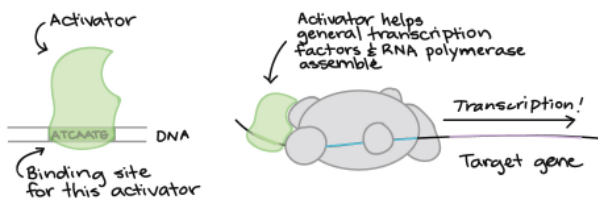
Initiation of eukaryotic transcription. Credit: Ravinesh rds, [CC BY-SA 4.0](#), via [Wikimedia Commons](#).

Enhancers are genetic sequences that can be recognized by activators, which are proteins with DNA-binding capability (transcription factors) that increase the rate of transcription from a specific gene upon binding to an activator sequence.

Other transcriptional regulatory elements include silencers and insulators. Silencers are distal genetic sequences that can repress gene transcription upon binding with proteins called repressors. Insulators are nucleotide sequences that function to block genes from being affected by the transcriptional activity of neighboring genes.

Transcription factors

In molecular genetics, a transcription factor (sometimes called a sequence-specific DNA-binding factor) is a protein that binds to specific DNA sequences, thereby controlling the flow (or transcription) of genetic information from DNA to mRNA. Transcription factors perform this function alone or with other proteins in a complex, by promoting (as an activator), or blocking (as a repressor) the recruitment of RNA polymerase (the enzyme that performs the transcription of genetic information from DNA to RNA) to specific genes. When binding to insulator sequences, transcription factors prevent co-transcription of neighboring genes along with the target gene.

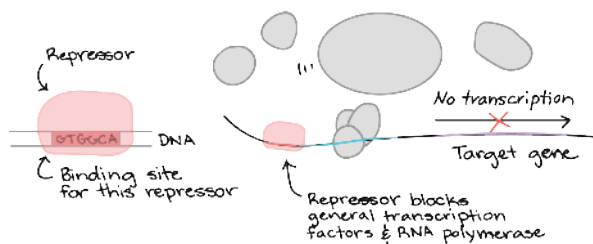


Transcription factors as activators.

Credit:

<https://www.khanacademy.org/science/ap-biology/gene-expression-and-regulation/regulation-of-gene-expression-and-cell-specialization/a/eukaryotic-transcription-factors>

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Transcription factors as repressors.
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Sometimes, all three functions can be accomplished by the same transcription factor. For example, a gene called CTCF encodes a transcriptional regulator that is able to bind different DNA target sequences and proteins. Depending upon the context of the site, the protein can bind a histone acetyltransferase (HAT)-containing complex and function as a transcriptional activator or bind a histone deacetylase (HDAC)-containing complex and function as a transcriptional repressor. If the protein is bound to a transcriptional insulator element, it

can block communication between enhancers and upstream promoters, thereby selectively regulating downstream gene expression.

Transcription factors directly control when, where, and the extent to which genes are expressed. Signal transduction pathways are responsible for either activating or inhibiting many of them. Transcription factors are also regulated by cofactors, forming complexes that can activate or inhibit transcriptional activity. Many transcription factors, such as nuclear receptors, reside in the cytoplasm and enter the nucleus upon activation (e.g., ligand binding). Posttranslational modifications and coregulating proteins provide additional layers of regulation. Transcription factors are involved in a wide variety of processes, such as development, stress responses, and immunity. Activation or inhibition of transcription factors is often dysregulated during oncogenesis. Transcription factors can also be dysregulated during developmental processes, promoting or inhibiting cellular differentiation. Analyzing the expression, regulation, activity, and sequence of transcription factor genes can help determine their relative importance to the biology of the cellular or disease processes under study.

A defining feature of transcription factors is that they contain one or more DNA-binding domains (DBDs), which attach to specific sequences of DNA adjacent to the genes that they regulate. Additional proteins such as coactivators, chromatin remodelers, histone acetylases, deacetylases, kinases, and methylases, while also playing crucial roles in gene regulation, lack DNA-binding domains, and, therefore, are not classified as transcription factors.

Transcription factors are essential for the regulation of gene expression and are, as a consequence, found in all living organisms. The number of transcription factors found within an organism increases with genome size, and larger genomes tend to have more transcription factors per gene.

There are approximately 2,600 proteins in the human genome that contain DNA-binding domains, and most of these are presumed to function as transcription factors. Therefore, approximately 10% of genes in the genome code for transcription factors, which makes this family the single largest family of human proteins. Furthermore, genes are often flanked by several binding sites for distinct transcription factors, and efficient expression of each of these genes requires the cooperative action of several different transcription factors. Hence, the combinatorial use of a subset of the approximately 2,000 human transcription factors easily accounts for the unique regulation of each gene in the human genome during development.

Transcription factors use a variety of mechanisms for the regulation of gene expression. These mechanisms include:

- Stabilize or block the binding of RNA polymerase to DNA.
- Recruit coactivator or corepressor proteins to the transcription factor DNA complex.
- Catalyze the acetylation or deacetylation of histone proteins. The transcription factor can either do this directly or recruit other proteins with this catalytic activity.

Many transcription factors use one or the other of two opposing mechanisms to regulate transcription:

- Histone acetyltransferase (HAT) activity – acetylates histone proteins, which weakens the association of DNA with histones, which make the DNA more accessible to transcription, thereby up-regulating transcription;
- Histone deacetylase (HDAC) activity – deacetylates histone proteins, which strengthens the association of DNA with histones, which make the DNA less accessible to transcription, thereby down-regulating

transcription.

Stages Of Genetic Expression

Transcription

Transcription is the process of creating a complementary RNA copy of a sequence of DNA. Transcription is the first step leading to gene expression. Genetic transcription converts the sequence of DNA nucleotides containing information about polypeptide structure into a corresponding sequence of RNA nucleotides. As opposed to DNA replication, transcription results in an RNA complement that includes uracil (U) in all instances where thymine (T) would have occurred in a DNA complement. Transcription takes place inside the nucleus.

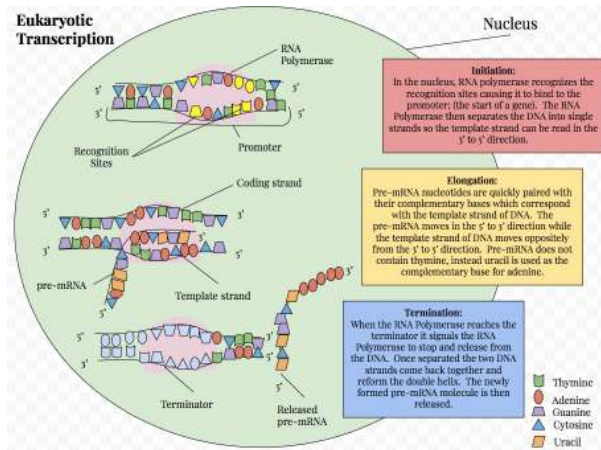
During transcription, the genetic information coded in the DNA is transferred to an RNA molecule. The RNA molecule produced in the course of transcription (called a messenger RNA or mRNA) can leave the nucleus (the site of DNA transcription in eukaryotes) and attach to the ribosomes, where free amino acids are joined together according to the nucleotide sequence of a particular mRNA molecule.

RNA nucleotides are linked together by the transcription enzyme RNA polymerase.

RNA polymerase synthesizes mRNA using the non-coding (anti-sense) strand of DNA as the template. The other DNA strand is called the coding strand, because its sequence is the same as the newly created RNA transcript (except for the substitution of uracil for thymine). The use of only the 3' → 5' strand produces a continuous mRNA molecule, thus eliminating the need for the Okazaki fragments seen in DNA replication. As in DNA replication, DNA is read from 3' → 5'

during transcription. Meanwhile, the complementary RNA is created from the 5' → 3' direction.

Transcription is divided into three main stages: initiation, elongation and termination.



Stages of eukaryotic transcription. Credit: Erinp.5000, CC BY-SA 4.0, via Wikimedia Commons.

Initiation

After TFIID binds to the TATA box via the TBP, five more transcription factors and RNA polymerase combine around the TATA box in a series of stages to form a preinitiation complex.

One transcription factor, DNA helicase, has helicase activity and so is involved in the separating of opposing strands of double-stranded DNA to provide access to a single-stranded DNA template. However, only a low, or basal, rate of transcription is driven by the preinitiation complex alone. Other proteins known as activators and repressors, along with any associated coactivators or corepressors, are responsible for modulating transcription rate.

Transcription factors mediate the binding of RNA polymerase

and the initiation of transcription. Only after certain transcription factors are attached to the promoter does the RNA polymerase bind to it. The completed assembly of transcription factors and RNA polymerase bind to the promoter, forming a transcription initiation complex.

Elongation

As transcription proceeds, RNA polymerase traverses the template strand and uses base pairing complementarity with the

DNA template to create an RNA copy. Although RNA polymerase traverses the template strand from 3' → 5', the coding (non-template) strand and newly-formed RNA can also be used as reference points, so transcription can be described as occurring 5' → 3'. This produces an RNA molecule from 5' → 3', an exact copy of the coding strand (except that thymines are replaced with uracils, and the nucleotides are composed of a ribose (5-carbon) sugar where DNA has deoxyribose (one less oxygen atom) in its sugar-phosphate backbone).

Messenger RNA transcription can involve multiple RNA polymerases on a single DNA template and multiple rounds of transcription (amplification of particular mRNA), so many mRNA molecules can be rapidly produced from a single copy of a gene.

Elongation also involves a proofreading mechanism that can replace incorrectly incorporated bases. In eukaryotes, this may correspond with short pauses during transcription that allow appropriate RNA editing factors to bind. These pauses may be intrinsic to the RNA polymerase or due to chromatin structure.

Termination

RNA transcription stops when the RNA polymerase encounters

a G-C-rich inverted repeat followed by a run of uracil nucleotides. The G-C rich repeat forms a hairpin (or stem-loop structure) via internal complementary binding within the inverted repeat sequence. When the hairpin forms, the mechanical stress breaks the weak rU-dA bonds, now filling the DNA-RNA hybrid. This pulls the poly-U transcript out of the active site of the RNA polymerase, in effect, terminating transcription. In the “Rho- dependent” type of termination, a protein factor called “Rho” destabilizes the interaction between the template and the mRNA, thus releasing the newly synthesized mRNA from the elongation complex.

Transcription termination in eukaryotes is less understood but involves cleavage of the new transcript followed by template- independent addition of As at its new 3' end, in a process called polyadenylation.



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Pre-mRNA modification – splicing

The mRNA splicing is an important step in creating the mRNA that is involved in protein synthesis, via the process of translation. Key factors in this process include: RNA, possessing introns and exons, and the spliceosome.

Several signals exist within the intron that are used in the splicing process. From the 5' end of the intron, these are, GU, the A branch site, a pyrimidine-rich region, and the 3' AG. The

AG and GU sequences define the beginning and end of the intron.

Splicing is mediated by the spliceosome, which consists of several protein-RNA complexes. The first step involves two complexes that bind near the GU sequence. The RNA is then looped, and three other protein-RNA complexes bind. This final complex then undergoes a conformation change. As the spliced mRNA is released from the spliceosome, the intron de-branches, and is then degraded.

Introns are non-coding RNA sequences that must be removed before translation. The process of removing the intron is called splicing. The intron is then cleaved at the 5' GU sequence and forms a lariat at the A branch site. The 3' end of the intron is next cleaved at the AG sequence, and the two exons are ligated together.

Selective removal of introns can add to the diversity of polypeptides that can be encoded on the same gene. If an intron is left in the mature mRNA molecule after splicing, it will be interpreted by the ribosome during translation as the end of translation sequence. Thus, an un-removed intron would function much like a stop codon during translation. Therefore, a selective removal of introns from the primary mRNA transcript can result in mature mRNA of different lengths, thus leading to the translation of polypeptides of different sizes.

Post-transcriptional mRNA processing also includes the addition of a 5' "cap" and a poly-adenine (poly-A) tail to the mRNA molecule.

5' End Capping

Post-transcriptional processing of the 5' end of the RNA product of DNA transcription comes in the form of a process called the 5' cap. At the end of transcription, the 5' end of the RNA transcript contains a free triphosphate group since it

was the first incorporated nucleotide in the chain. The capping process replaces the triphosphate group with another structure called the “cap”. The cap is added by the enzyme guanyl transferase. This enzyme catalyzes the reaction between the 5' end of the RNA transcript and a guanine triphosphate (GTP) molecule.

The Poly A Tail

Post-transcriptional RNA processing at the opposite end of the transcript comes in the form of a string of adenine bases attached to the end of the synthesized RNA chain. This string of adenine is called the “poly A tail”. The addition of the adenines is catalyzed by the enzyme poly (A) polymerase, which recognizes the sequence AAUAAA as a signal for the addition. The reaction proceeds through mechanism similar to that used for the addition of nucleotides during transcription.



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Translation

Genetic translation is a conversion of nucleotide sequence of mRNA into a sequence of amino acids joined together via peptide bonds to form a polypeptide. Translation takes place in the cytoplasm and involves mRNA, a special type of RNA mole-

cules called transport RNA (tRNA), ribosomes, as well as free amino acids found in the cytoplasm.

Translation takes place on the ribosomes in the cytoplasm. (RNA is also a part of the ribosomal structure! It's called the ribo- somal RNA, or rRNA.)

A special RNA molecule called transfer RNA or tRNA, mediates the translation of the triplet code of amino acids into the amino acid sequence.

As the mRNA passes through the ribosomes during translation, the tRNA molecules recognize the codons on the mRNA and bring the corresponding amino acids to the assembly site. Each following amino acid in the polypeptide sequence is attached to the previous one by a peptide bond.

When the translation machinery reaches a stop codon on the mRNA, the translation stops and the newly formed polypeptide separates from the ribosomes and dispatches into the cytoplasm for further processing. Polypeptide chain coils and folds spontaneously, due to the nature of its chemical bonds, forming a functional protein of specific conformation.



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Key Takeaways

- The Central Dogma of Molecular Biology states that genetic information flows only in one direction, from DNA, to RNA, to protein.
- Eukaryotic gene expression is regulated by promoter regions, enhancer, silencer and insulator DNA sequences and transcription factors that bind to these sequences.
- Transcription factors bind to enhancer, silencer, insulator, or promoter regions of DNA adjacent to the genes that they regulate. Depending on the transcription factor and the DNA region the transcription factor binds to, the transcription of the adjacent gene is either up- or down-regulated. Such as, transcription factor binding to an enhancer sequence of DNA regulating the expression of gene X, up-regulates gene expression by enhancing transcription from gene X. The same transcription factor binding to a silencer DNA sequence of the same gene shuts down DNA expression by preventing transcription from taking place.

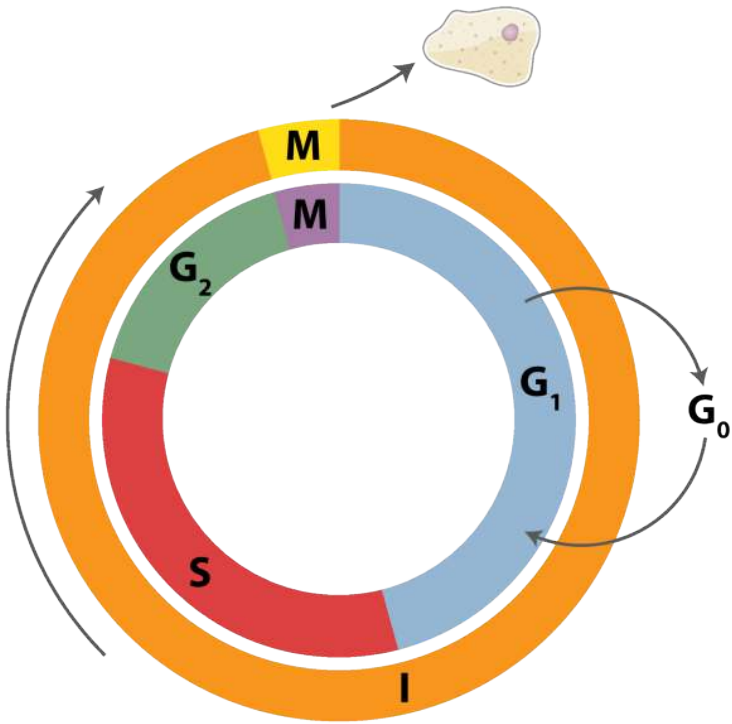
Chapter 7. Cell Cycle and Cell Division

The Cell Cycle

Dividing somatic cells go through a cell cycle. The cell cycle consists of the actual cell division stages (Mitosis) and the period between successive cell divisions, called the Interphase. The Interphase is subsequently divided into three sub-phases: Gap 1 (G1), Synthesis (S), and Gap2 (G2). G1 is the period between mitosis and the initiation of nuclear DNA replication. The cell actively performs its functions only during the G1 phase of the cell cycle.

The S phase is the period of nuclear DNA replication.

G2 involves duplication of cellular organelles and an increase in cellular size. G2 is the period between the completion of nuclear DNA replication and mitosis. G2 phase involves synthesis of proteins essential for cell division; it's a preparation phase for mitosis.



Cyclins, cyclin-dependent kinases, and the cell cycle check- points

There are three major checkpoints (restriction stages) at which the cell assess its needs (and ability) to divide. The main decision to divide takes place at the G_1 checkpoint. The G_2 checkpoint assesses the fidelity of DNA replication. The M checkpoint insures the chromosomes have been properly separated.

Multiple checkpoints in the eukaryotic cell cycle ensure that division occurs only after sufficient growth and faithful DNA replication, and only when favorable conditions exist. At each

checkpoint, numerous proteins engage in a series of carefully coordinated biochemical reactions. This complexity allows for precise regulation of all steps in the cell cycle and it is essential to preventing the devastating consequences of cell division gone awry.

If after the G1 checkpoint the cell is going on to mitosis, it begins to produce histones, which are critical for organizing and condensing chromosomes during mitosis. As the production of histones peaks, the S (synthesis) phase of the interphase begins.

As a cell enters the G1 stage, proteins called cyclins and cyclin- dependent kinases (CDKs) are synthesized. CDKs, which exist at constant concentrations throughout the cell cycle, can only become enzymatically active if they form associations with cyclins. Cyclins are a family of proteins that have no enzymatic activity of their own but activate CDKs by binding to them. The cyclin concentration in a cell varies throughout the cycle, depending on the signals from the cellular environment. If the cyclin concentration at G1 is low, the cell goes into a non-dividing state (G0 phase).

CDKs signal the cell to go into the next stage of the cell cycle by phosphorylating (supplying with energy) various other proteins involved in the cell cycle process. CDKs are regulated by changing concentrations of the cyclins.

The cell cycle and the signal transduction pathway

Cells are stimulated to divide by a variety of mechanisms, both internal and external. External signals are processed through a signal transduction pathway (STP). Signal transduction is the transmission of molecular signals from a cell's exterior to its

interior. Signals received by cells must be transmitted effectively into the cell to ensure an appropriate response.

The STP contains the following components: growth (stimulating) factors, growth factor receptors, relay molecules, and transcription factors.

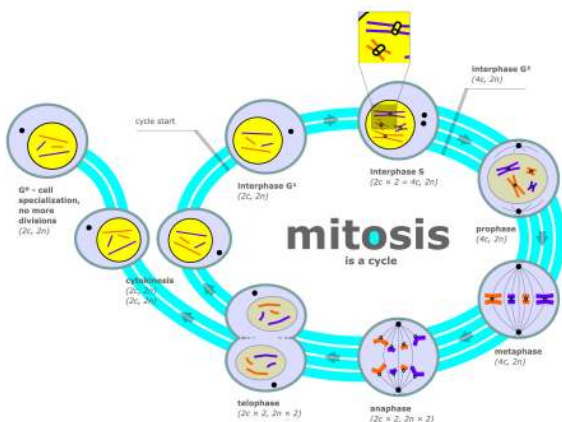
A growth factor is a naturally occurring substance capable of stimulating cellular growth, proliferation and cellular differentiation. It is usually a protein or a steroid hormone. Common growth factors include Platelet-derived growth factor (PDGF) and Insulin-like growth factor (IGF). The binding of growth factors to specific receptors on the plasma membrane is usually necessary to initiate cell division.

The p53 protein senses DNA damage and can halt progression of the cell cycle in both G1 and G2. Both copies of the p53 gene must be mutated for this to fail so mutations in p53 are recessive, and the p53 gene qualifies as a tumor suppressor gene.

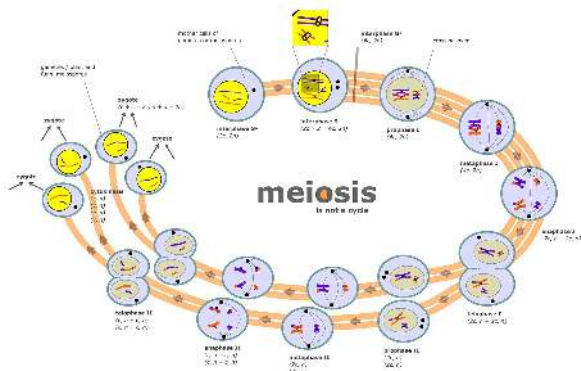
MAD ("mitotic arrest deficient") is a protein that assists in proper chromosome distribution to two daughter cells during mitosis. Cells deficient for MAD produce daughter cells with too many or too few chromosomes (aneuploidy, one of the hallmarks of cancer). Damage to nuclear DNA can activate p53 which, in turn, shuts down the cell cycle.

Mitosis, Meiosis

Mitosis is a type of cellular reproduction where a cell will produce an identical replica of itself with the same number and patterns of genes and chromosomes. Meiosis, on the other hand, is a special process in cellular division where cells are created containing gene patterns of different types and combinations with 50% of the number of chromosomes (one of the two sets) of the original cell.



A diagram
of Mitosis.
Credit:
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A diagram
of Meiosis.
Credit:
Wikimedia
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Meiosis is used in sexual reproduction of organisms to combine male and female, through the spermatozoa and ova, to create a new, singular biological organism.

Cell Cycle Control Breakdown And Cancer

When the cell cycle control system breaks down, a normal cell turns into a cancerous (tumor) cell.

Benign tumor cells are localized in a particular organ/tissue; malignant tumor cells can spread to nearby tissues.

The spread of cancer cells beyond their original site is called metastasis.

Genes that stimulate normal cell division, in their normal state, are called proto-oncogenes. When mutated, they are called oncogenes. Most oncogenes behave as dominant alleles (ex.: the ras proto-oncogene).

Genes that suppress normal cell growth and division are called tumor suppressor genes. Mutations in these genes release

cells from their controlling functions, such as DNA repair, cell adhesion, or cell-cycle inhibition. Tumor-suppressor genes usually behave as recessive alleles (ex.: BRCA1 and BRCA2).

An important class of tumor suppressor genes is the DNA repair genes. Mutations in these genes impair the ability of the cell to repair any further mutations, thus accelerating the progression of cancer.

All cancers are caused by the accumulation of genetic mutations that cause, or allow, a cell to grow out of control and become a tumor. Some mutations occur during somatic DNA replication when cells in the body copy their genetic material and an error is introduced into a gene (spontaneous mutations). Some mutations are caused by environmental factors. Some mutations occur in reproductive cells and then subsequently can be passed from parent to child, from one generation to the next. The latter type of mutation is said to be hereditary and may cause an inherited (familial) form of cancer.

Carcinogens and mutagens

Cancer is overwhelmingly a result of environmental factors. Exposure to mutagens in the environment increases mutation rate, thus increasing the chance of cancer. At least 50% of cancers are environmentally induced. Recent studies indicate that this can actually be up to 90%.

Carcinogens are mutagens that cause mutations which lead to cancer. Nearly 75% of the risk of colorectal cancer is due to diet. 75% of chance of developing head and neck cancers is because of tobacco and alcohol. 86% of cases of skin cancer are due to sun exposure. Direct continuous exposure to UV light (a carcinogen) leads to formation of thymine dimers (abnormally paired thymines), which interfere with the fidelity of DNA replication. Xeroderma pigmentosum is a form of skin cancer that results from mutations in DNA repair enzymes responsible for the removal of thymine dimers.

Two classes of genes appear to play major role in triggering cancer. Oncogenes can transform a normal cell into a cancerous cell by continuously inducing the cell to divide. Tumor suppressor genes are genes that normally suppress cell growth, but, when affected by a mutation, permit the cell to grow without restraint.

DNA repair genes ensure that each DNA strand is replicated (during the S phase of the cell cycle) without mistakes. DNA repair genes are classified as tumor suppressor genes.

Genes Involved In Human Cancer

Oncogenes

Oncogenes are mutated forms of proto-oncogenes. Normally, proto-oncogenes function as growth factors, growth factor

receptors, relay molecules and transcription factors. Proto-oncogenes regulate cell cycle by promoting cell division in response to an extracellular signal (hormones and other growth factors etc.) via the signal transduction pathway. Gain-of-function mutations in proto-oncogenes may lead a proto-oncogene to be permanently turned on. Such mutations in proto-oncogenes are usually dominant in nature (the product of an oncogene must be PRESENT in order to override the normal regulation of cell division). Mutations that permanently switch proto-oncogenes “on” result in abnormally accelerated cell division, which may lead to cancer.

Major mutational events that lead to proto-oncogenes changing to oncogenes are chromosomal translocations (in this case, proto-oncogene would be in the vicinity of the breakpoint) and amplifications of chromosomal regions harboring proto-oncogenes.

To date, more than 50 proto-oncogenes have been discovered. Examples of proto-oncogenes:

PDGF codes for a protein called platelet-derived growth factor (involved in some forms of brain cancer).

Ki-ras codes for a protein involved in a stimulatory signaling pathway (involved in lung, ovarian, colon, and pancreatic cancer).

MDM2 codes for a protein that is an antagonist of the p53 tumor-suppressor protein (involved in certain connective tissue cancers)

Tumor-Suppressor Genes

Tumor suppressor genes (TSGs) tell the cell to stop dividing. TSGs normally function in the cell by suppressing cell division. Loss-of-function mutations in TSGs may lead to an unregulated cell division (TSG gene products must be ABSENT in order for

cell division to take place) and to cancer. Mutations in TSGs are usually recessive, but can show dominant pattern of inheritance, ex., p53. Most familial (inherited) cancers are related to defects in TSGs.

Examples of TSGs:

- NF-1 codes for a protein that inhibits a stimulatory protein (involved in myeloid leukemia).
- RB codes for the pRB protein, a key inhibitor of the cell cycle (involved in retinoblastoma and bone, bladder, and breast cancer).
- BRCA1 codes for a DNA repair protein associated with the breast and ovarian cancers).

BRCA Gene, Should You Be Worried?

By Tianna McCormick | August 22, 2018

<https://raisevegan.com/brca-gene-should-you-be-worried/>

Everyone has a BRCA gene, and for most people, they function just fine. Without reading this or other articles, you probably wouldn't even be aware of them, or have any reason to be aware of them. These two BRCA genes do have a specific and important function though, since they produce a type of protein called Tumor Suppressor Protein. Sounds important right? Yes, we definitely want a lot of tumor-suppressing going on.

What is the BRCA Gene?

BRCA1, a gene on our 17th chromosome and BRCA2, a gene on our 13th chromosome, are both tumor suppressor genes that produce a tumor suppressor protein called Breast Cancer Type 1 Susceptibility Protein and Breast Cancer Type 2 Susceptibility Protein, respectively. Mutations in these genes can be inherited from either parent with a 50/50 chance of inheritance. So what happens when there is a mutation in one of these genes and what do people need to do about it? Let's understand the cell division process a little.

Human cells are always dividing through a process called mitosis. During the cell division process and the time before a cell dies, there are many different biochemical reactions taking place that direct the cell on DNA replication, performance, and even its death. The direction of these processes comes from the many of other proteins living inside the cell. One such protein, called a tumor suppressor protein, has a few goals, with the main purpose of controlling cell growth. They are able to do this by helping to repair damaged DNA.

When a cell divides, it replicates its own DNA. During replication, the double strand of DNA often breaks or has mistakes in the coding, leading to DNA mismatching. The role of the tumor suppressor is to repair these breaks and/or mismatches. If a mismatch or break is unable to be repaired, the tumor suppressor works with other cell proteins to induce programmed cell death, a process called apoptosis. This is a biochemical mechanism that occurs in our body on a

regular basis. And generally speaking, it works very well...until our tumor suppressor proteins don't work.

When a BRCA gene is mutated, the protein that is produced is non-functional. So not only is damaged DNA unable to be corrected, but also, the ability to induce apoptosis can also be inhibited, meaning that the cell can replicate with broken and mismatched DNA. As more of these damaged cells replicate, they will begin to form unhealthy tissue, leading to malignant tumors.

People with mutations still have one healthy functioning gene (because we are born with two of each gene) so there isn't a 100% chance that malignancy will occur. However, studies show that people with specific BRCA1 and BRC2 mutations have a 69% and 72% chance of developing breast cancer by the age of 80, respectively, and the risk of developing ovarian cancer is 44% and 17%, respectively. Considering that the general population has a 12% chance of developing breast cancer and a 1.3% chance for ovarian cancer, those numbers are considerable. So what does one do?

Enhanced screening through various scans and exams can catch cancers in the very early stages however there are limitations, especially with ovarian cancer, which has no proven effective screening tool. Chemoprevention and oral contraceptives are also options to lower breast and ovarian cancer respectively. People are also advised to have prophylactic or risk-reducing surgery. This includes bilateral

salpingo-oophorectomy, removal of both ovaries and fallopian tubes, and mastectomy, the removal of one or both breasts. However, the decision to have such invasive surgery is very personal and definitely not a choice all people are willing to make.

So what happens when you learn that you have a mutation in one of a BRCA gene? Learn everything you possibly can about your mutation and what your personal risks are. It is important to note that not all mutations are bad, so be sure to work closely with a medical care team who understands not only your mutation, but also your risk based on family history as well.

p53 tumor suppressor gene

p53 is “the guardian of the genome” because it causes cells that have DNA breaks and other DNA damage (“genotoxic injury”) to arrest in the G1 interval of the cell cycle or undergo programmed cell death, thereby preventing replication of the DNA breaks, which could result in broken chromosomes and progressive genetic instability that leads to cancer.

Inactivation of the p53 tumor suppressor is a frequent event in tumorigenesis. In most cases, the p53 gene is mutated, giving rise to a stable mutant protein whose accumulation is regarded as a hallmark of cancer cells. Mutant p53 proteins not only lose their tumor suppressive activities, but often gain

additional oncogenic functions that endow cells with growth and survival advantages.

The p53 tumor suppressor protein binds to DNA as a dimer or tetramer to regulate transcription of genes that mediate responses to cellular stress. Mutant p53 protein, which takes on an abnormal conformation, is more stable than the wild-type (half-life of several hours compared to 20 minutes for the wild type p53). Furthermore, it appears that the mutant p53 protein can dimerize with the wild-type p53 and inhibit its normal function. This loss of a function of a product from the “good” copy of the p53 gene in the presence of the product from a defective p53 is referred to as the loss of heterozygosity (LOH). LOH refers to a loss of the function of the wild type allele in a cell line of an individual who carries a mutation in a gene associated with cancer formation. Thus, the cell becomes essentially homozygous for the defective allele due to LOH, losing the ability to control (restrict) its cell division.

In familial p53-related cancers, such as the Li-Fraumeni syndrome, one defective copy of a gene is inherited. Sometimes a mutation in one the p53 alleles can occur spontaneously in somatic cells (sporadic cancers).

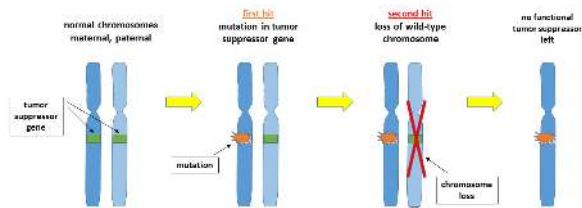
Genetic Pathways To Cancer

The Two-Hit hypothesis (A.G. Knudson, 1971)

Both alleles of a tumor suppressor gene must be inactivated to get cancer (one active allele is enough to provide cell cycle regulation). One copy of defective gene is inherited and the other one is inactivated somatically (through LOH or chromosome loss). This hypothesis explains inherited predisposition to cancer.

Example: retinoblastoma.

Children inherit one mutated pRB gene (one hit). Somatic mutation inactivates the second pRB gene (two hits) leading to retinoblastoma.



Loss of a tumor suppressor gene locus according to the two-hit model: In the first hit, the tumor suppressor gene on one of the two chromosomes is affected by a mutation that makes the gene product non-functional. This mutation may arise spontaneously as a DNA replication error or may be induced by a DNA damaging agent. The second hit removes the remaining wild-type chromosome, for example through a mitotic nondisjunction event. There are several other potential

mechanisms for each of the two steps, for example an additional mutation, an unbalanced translocation, or a gene deletion by recombination. As a result of the double lesion, the cell may become malignant because it is no longer able to express the tumor suppressor protein.

Credit: Wpessner, CC BY-SA 3.0, via Wikimedia Commons

The Mutation Accumulation Hypothesis

No single event is enough to turn a cell into a cancerous cell. The accumulation of damage to a number of genes ("multiple hits") across time leads to cancer.

Cancer results from an accumulation of somatic mutations in both TSGs and proto-oncogenes. Each new somatic mutation creates changes in the cell which contribute to the malignant phenotype.

Some mutations are common to many cancers, while others are specific.

Example: colorectal cancer.

Inactivation of APC tumor suppressor gene leads to benign tumors (polyps). Activation of k-ras oncogene and inactivation of 18q and TP53 tumor suppressor genes leads to intestinal carcinoma.

Mutation of more oncogenes and tumor suppressor genes leads to metastatic colorectal cancer.

The Immune Surveillance Theory

This theory proposes that oncogenic transformation of normal cells is a frequent occurrence but the immune system clears them as they emerge.

Cancer cells "give themselves away" by displaying cell specific signals on their cell surface (ex., prostate-sensitive antigen, PSA) that are recognizable by the immune system. During prolonged periods of stress (chronic stress) the immune system is suppressed and, thus, is not functioning at its highest

capacity and is unable to detect some of the tumor cells, which subsequently become cancerous. The suppression of the immune system function with advanced age lets cancer cells escape detection, resulting in an increased incidence of cancers in older people.

Recent studies on animal models as well as human cancer studies provide compelling evidence for the existence of tumor surveillance cells (Swann and Smyth, *J. Clin. Invest.* 117(5): 1137-1146 (2007)).

Chapter 8. Sex Determination and Sex Linkage

The Genetics of Sex Determination

Sex determination is a complicated series of physiological events and patterns are often controlled through hormonal expression.

Sexual characteristics can also be triggered by a change in hormonal balance, sometimes determined by an environmental change.

In some organisms (fruitflies) sex is determined by a ratio of X chromosomes to the number of autosomal sets. A 1:1 ratio produces a female, any deviation (1:2 etc.) produces a male.

For higher organisms, like humans, sex is determined by genes on the sex chromosomes. In humans, as in other animals, sex determination involves the XY system, in which sex is determined by the presence of the Y chromosome. The sex in which the sex chromosomes are not the same is referred to as heterogametic, or heteromorphic. Thus, in the case of humans, females are referred to as the homomorphic sex, and males are heteromorphic. Normal female karyotype contains two X chromosomes. Variations of the XX state include XO (Turner Syndrome), as well as XXX, XXXX (all are females).

The normal human male karyotype is XY males. Sex chromosome variations in males include Klinefelter Syndrome (XXY), as well as XXXY, XXXXY (are all males).

In humans, the presence of the Y chromosome leads to male

sex. The female sex is the default setting of development. Approximately six weeks into development the gonads become either the ovaries or the testes.

A chromosome in a diploid organism is called hemizygous when only one copy is present. Hemizyosity is observed in the heterogametic sex when a gene is located on a sex chromosome.

Sex-linked inheritance

Sex linkage is the phenotypic expression of an allele related to the chromosomal sex of the individual. This mode of inheritance is in contrast to the inheritance of traits on autosomal chromosomes, where both sexes have the same probability of inheritance. Since humans have many more genes on the X than the Y, there are many more X-linked traits than Y-linked traits.

X-Linked Recessive Disorders

X-linked recessive traits are expressed in all heterogametics, but are only expressed in those homogametics that are homozygous for the recessive allele. For example, an X-linked recessive allele in humans causes hemophilia.

Hemophilia is caused by a deficiency in blood clotting factors, which impairs the body's ability to control blood clotting or coagulation – a necessary function to control bleeding if a blood vessel is ruptured. Hemophilia A (clotting factor VIII deficiency) is the most common form of the disorder, present in about 1 in 5,000–10,000 male births. Hemophilia B (factor IX deficiency) occurs in around 1 in about 20,000–34,000 male births. A non- sex-linked hemophilia C due to coagulant

factor XI deficiency, which can affect either sex, is more common in Jews of Ashkenazi (east European) descent but rare in other population groups.

Hemophilia is much more common in males than females because males are hemizygous and therefore express the trait when they inherit one mutant allele. In contrast, a female must inherit two mutant alleles, a less frequent event since the mutant allele is rare in the population. Tsarevich Alexei of Russia was the most famous sufferer of X-linked hemophilia, and his disease may have played an important role in the overthrow of the imperial regime in 1917, which changed the course of his- tory for millions of people.

Other examples of X-linked recessive disorders include the Duchenne Muscular Dystrophy (progressive weakness, proximal > distal, pseudohypertrophy due to replacement of muscle by fat and fiber) and Red-green color blindness (the genes for the red and green color receptors are located on the X chromosome).

X-Linked Dominant Disorders

A male or female child of a mother affected with an X-Linked dominant trait has a 50% chance of inheriting the mutation and thus being affected with the disorder. All female children of an affected father will be affected (daughters possess their fathers' X-chromosome). No male children of an affected father will be affected (sons do not inherit their fathers' X-chromosome).

Examples of X-linked dominant disorders include Incontinentia Pigmenti and vitamin D resistant rickets. Incontinentia Pigmenti is exclusively seen in females (presumed lethal in males). Characterized by swirling patterns of melanin,

inflammatory skin lesions, missing teeth. The symptoms of vitamin D resistant rickets include growth failure, fractures very difficult to heal, spinal cord compression, hearing loss.

Mitochondrial Inheritance

Mitochondria contain circular DNA, thought to be derived from an ancient symbiosis with bacteria. All mitochondria are inherited from the mother. Thus, the egg contributes both genetic material in the nucleus and genetic material from the mitochondria.

Mitochondria are key to energy management for cells. Mutations in mitochondrial DNA lead to aberrations in energy management. Brain, eye, and muscle are high users of energy and so are the organs most often affected by mitochondrial disorders.

Sex Determination

In the animal world, environmental (rather than genetic) determination of sex is common.

Some organisms maintain both sexes in the same body.

By the 6th week of development in humans, both 46, XX and 46, XY embryos have the same sex organs.

By the 7th week of development, the SRY gene on the Y chromosome in 46, XY embryos produces testis-determining factor

(TDF) that triggers testes development. TDF is a testis determining factor, a protein encoded by the SRY gene on the Y chromosome. This protein is a transcription factor, which means it attaches (binds) to specific regions of DNA and helps control the activity of particular genes. TDF causes a fetus to develop as a male.

The presence of the SRY gene product activates other genes, found on the X chromosome and the autosomes, which further stimulate the development of male sexual organs.

By the end of the 4th month of normal development, the sex of the fetus becomes evident.

On the basis of appearance of external genitals, a newborn's sex is assigned and recorded on the birth certificate.

Sometimes, male-female assignment is not straightforward.

Gonadal determination during normal human development

True hermaphrodites are individuals with complete sets of male and female genitals.

Genetically, true hermaphrodites are either true 46, XX or sex chromosome mosaics (XXXV/XY, XX/XY etc.).

Sometimes, karyotypic males can be phenotypically females (and vice versa) due to a variety of genetic (and environmental) abnormalities:

- Androgen insensitivity syndrome
- Gonadal dysgenesis
- 5 alpha-reductase deficiency.

Complete Androgen Insensitivity Syndrome (CAIS)

The principal mammalian androgens are testosterone and its more potent metabolite, dihydrotestosterone (DHT). The andro- gen receptor (AR) is a large nuclear receptor protein of at least 910 amino acids. Each molecule consists of a portion which binds the androgen, a zinc finger portion that binds to

DNA in steroid sensitive areas of nuclear chromatin, and an area that controls transcription.

Testosterone diffuses from the circulation into the cytoplasm of a target cell. Some is metabolized to estradiol, some reduced to DHT, and some remains as testosterone (T). Both T and DHT can bind and activate the androgen receptor, though DHT does so with more potent and prolonged effect. As DHT (or T) binds to the receptor, a portion of the protein is cleaved. The AR-DHT combination dimerizes by combining with a second AR-DHT, both are phosphorylated, and the entire complex moves into the cell nucleus and binds to androgen response elements on the promoter region of androgen-sensitive target genes. The transcription effect is amplified or inhibited by coactivators or corepressors.

Although testosterone can be produced directly and indirectly from ovaries and adrenals later in life, the primary source of testosterone in early fetal life is the testes, and it plays a major role in human sexual differentiation. Before birth, testosterone induces the primary sex characteristics of males. At puberty, testosterone is primarily responsible for the secondary sex characteristics of males.

CAIS is a recessive sex-linked disorder (a form of male pseudo-hermaphroditism) affecting 1 in 20,000 males. CAIS is caused by a mutation in the Androgen Receptor gene (Xq11-12) that leads to lack of development of most of male physical characteristics and development of female genitalia. Individuals with CAIS have undescended testes that produce hormones inhibiting the development of uterus and fallopian tubes. The scientific and medical world accepts individuals with CAIS as females, though karyotypically they are males.

Gonadal Dysgenesis: Swyer Syndrome and True GD (46, XY and 46, XX Sex Reversal)

Gonadal dysgenesis is a reproductive system development disorder.

Swyer syndrome, or XY gonadal dysgenesis, is condition in which no functional testes are present in an externally female person whose karyotype is then found to be XY. Frequency: 1 in 30,000.

Swyer syndrome is a condition in which individuals with one X chromosome and one Y chromosome in each cell, the pattern normally found in males, have a female appearance. People with this disorder have female external genitalia and a normal uterus and Fallopian tubes. However, they do not have functional gonads (ovaries or testes). Instead, they have undeveloped clumps of tissue called streak gonads. These abnormal gonads often become cancerous, so they are usually removed surgically early in life. Some patients have no evidence of masculinization and have a female phenotype with the somatic signs of Turner syndrome. The presence of unilateral testis or a streak gonad and chromosomal mosaicism with both XO and XY cell lines (45,X/ 46, XY mosaicism) indicate the mixed gonadal condition.

SRY mutations that cause Swyer syndrome prevent production of the sex-determining region Y protein or result in the production of a nonfunctioning protein. A fetus whose cells do not produce functional sex-determining region Y protein will develop as a female despite having a Y chromosome.

People with Swyer Syndrome have streak gonads instead of testes (no germ cells or hormone-producing cells); they lack ovaries but have an otherwise normal female anatomy.

Mutations in the SRY gene have been identified in 15-20% of individuals with Swyer syndrome. Other genes involved:

-NR5A1 – autosomal (dominant mutation), hormonal production pathway;

-NR0B1 – X-linked, hormone pathway, endocrine function.

True sex reversal (complete gonadal dysgenesis) is found in phenotypic females with 46, XY genotype and phenotypic males with 46, XX genotype (frequency 1 in 20,000 for each type). The origins of true GD are traced to an illegitimate crossing over in the SRY region during spermatogenesis or mutations of the SRY gene on the Y chromosome.

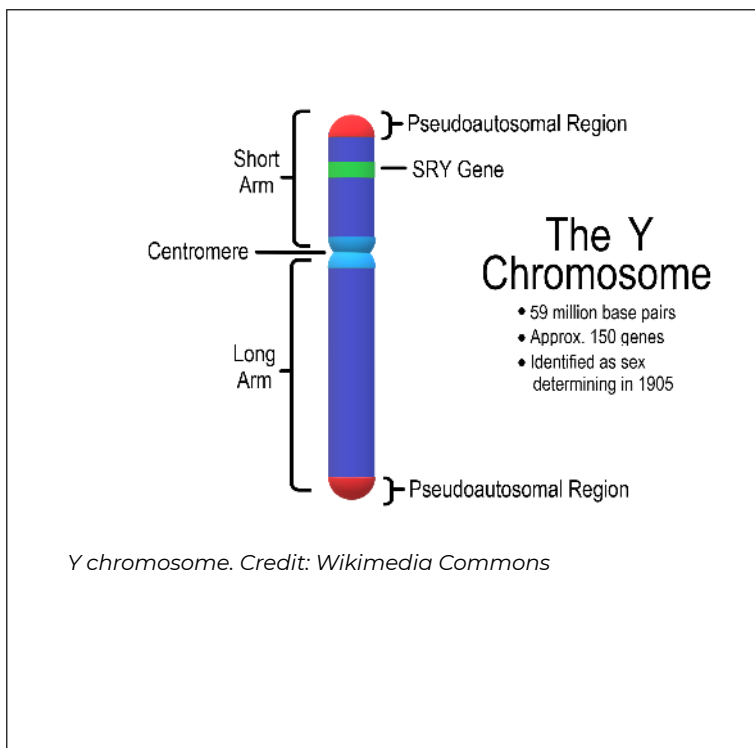
The Y Chromosome

The tips of both arms of the X and Y chromosomes are homologous (pseudoautosomal), which allows X and Y to pair during meiotic divisions of gametogenesis.

Pseudoautosomal regions of X and Y undergo normal crossing over in prophase I of spermatogenesis.

Rarely, an “illegitimate” crossing over occurs within the region just below the pseudoautosomal region, where the SRY gene is located.

The resulting Yp translocation leads to SRY loss in 46, XY females and a gain of SRY in 46, XX males.



5 alpha Reductase Deficiency

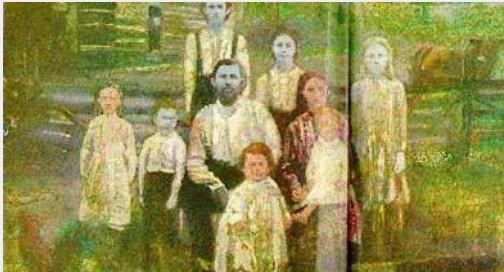
5 alpha reductase deficiency is an autosomal (chromosomes 2 or 5) recessive condition.

The enzyme 5 α -reductase converts testosterone to dihydrotestosterone (DHT), which exerts an even stronger effect on the developing body than testosterone.

In this syndrome, the internal genitalia of males develop normally but the external genitalia resemble that of females.

The infant is typically raised as a girl but then at the next testosterone peak at puberty the external genitals suddenly become obviously male.

Chapter 9. Pedigree Analysis



The Blue Fugates are shown in this colorized black and white photo. Date unspecified. Via ABC News.

Fugates of Kentucky: Skin Bluer than Lake Louise

By SUSAN DONALDSON JAMES | Good Morning America

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Benjamin “Benjy” Stacy so frightened maternity doctors with the color of his skin — “as Blue as Lake Louise” — that he was rushed just hours after his

birth in 1975 to University of Kentucky Medical Center. As a transfusion was being readied, the baby's grandmother suggested to doctors that he looked like the "blue Fugates of Troublesome Creek." Relatives described the boy's great-grandmother Luna Fugate as "blue all over," and "the bluest woman I ever saw."

In an unusual story that involves both genetics and geography, an entire family from isolated Appalachia was tinged blue. Their ancestral line began six generations earlier with a French orphan, Martin Fugate, who settled in Eastern Kentucky. Doctors don't see much of the rare blood disorder today, because mountain people have dispersed and the family gene pool is much more diverse. But the Fugates' story still offers a window into a medical mystery that was solved through modern genetics and the sleuth-like energy of Dr. Madison Cawein III, a hematologist at the University of Kentucky's Lexington Medical Clinic.

Cawein died in 1985, but his family charts and blood samples led to a sharper understanding of the recessive diseases that only surface if both parents carry a defective gene. The most detailed account, "Blue People of Troublesome Creek," was published in 1982 by the University of Indiana's Cathy Trost, who described Benjy's skin as "almost purple."

The Fugate progeny had a genetic condition

called methemoglobinemia, which was passed down through a recessive gene and blossomed through intermarriage. “It’s a fascinating story,” said Dr. Ayalew Tefferi, a hematologist from Minnesota’s Mayo Clinic. “It also exemplifies the intersection between disease and society, and the danger of misinformation and stigmatization.”

Methemoglobinemia is a blood disorder in which an abnormal amount of methemoglobin — a form of hemoglobin — is produced, according to the National Institutes for Health. Hemoglobin is responsible for distributing oxygen to the body and without oxygen, the heart, brain and muscles can die. In methemoglobinemia, the hemoglobin is unable to carry oxygen and it also makes it difficult for unaffected hemoglobin to release oxygen effectively to body tissues. Patients’ lips are purple, the skin looks blue and the blood is “chocolate colored” because it is not oxygenated, according to Tefferi. “You almost never see a patient with it today,” he said. “It’s a disease that one learns about in medical school and it is infrequent enough to be on every exam in hematology.”

The disorder can be inherited, as was the case with the Fugate family, or caused by exposure to certain drugs and chemicals such as anesthetic drugs like benzocaine and xylocaine. The carcinogen benzene and nitrites used as meat

additives can also be culprits, as well as certain antibiotics, including dapsone and chloroquine.

The genetic form of methemoglobinemia is caused by one of several genetic defects, according to Tefferi. The Fugates probably had a deficiency in the enzyme called cytochrome-b5 methemoglobin reductase, which is responsible for recessive congenital methemoglobinemia. Normally, people have less than about 1 percent of methemoglobin, a type of hemoglobin that is altered by being oxidized so is useless in carrying oxygen in the blood. When those levels rise to greater than 20 percent, heart abnormalities and seizures and even death can occur. But at levels of between 10 and 20 percent a person can develop blue skin without any other symptoms. Most of blue Fugates never suffered any health effects and lived into their 80s and 90s. “If you are between 1 percent and 10 percent, no one knows you have an abnormal level and this might be the case in a lot of unsuspecting patients,” he said. Many other recessive gene diseases, such as sickle cell anemia, Tay Sachs and cystic fibrosis can be lethal, he said. “If I carry a bad recessive gene with a rare abnormality and married, the child probably wouldn’t be sick, because it’s very rare to meet another person with the [same] bad gene and the most frequent cause therefore is in-breeding,” Tefferi said. Such was the case with the Fugates.

Martin Fugate came to Troublesome Creek from

France in 1820 and family folklore says he was blue. He married Elizabeth Smith, who also carried the recessive gene. Of their seven children, four were reported to be blue. There were no railroads and few roads outside the region, so the community remained small and isolated. The Fugates married other Fugate cousins and families who lived nearby, with names like Combs, Smith, Ritchie and Stacy. Benjy's father, Alva Stacy showed Trost his family tree and remarked, "If you'll notice — I'm kin to myself," according to Trost. One of Martin and Elizabeth Fugate's blue boys, Zachariah, married his mother's sister. One of their sons, Levy, married a Ritchie girl and had eight children, one of them Luna. Luna married John E. Stacy and they had 13 children.

Pedigrees: Family Trees

A pedigree is used to trace inheritance of a trait over several generations

Three primary patterns of inheritance:

1. Autosomal dominant
2. Autosomal recessive
3. Sex-linked (X-chromosomal)

Other patterns of inheritance: Y-linked genes, mitochondrial genes.

Pedigree analysis

In pedigrees, squares symbolize males and circles represent females. Two parents are joined by a horizontal line, with offspring listed below in their order of birth, from left to right. Shaded symbols stand for individuals with trait being traced.

Pedigree chart symbols to represent male and female individuals		Some other symbols used in pedigree chart
 Male	 Female	 Unknown sex
 Three males	 Three females	 Lived one day
 Proband male	 Proband female	 Still birth
 Deceased male	 Deceased female	 Miscarriage
 Affected male	 Affected female	 Pregnancy
 Examined male	 Examined female	 Marriage
 Male fraternal twins	 Carrier	 Consanguineous marriage
 Male identical twins	 Female fraternal twins	 Extramarital mating
 Adopted male	 Female identical twins	 Divorce
	 Adopted female	 No children

Pedigree uses standard symbols and formatting. Image from <https://biologyreader.com/pedigree-analysis-chart.html>. The use of the image adheres to the Fair Use application of 17 U.S. Code § 107, <https://www.copyright.gov/title17/92chap1.html#107>)

Autosomal dominant trait pedigree

Dominant: Affected individuals can appear in every generation.
Autosomal: Gene is on one of the autosomes. Male and female offspring equally likely to inherit trait.

The trait doesn't appear in the descendants of two unaffected parents.

A trait that appears in successive generations is normally due to a dominant allele.

Autosomal recessive trait pedigree

Recessive: If neither parent has the characteristic phenotype (disease) displayed by the child, the trait is likely to be recessive.

Matings between an affected and an unaffected parent will usually yield all unaffected offspring (unless the unaffected parent is heterozygous for the trait).

Autosomal: Male and female offspring equally likely to inherit trait.

Note: Parents of affected children in pedigree shown are first cousins.

Sex-linked trait pedigrees

Male to male transmission rules out X-linkage.

X-linked recessive: The trait is preferentially seen in males, who are hemizygous. Females are heterozygous "carriers."

Affected fathers and homozygous "wild type" mothers produce only unaffected sons and carrier daughters.

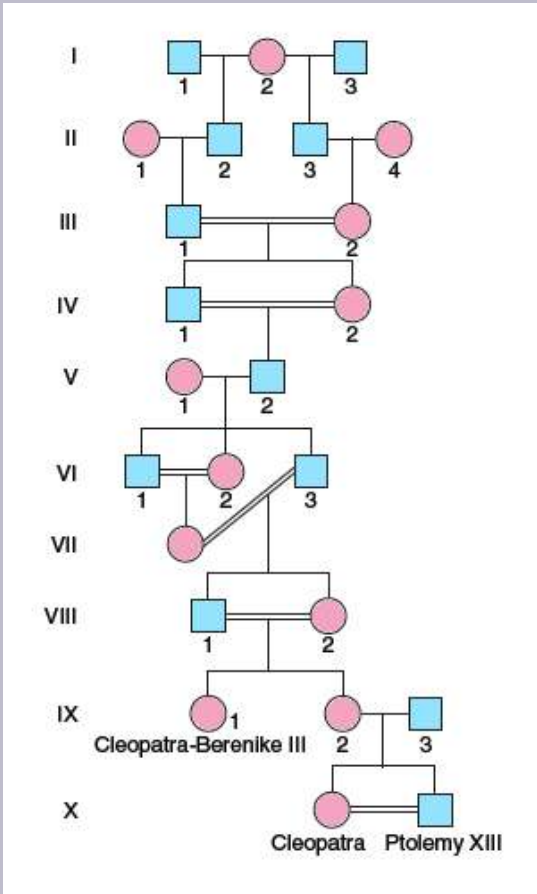
100% of daughters of affected males are carriers; none of their sons are affected.

Occasionally, a carrier female expresses some characteristics of an X-linked trait due to mosaicism.

Most X-linked traits are recessive.

100% of daughters of affected males are affected, none of their sons.

Another example of consanguinity: the Ptolemy dynasty of Egypt



The Ptolemy Dynasty of Egypt pedigree. Image from https://www.mun.ca/biology/scarr/Sib-Sib_marriages.html. The use of the image adheres to the Fair Use application of 17 U.S. Code § 107, <https://www.copyright.gov/title17/92chap1.html#107>

Chapter 10. Mutations

Mutations

A mutation is any change in the nucleotide sequence of a gene. A mutation occurs when genetic material is damaged or changed in such a way as to alter the genetic message carried by that gene. Mutations can be induced by a variety of factors called mutagens.

Mutations and their functional consequences

In terms of functional changes to gene product, mutations can be classified as loss of function and gain of function.

Loss of function mutations are usually recessive at the genetic level, because, most of the time, a functional product is still made by the undamaged copy of the gene (the wild type allele).

Gain of function mutations are usually dominant, because a new abnormally active product is made which may cause unregulated cellular changes.

A **mutagen** is an agent or substance that can bring about a permanent alteration to the physical composition of DNA

such that the genetic message is changed. Once the gene has been damaged or changed the mRNA transcribed from that gene will now carry an altered message. The polypeptide made by translating the altered mRNA will now contain a different sequence of amino acids. The function of the protein made by folding this polypeptide will probably be changed or lost. Consequently, the phenotype of the organism carrying the mutation will be changed as well.

Mutations in the DNA of somatic cells may lead to a change in an individual phenotype, but will not be passed on to the next generation. Only mutations in reproductive cells are inherited.

Types of mutagens:

- Spontaneous (unrepaired errors in DNA replication).
- Chemical (caused by a chemical agent, such as benzene or hormones (estrogen, testosterone)).
- Physical (caused by a physical agent, such as radiation).
- Biological (caused by transposable elements and viruses).

Environmental Factors: Radiation

Radiation is a process by which energy travels through space. Exposure to radiation is unavoidable and comes from a wide variety of sources both natural occurring and due to human activity.

The Chernobyl Nuclear Accident

On April 26, 1986, a reactor at the Chornobyl nuclear power plant in Ukraine experienced a catastrophic failure, which led to a release of vast amounts of radiation into the atmosphere. Radiation traveled around the world and within 24 hours was reported as far away as Canada. The bulk of radioactive material fell on the 20-km territory in the immediate vicinity of the plant. This “exclusion zone” became permanently unsuitable for human habitation. People traveling to the zone have to wear protective gear patterned after an astronaut suit, since it is as dangerous for a human body to be in the zone as being in an outer space. The most prominent health result of exposure to Chornobyl radiation is a 1,000-fold increase in thyroid cancer rates in children and adolescents who were exposed to radiation effects at the time of the accident. Elevated rates of hydrocephaly (see picture), leukemia, and various birth defects have been reported in the affected population. Cancer rates are projected to increase in the larger population in the affected territory for another 20 years.



Chornobyl's exploded 4th reactor, 1986. Credit: Joker345, CC BY-SA 4.0, via Wikimedia Commons

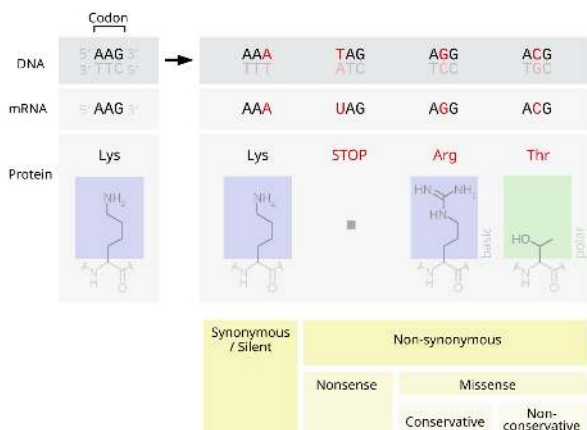
Point mutations

Point mutations are single base (nucleotide) changes (additions, deletions, substitutions) in a sequence of DNA that specifies a cellular product.

Base substitutions are classified as follows.

A nonsense mutation creates a stop triplet where none previously existed.

A missense mutation changes the triplet code of one amino acid for a different one. A silent mutation has no effect on the code.



Point mutations of a codon, classified by their impact on protein sequence.
 By Jonsta247 – Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=12481467>

Within a gene, deletions or insertions of one or two nucleotides result in a frame shift. Larger deletions or insertions (≥ 2 nucleotides) may result in cessation of transcriptional activity in that gene.

Mutations with Expanded Trinucleotide Repeats

Trinucleotide repeats are a class of mutations associated with a number of genetic disorders. Trinucleotide repeats can expand in successive generation, leading to an increase in the size of genes in which these repeats are found and, in some cases, lead to an inherited genetic disorder. Disease severity may be related to the size of the expanded gene. The expansion of nucleotide triplets causes the type of inheritance called allelic expansion.

Allegan County Sheep Could Hold Key To Treating Huntington's Disease

Joe LaFurgey and 24 Hour News 8 web staff

Published: May 6, 2016, 6:34 pm

<https://www.woodtv.com/news/allegan-county-sheep-could-hold-key-to-treating-huntingtons-disease/>

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HOPKINS, Mich. (WOOD) – Research coming out of an Allegan County farm could bring hope to the 30,000 people who suffer from Huntington’s disease and the 200,000 people at risk of getting it. Mike Ludlam’s sister, Dori was part of that statistic. He lost her to Huntington’s disease four years ago.

“You basically just watch somebody deteriorate in front of your eyes. [They] go from such a strong person... [to] being helpless,” said Ludlam, who owns Windswept Farms near Hopkins.

Huntington’s disease is a genetic disorder that damages brain cells and patients’ mental and physical abilities, eventually taking their life. But an article Ludlam recently came across could be a game-changer.

The report described a genetic disorder in some

sheep which cause the animals to overproduce the GM1 ganglioside, which is a substance that helps protect brain cells. It's also a substance lacking in people with Huntington's disease. Suddenly, the Ludlams' decision years ago to raise sheep took on new meaning. "We have sheep. We're sheep farmers. This is a no brainer," said Heather Ludlam, who is also a veterinarian. The couple helped jumpstart research involving breeding sheep with the genetic disorder that produces the additional GM1 and extracting the substance for use in treating Huntington's.

GM1 isn't a cure, but it could bring an effective treatment for the disease. The early results are promising and piqued the interest of U.S. Rep. Fred Upton, who toured the Ludlam's farm Friday. "Do they have a time line of when they finish that first study?" Upton asked Heather Ludlam. "They're hoping, I'd say, between two and three years. It could come down to funding," she answered.

Upton is the primary sponsor of the 21st Century Cures Act. The act would provide additional funding for the National Institutes of Health, as well as streamline the process of getting treatments to patients quicker.

For Mike Ludlam, Upton's visit was one more step in the process. "We've kind of figured with this project, we don't have anything to lose," Mike Ludlam added. "We're just going to keep knocking on doors and trying to get the story out... to get it in front of that one person."

The Position of the Mutation in a Gene Affects the Phenotype

Cystic fibrosis (CF) is an autosomal recessive genetic disorder affecting the lungs, pancreas, liver, intestine, and reproductive tract. It is characterized by abnormal transport of chloride ions across epithelium, leading to thick, viscous secretions.

The cystic fibrosis transmembrane conductance regulator (CFTR) gene is located on chromosome 7, position 7q31. A defect in the CFTR gene leads to clogging of bronchial tubes with thick mucus.

In normal cells, the CFTR protein acts as a channel that transports chloride into and out of cells, and regulates the passage of salts. It is made up of five domains: two membrane-spanning domains that form the chloride ion channel, two nucleotide-binding domains (NBDs) that bind and hydrolyze ATP and a regulatory domain (<http://www.cfgenetherapy.org.uk/>).

There are currently around 1400 mutations known to be associated with CF. The most common mutation that leads to the CF disease is a P508 deletion (delta508). The removal of phenylalanine 508 from the N-terminal region of the NBD1 leads to the disruption of the proper folding of the N-terminal, leading to its inability to bind ATP. Thus, the P508 deletion affects the function of the entire CFTR protein.

The P508 mutation is considered to be associated with overdominance. Studies in mice and human cell cultures show that cells homozygous for the delta508 mutation make cells impermeable to *Salmonella typhi*, the bacterium that causes typhoid fever in humans. The protective effect of delta508 against *S. typhi* manifests even when only one delta508 allele is present. Gastrointestinal submucosa cells of the heterozygous for delta508 mice reduce the translocation ability of *S. typhi* by 86% ([Pier et al., 1998](#)).

Biological Mutagens: Transposable elements

Transposable genetic elements (TEs, also called transposons or mobile genetic elements) comprise a diverse family of DNA sequences that are capable of moving to new sites in genomes either directly by a cut-and-paste mechanism (transposons) or indirectly via an RNA intermediate (retrotransposons). Transposons were first discovered by a Ukrainian geneticist Serhij Hershenson in the 1920s in *Drosophila* (Hershenson, 1928, 1939). The worldwide scientific community did not appreciate this discovery for nearly two decades, until TEs were rediscovered in maize plants by the American geneticist Barbara McClintock in the mid-1940s. For several decades, transposons were considered something of a genetic oddity. In the 1980s, TEs were labeled as “junk” DNA, along with other non-coding sequences in the human genome. In the 1980s, the biological importance of TEs started to come into focus.

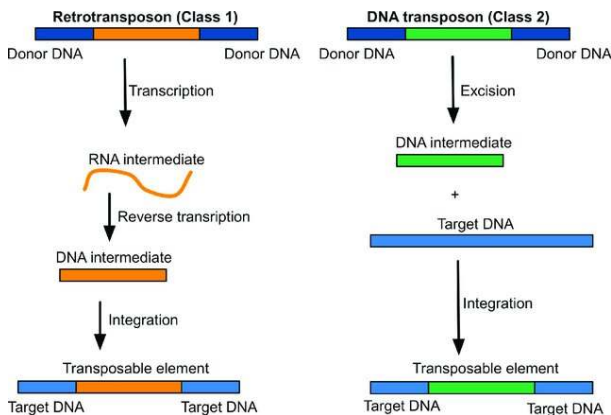
Transposable elements are found in genomes of all living things. **Transposons are the single largest component of the genetic material of most eukaryotes.** Mobile genetic elements constitute more than half of the DNA in most higher eukaryotes.

They play a major role in spontaneous genetic damage, including various gene mutations, modifications in gene expression, alterations of gene sequence, site-specific recombination, deletion and duplication of DNA sequences, chromosomal aberrations and rearrangements, and chromosomal breakage. In organisms with active TE transposition, such as in *Caenorhabditis elegans* or *Drosophila*, TEs can account for the great majority of spontaneous mutations. It has been shown that TEs are directly involved in expression and determination of quantitative genetic characters in *Drosophila*, including

fitness and its components. On the basis of their effect on quantitative traits, it has been suggested that TEs may represent components of an autonomous genetic regulation system that modifies the function, expression and evolution of the genome (Nikitin and Shmookler Reis, 1997).

There are two types of TEs:

- Class II transposons. These consist of DNA that moves directly from place to place (cut-and-paste).
- Class I transposons. These are retrotransposons that first transcribe the DNA into RNA and then use [reverse transcriptase](#) to make a DNA copy of the RNA to insert in a new location (copy-and-paste).



Transposable elements self-replicate through two main mechanisms: via an RNA intermediate ("copy-and-paste"; class 1) or straight excision-insertion ("cut-and-paste"; class 2).
<https://doi.org/10.1371/journal.pgen.1007700.g004>, CC BY 4.0.

Human retrotransposons include autonomous (can move on their own) Long Interspersed Elements – LINEs (representing 17% of the genome), and non-autonomous *Alu* elements (11% of human genome). LINEs and *Alu* elements are the only TEs in the human genome that are still active.

Sequences derived from TEs are found within introns of most mammalian genes. The proliferation and evolution of retrotransposons have had multiple impacts on organismal genomes. Retrotransposition can destabilize the genome, shaping genomic landscapes by insertional mutagenesis, deletions and gene rearrangements (see Chapter 16).

Transpositional activity of DNA (class II) TEs is predominantly deleterious and mostly restricted in its effect to the somatic genome ([Nikitin & Shmookler Reis, 1997](#)). Transpositional activity of RNA (class I) TEs (retrotransposons) can have major evolutionary consequences, exerted through facilitation of ectopic recombination and changes in gene expression.

In humans, *Alu* sequences are a major factor contributing to non-allelic homologous recombination and insertional mutagenesis. The table below provides examples of insertional mutagenesis caused by *Alu* sequences linked to human diseases.

Gene	Disorder	Element	Mechanism	Reference
NF1	Neurofibromatosis	AluYa5	Intron insertion/ frameshift	Wallace et al. 1991
ChE	Acholinesterasemia	AluYb8	Exon insertion	Muratani et al. 1991
F9	Hemophilia B	AluYa5	Exon insertion	Janicic et al. 1995
CASR	Familial hypocalciuric hypercalcemia	AluYa4	Exon insertion	Bai et al. 1997
ADD1	Huntington disease	Alu	Intron	Hutchinson et al. 1993

Chapter 11.

Chromosomal Abnormalities

The Spectrum of Human Genetic Disorders

Human diseases consist of those which are predominantly acquired (environmental, such as bacterial infections), multifactorial (have both a genetic and an environmental component), and those which are mostly genetic.

Human genetic disorders can be classified as follows:

- Chromosomal numerical (those that are caused by mistakes in cell division and the resulting unequal distribution of the chromosomes to newly formed cells);
- Chromosomal structural (other chromosomal abnormalities);
- Molecular (those that are caused by changes in the nucleotide structure of DNA).

Changes In Chromosomal Number

Cells with normal chromosome sets (two sets of 23 chromosomes in humans) have euploid (normal) karyotype.

A change in normal chromosomal number involving single chromosomes within a diploid set leads to an aneuploid karyotype.

Aneuploid karyotype is observed in 1 out of 250 live births in humans.

Aneuploid individuals possess a genetic imbalance.

Aneuploidy can occur in the result of either meiotic or mitotic non-disjunction.

Most chromosomal abnormalities produce such serious changes in the phenotype that they are not compatible with life. Any polyploidy (extra copies of all chromosomes, i.e., triploidy, tetraploidy, etc.) higher than diploid results in early death in utero. For example, triploidy (69, XXX or 69, XXY or 69, XYY) occurs in the result of double fertilization of an ovum (dispermy). The extra set of paternal chromosomes predisposes to formation of a partial mole, features of which may or may not be grossly or microscopically apparent.

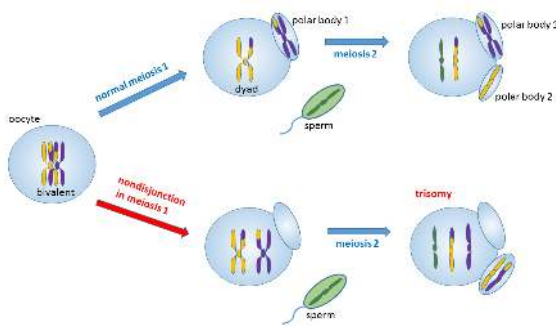
Extra copies of a single chromosome (i.e., chromosome 21) is referred to as trisomy. Trisomy, except for chromosomes 13, 18, or 21, is not compatible with life. Trisomy 13 and Trisomy 18 each lead to early death, usually in the neonatal period of development, if not *in utero*.

Mechanisms of aneuploidy: Meiotic and mitotic nondisjunction

Nondisjunction (“not coming apart”) is the failure of chromosome pairs to separate properly during cell division. This could arise from a failure of homologous chromosomes to separate in Meiosis I, or the failure of sister chromatids to separate during Meiosis II or Mitosis. The result of this error is a cell with an imbalance of chromosomes.

In meiotic nondisjunction during Meiosis I, a pair of duplicated chromosomes (a tetrad on the Metaphase I plate) fails to separate in subsequent stages of Meiosis I division, resulting in

the products of Meiosis I both being aneuploid (one resulting cell has one extra chromosome and the other one lacks a chromosome). In Meiosis II division, the aneuploid products of Meiosis I each produce two aneuploid gametes, two with an extra chromosome and two with a missing chromosome.



Trisomy due to nondisjunction in maternal Meiosis I.
Credit: Wpeissner, CC BY-SA 3.0, via Wikimedia Commons

In a nondisjunction event during Meiosis II, one pair of sister chromatids in one of the two cells produced in Meiosis I fails to separate after Metaphase II, leading to the resulting products of that cell's division being aneuploid (one having one less chromosome and one having an extra chromosome).

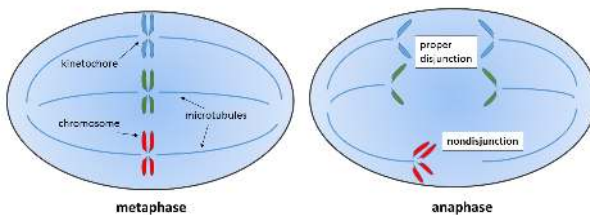
Thus, all four products of a meiotic cell division experiencing a chromosomal non-disjunction event during Meiosis I will be aneuploid, while only two out of four resulting meiotic products of a cell affected by a non-disjunction event during Meiosis II will be aneuploid. A fertilization event involving a gamete with normal haploid chromosomal complement and an aneuploid gamete will result in an aneuploid embryo.

Meiotic nondisjunction in either the first or second meiotic division is the major cause of chromosomal aneuploidy. The greatest percentage of these (75%) occurs in oogenesis, where the probability of non-disjunction increases with maternal age.

AI- most 80% occur in the first meiotic division. Meiotic non-disjunction may affect as many as 25% of all ova and 2% of all sperm.

Nondisjunction of sex chromosomes is better tolerated than non-disjunction of autosomes. An excess of chromosomes is tolerated better than a deficiency.

Mitotic nondisjunction occurs when two sister chromatids fail to separate during somatic cell division (see an example in the figure below). Mitotic non-disjunction can happen during development, but it also known to be associated with cancer.



*Mitotic
nondisjunction.
Credit:
Wpeissner,
CC BY-SA
3.0, via
Wikimedia
Commons.*

An individual with a mixture of cells with different chromosome number is called a mosaic. If two independent cell lines (from two zygotes) fuse early in development, such an individual will be referred to as a chimera.

If the somatic nondisjunction occurred early in development, it may lead to germinal mosaicism.

Nondisjunction Of Autosomes

Nondisjunction of autosomes leads to extra or missing chromosomes. As a rule, autosomal nondisjunction is not tolerated. Only three autosomal aneuploids survive after birth in humans and all three are trisomies.

Trisomy 21 (47, XY, +21)

Trisomy 21 is dubbed Down syndrome after the superintendent of an asylum for children with mental retardation in Rurrey, England, named John Langdon Down, who first studied the condition and published an essay on it in 1866.

There are two types of trisomy 21. One results from chromosomal non-disjunction, and another one is produced by a chromosomal translocation and is an inherited form on trisomy 21 (to be discussed in the next section).

The type of trisomy 21 caused by a non-disjunction cannot be inherited. People with this condition are not able to reproduce. Around 80% of all cases of trisomy 21 are caused by a non-disjunction during the first meiotic division of oogenesis, which shows strong maternal-age association. Maternal meiosis II errors constitute 20 to 24% of maternal errors and 18 to 20% of all cases of non-inherited trisomy 21 (see Robertsonian translocations in Part II for the inherited type of Down syndrome).

Trisomy 18: Edward's Syndrome (47,X,Y +18)

Trisomy 18 results mostly from maternal non-disjunction in meiosis II (during ovulation).

Frequency: 1/6,000 live births. 80% are female; nearly all affected individuals die in their first year. Features include micrognathia (small lower jaw), overlapping fingers, horseshoe kidney, rocker bottom feet, cardiac defects, diaphragmatic hernia, omphalocele (defect in muscles of the abdominal wall).

Trisomy 13: Patau's Syndrome (47, XX, +13)

Frequency: 1/12,000 live births; mean survival time is 3 months.

Features include microcephaly (small head), cleft lip and/or palate, postaxial polydactyly, cardiac defects, deafness, holoprosencephaly (malformation of the cerebrum).

Trisomy 16

Trisomy 16 is one of the most common chromosomal abnormalities, but the affected embryos or fetuses never survive past the first trimester. Many first trimester fetuses are lost in this fashion (many are “silent” miscarriages).

Nondisjunction Of Sex Chromosomes

Sex-chromosome abnormalities may also be caused by nondisjunction of one or more sex chromosomes. The majority of sex-chromosomal aneuploidies carry extra chromosomes. Only one sex chromosomal aneuploidy is a monosomy.

Any combination (up to XXXXY) produces maleness. Males with more than one X are usually underdeveloped and sterile.

XXX and XO women are known. XXX females are fertile, although, in most cases, XO females are sterile.

Klinefelter’s Syndrome (KS) (47, XXY)

Frequency: 1.5/1,000.

KS males have an extra X chromosome resulting in XXY.

As many as four X chromosomes may be present.

An extra Y chromosome may be present on some occasions giving XXYY.

Associated conditions:

Eunuchoid (tall) body proportions.

Sterility due to azoospermia (failure to produce sperm).

Gynaecomastia (swollen breasts). Male breast cancer is estimated to affect some 3% of KS men, about 20 times what one would expect to find in ordinary 46,XY males).

Mild mental retardation is present in some KS cases.

Respiratory disorders such as asthma seem to be much more common in individuals with KS.

Jacobs Syndrome (47, XYY)

Jacobs syndrome is an chromosomal condition in which a male has an extra Y chromosome. There are usually few symptoms, such as being taller than average, but the person is generally otherwise normal.

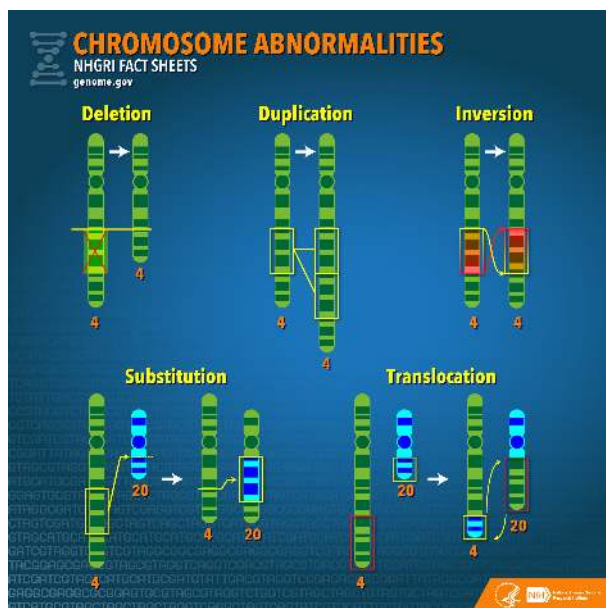
Turner's Syndrome (TS) (45, X)

Synonym: XO gonadal dysgenesis. The only surviving monosomy of all human chromosomal aneuploids.

Frequency: 1/ 2,500; most common karyotype among miscarriages.

Characteristics: short stature, the lack of sexual development at puberty (underdeveloped ovaries and lack of estrogen), heart defects, kidney abnormalities, cystic neck hydromas etc.

Structural Chromosomal Abnormalities



Structural chromosomal abnormalities.

Credit: National Human Genome Research Institute, CC BY 2.0.

Structural chromosomal abnormalities include deletions, duplications, inversions, transpositions, and translocations.

Chromosomal deletions

Chromosomal deletion is characterized by the loss of a chromosomal segment. A number of human conditions (syndromes) occur when parts of chromosomes go missing. These conditions are usually associated with severe congenital abnormalities. Examples of human conditions resulting from deletions include Cri-Du-Chat, Williams and Prader-Willi/Angelman syndromes.

Cri-Du-Chat Syndrome cytology: loss of much of the short arm of chromosome 5 (5p). Symptoms: severe mental retardation.

Cri-Du-Chat Syndrome frequency: 1/50,000 live births.

Williams (7q11.23): a neurodevelopmental disorder caused by a deletion of about 26 genes, ~ 1/14,000 frequency.

Prader-Willi/ Angelman (15q partial deletion, ~1/12,500 frequency): developmental delay, low muscle tone in PWS, distinct personality features (happy disposition) and movement discoordination in Angelman.

Chromosomal duplications

In chromosomal duplications, extra copies of a chromosomal region are formed, resulting in different copy numbers of genes within that area of the chromosome.

Example of a chromosomal duplication in humans: Charcot-Marie Tooth syndrome

Frequency: 1/2,500.

Cytology: duplication of the 17p12 chromosomal region.

Symptoms: CMT is a progressive neurological disorder. CMT patients slowly lose normal use of their feet/legs and hands/arms as nerves to the extremities degenerate.

Chromosomal inversions

Chromosomal inversions occur when a segment breaks off and reattaches within the same chromosome. Chromosomal inversions are classified as pericentric, if the inverted fragment includes the centromere, and paracentric, if the centromere is not included. Pericentric inversions are most frequent, often reported for chromosomes 1, 2, 3, 5, 9, 10 and 16. The frequency in the general population of chromosome inversions is 1–5 in

10,000 for paracentric inversions and 1–7 in 10,000 for pericentric inversions.

45 percent of severe hemophilia A cases are due to inversions. Most inversions originate almost exclusively in male germ cells.

Inverted duplications – proximal spinal muscular atrophy (SMA), a recessive-lethal condition affecting 1/10,000 births, is associated in ~18% of cases with rearrangements in an inverted duplication in 5q13.

Transpositions

Insertions of mobile genetic elements or transposons (see Chapter 1) contribute to disease by either disrupting a coding region or a splice signal. Although insertions of such human transposons as Alu causing disease are broadly spread throughout the genome, some genes seem more prone to disease-causing insertions of this type. Fourteen new Alu insertions inactivating the NF1 gene have been reported, representing 0.4% of known mutations in this gene. Similarly, many diseases caused by non-allelic homologous recombination between Alu elements have been reported as well. Although these events are also broadly spread throughout the genome, some regions, such as the MSH2, VHL and BRCA1 genes, are much more subject to this instability than others. Most Alu-related genomic instability events will either have no major functional consequence, and over many generations simply be lost from the human population gene pool through random fixation, or be deleterious and therefore lost through negative selection. Thus, the events described above represent only a tiny proportion of the overall genetic instability in the human population caused by such elements.

Translocations: 14q/21q Robertsonian translocation and Down syndrome

Reciprocal translocations are an exchange of material between nonhomologous chromosomes (1/625 newborns).

In Robertsonian translocations, the participating chromosomes break at their centromeres and the long arms fuse to form a single, large chromosome with a single centromere. The short arms also join to form a smaller reciprocal product, which typically contains nonessential genes and is usually lost within a few cell divisions.

About 9% of Down syndrome children born to mothers who are less than 30 years of age are the result of Robertsonian translocation.

Robertsonian translocations are passed from generation to generation, and with this type of inheritance Down syndrome may “run in families.”

Prenatal Diagnostics

Prenatal diagnosis employs a variety of techniques to determine the health and condition of an unborn fetus. Without knowledge gained by prenatal diagnosis, there could be an untoward outcome for the fetus or the mother or both. Congenital anomalies account for 20 to 25% of perinatal deaths. Specifically, prenatal diagnosis is helpful for:

- Managing the remaining weeks of the pregnancy.
- Determining the outcome of the pregnancy.

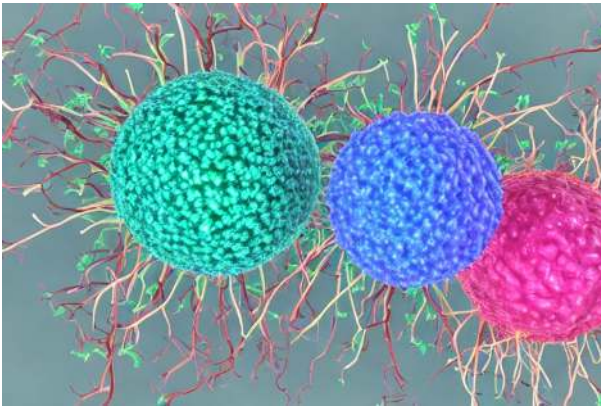
- Planning for possible complications with the birth process.
- Planning for problems that may occur in the newborn infant.
- Deciding whether to continue the pregnancy.

- Finding conditions that may affect future pregnancies.

There are a variety of non-invasive and invasive techniques available for prenatal diagnosis. Each of them can be applied only during specific time periods during the pregnancy for greatest utility. The techniques employed for prenatal diagnosis include:

- Ultrasonography
- Amniocentesis
- Chorionic villus sampling
- Fetal blood cells in maternal blood
- Maternal serum alpha-fetoprotein
- Maternal serum beta-HCG (human chorionic gonadotropin).
- Maternal serum estriol

Chapter 12. Genetics of Immunity



<https://deepai.org>

The Importance of Cell Surfaces

Foreign antigens (molecules) elicit an immune response from a host.

Antibodies and cytokines produced by the immune system attack foreign antigens.

Blood Groups

Early blood transfusions were oftentimes unsuccessful. With the discovery of human blood groups systems, properly matched donor and recipient types resulted in successful transfusions.

Blood types, including the ABO groups (Chapter 2), Rh factor, and others, result from self antigen patterns on red blood cells. Rh incompatibility may put a fetus at risk.

The Human Leukocyte Antigens

In addition to blood group antigens, a large number of cell surface molecules are recognized by the immune system.

Many of the cell surface proteins that help to establish immunity are encoded by the approximately seventy genes of the major histocompatibility complex (MHC) located on chromosome 6 in humans.

Class I and II genes of the MHC encode human leukocyte antigens (HLA).

Class III genes produce plasma proteins involved in the immune response.

HLA antigens on leukocytes are involved in processing of foreign antigens.

HLA proteins are each encoded by several genes with many alleles and as a result only 2 in every 20,000 unrelated people match for the six major HLA genes by chance.

Individuals with certain HLA combinations have an increased risk of developing certain HLA-linked diseases.

The Immune System

At the cellular level the immune system consists of various types of lymphocytes and macrophages.

The immune response consists of an immediate, generalized, innate immunity and a slower, more specific adaptive immunity.

Physical Barriers and the Innate Immune Response

Skin, mucous membranes, tears, and saliva are examples of physical and chemical barriers that keep pathogens from entering the body.

Pathogens that breach this barrier encounter an innate immune response consisting of inflammation, phagocytosis, complement, collectins, and cytokines.

The Adaptive (Acquired) Immune Response

The innate response is rapid (minutes) while the acquired immune response must be stimulated to action and takes days to respond.

The acquired immune response is divided into the humoral and cellular immune responses. Both of these responses differ from the innate immune response in that they are more complex, highly specific and each has a cellular component with memory.

The humoral response involves B cells that secrete antibodies in order to neutralize, clump, and stimulate the destruction of pathogens by recognizing and binding specific foreign antigens.

Antibodies are made of Y-shaped polypeptides consisting of constant and variable regions.

The astounding diversity of antibody binding activities is due to a shuffling of gene pieces (exons) encoding antibody polypeptide products in B cells.

In the cellular immune response, helper T cells stimulate B cells to manufacture antibodies and cytotoxic T cells to secrete cytokines.

Some T cells bind to nonself cells and virus-covered cells and burst them. Other T cells function to coordinate the immune response.

Abnormal Immunity Inherited Immune Deficiencies

Inherited immune deficiencies represent defects in the genes that encode proteins involved in immunity.

Acquired Immune Deficiency Syndrome

Acquired immune deficiency syndrome is caused by HIV.

HIV replicates very rapidly, and T cell production matches it until the immune system is overwhelmed and AIDS begins.

HIV is a retrovirus that injects its RNA into host cells by binding receptors. Reverse transcriptase then copies viral RNA into DNA. HIV uses the cell's protein synthesis machinery to mass produce itself; then the cell bursts, releasing virus.

Reverse transcriptase and protease inhibitors have been effective in slowing down HIV.

Autoimmunity

In autoimmune disorders, autoantibodies attack healthy tissue. These conditions may be caused by a virus that borrows a self antigen, T cells that never learn to recognize self, or healthy cells bearing antigens that resemble nonself antigens.

Some conditions thought to be autoimmune may actually reflect an immune system response to retained fetal cells.

Mutations in some genes may present the symptoms of an auto-immune disease.

Allergies

An overly sensitive immune system causes allergies.

In an allergic reaction, allergens bind to IgE antibodies on mast cells, which release allergy mediators.

A subset of helper T cells secrete cytokines that contribute to allergy symptoms.

Altering Immune Function Vaccines

Vaccines are disabled pathogens or their parts that elicit an immune response, protecting against infection by the active pathogen.

Immunotherapy

Immunotherapy uses immune system components to fight disease.

Hybridomas are artificial cells that consist of a B cell fused with a cancer cell and produce monoclonal antibodies (Mabs) that can target specific antigens.

Cytokines boost immune function and destroy cancer cells.

Transplantation

Autografts transfer tissue from one part of a person's body to another.

Isografts are tissue transfers between identical twins.

Allografts involve tissue transfers between members of the same species. These transplants can cause tissue rejection reactions.

A Xenograft is a cross species transplant. A danger of these transplants is that they can set off a hyperacute rejection.

Graft-versus-host disease involves a rejection of recipient tissues by transplanted bone marrow.

The success rate of transplants is improved by the use of immunosuppressive drugs, by stripping antigens from donor tissue, and by matching donor to the recipient.

A Genomic View of Immunity – The Pathogen's Perspective

Analyzing the genomes of pathogens may reveal the molecular basis of pathogenesis.

Crowd Diseases

Crowd diseases readily spread through populations and can cause epidemics.

Native populations with no immunity can be devastated by introduction of new diseases.

Bioweapons

Microorganisms like viruses, bacteria or fungi, and the toxic substances produced by them and other living organisms, can be modified and enhanced to produce agents that, when released deliberately, cause catastrophic disease and death in

humans, as well as other living things. A variety of pathogens have been adapted to military use in the USA, the former Soviet Union, and other countries around the world. Biological agents like anthrax, botulinum toxin and bubonic plague can produce a public health emergency causing large numbers of deaths in a short amount of time. Biological agents which are capable of secondary transmission can lead to epidemics. An attack involving a biological agent may mimic a natural event, which may complicate the public health assessment and response.

Between the 1950-1980s, the Soviet Union had developed a wide range of weaponized agents, based on genetically modified agents, capable of causing devastating pandemics. Among these were antibiotic-resistant strains of plague, anthrax (called Siberian Ulcer in Russian), tularemia, and glanders. The full extent of the capabilities of Soviet bioweapons is not known. The Russian Federation that inherited the Soviet bioweapon stockpile, has the potential to use these agents, considering the reckless political behavior of the Russian state, particularly towards its immediate neighbors such as Ukraine, as well as the world at large.

The Epidemiology of the COVID-19 coronavirus infection

The severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) virus, better known as COVID-19-causing virus, or just COVID-19, emerged at the end of 2019 from a yet unidentified source, potentially in the city of Wuhan, China, causing a crowd infection that spread rapidly across the world. In the absence of verifiable source of the COVID-19 infection, COVID-19 can also be considered a potential bioweapon. By March 2020, COVID-19 caused a global pandemic. According to the [World Health Organization \(WHO\)](#), as of May 2023 one tenth of the planet's

population contracted COVID-19 and there have been nearly 7 million deaths from COVID-19 worldwide.

SARS-CoV-2 belongs to the coronavirus family of viruses and is related to the severe acute respiratory syndrome (SARS) and Middle East respiratory syndrome (MERS) coronaviruses. The COVID -19 virus uses the human ACE2 receptor to infect epithelial cells in the nasopharynx, where it replicates and subsequently enters the distal alveolar space. Human immune system recognizes the virus through cell receptors, primarily of the Toll-like receptor group such as TLR3 and TLR7, which induce the synthesis of type I interferons.

Individual response to the COVID-19 infection varies significantly among people. For some, the infection is asymptomatic, while others suffer symptoms varying in strength and proliferation. In 2020, several studies attempted to identify the genetic foundation of COVID-19 response. Several regions on chromosome 3 and 9 have been linked with the severe COVID-19 response. A further investigation determined that the regions associated with severe COVID-19 response on chromosome 3 appear to display high Linkage Disequilibrium (LD) ([Zeberg, Pääbo, 2020](#)). This means that these regions are highly associated with each other in the genome, resisting being separated by genetic recombination, which tends to dissociate linked genes (such as genes on the same chromosome), unless strong selective pressure favors these genes to remain linked. It was further determined that these linked genetic segments introgressed into human genome from Neanderthals (more on genomic introgression between humans and Neanderthals in Chapter 16).

It was determined that European and Siberian Neanderthals had some of these genetic sequences in homozygous state in their genome, and these sequences were absent in the Denisovans, a sister hominin group to Neanderthals (see Chapter 16 for a discussion on human extinct relatives). It was further determined that these sequences are unlikely to have

existed in the common ancestor of humans and Neanderthals and that their appearance in the human genome was likely due to an interbreeding event between the two hominin groups.

Of the human populations living today, the highest rate of the severe COVID-19 risk haplotype is registered in South Asia, particularly in Bangladesh, where more than half of the population carries these Neanderthal genetic variants. Surprisingly, the risk alleles are nearly absent in East Asian populations. The reason for these differences in the risk allele frequency between the two parts of the same continent is not entirely clear. In all likelihood, this is the result of a strong selection event in the past, except that many epidemiologists consider COVID-19 to be a novel virus to humans, making human encounters with COVID-19 prior to 2019 unlikely. There has to be something other than the response to COVID-19 in these genomic sequences that induced a strong selective response in the past.

A closer look at the genomic region on chromosome 3 associated with the COVID-19 risk haplotype reveals that the region harbors several genes vital to human immune response, specifically, the CCR gene cluster.

The CCR gene cluster encodes members of the beta chemokine receptor family. The ligands of these receptors include macrophage inflammatory protein 1 alpha (MIP-1 alpha), regulated on activation normal T expressed and secreted protein (RANTES), monocyte chemoattractant protein 3 (MCP-3), and myeloid progenitor inhibitory factor-1 (MPIF-1). Chemokines and their receptors mediated signal transduction are critical for the recruitment of effector immune cells to the site of inflammation. Knockout studies of the mouse homolog suggested the roles of this gene in host protection from inflammatory response, and susceptibility to virus and parasite. The chemokine receptor genes, including CCR2, CCRL2, CCR3,

CCR5, CCR9 and CXCR1, are found to form a gene cluster on chromosome 3p.

The COVID 19-associated region contains the CCR9 gene. CCR9 is differentially expressed by T lymphocytes of small intestine and colon, suggesting a role in the thymocytes recruitment and development that may permit functional specialization of immune responses in different segment of the gastrointestinal tract.

The rs35044562 (A-G) single nucleotide polymorphism (SNP), which is at strong LD with 12 other SNPs within the COVID 19 risk region in non-African populations, is at the very beginning of the COVID- 19 associated region on chromosome 3. The same region harbors the gene called the LZTFL1 transcription factor (when read on the opposite DNA strand in the 5'-3' direction). LZTFL1 in a putative tumor suppressor region on chromosome 3. The COVID 19 risk haplotype is almost entirely within the LZTFL1 gene.

It is easy to imagine that the combination of CCR9 and LZTFL1 would be vitally important to modern humans as they ventured out of Africa into Eurasia (see Chapter 16). Thus, it should be of no surprise that a chromosomal region carrying an immunity-boosting and cancer-suppressing gene sequence introgressed from the locally adapted Neanderthals as humans were making their way into uncharted new territory would be a vital aid to survival and would be strongly favored by natural selection. It is likely an unfortunate coincidence that this same region would confer a severe response to a novel viral infection. At the same time, this response may be reflecting a discrete evolutionary adaptation whereby the immune response genes being adapted to known viruses produce an overwhelming and ultimately negative response to a new viral agent.

Decoding the Genetics Behind COVID-19 Infection

Researchers join forces to unravel a biological mystery

February 17, 2022

<https://covid19.nih.gov/news-and-stories/decoding-genetics-behind-covid19-infection>

Since the start of the pandemic, nearly everyone has wanted to know the answer to one question: What is responsible for the different ways people's bodies respond to SARS-CoV-2 infection? Some people infected with SARS-CoV-2, the virus that causes COVID-19, are asymptomatic; some develop moderate but manageable COVID-19; and some, including individuals who are young and otherwise healthy, become severely ill or die from the disease. Some people may never test positive for COVID-19, even while living with an infected person.

Scientists think that for some people the answer lies in the genes, the blueprints we inherit from our parents that play a role in determining everything from our eye color to our risk of having a heart attack. Our

genes also play a role in how our immune system responds to invaders and the characteristics of our lung cells, which can be attacked during the early stages of SARS-CoV-2 infection.

In two studies supported in part by the National Institute of Allergy and Infectious Diseases (NIAID) and the National Human Genome Research Institute (NHGRI), researchers working separately with COVID-19 patients and human lung cells are studying the genetics behind why people are less or more likely to develop COVID-19. These scientists have made unprecedented progress in record time and are laying the groundwork for drug and vaccine developers to create new treatments and vaccines, as well as improve existing ones.

A Misguided Immune Response to COVID-19

“In 18 months, we’ve been able to determine that genetics account for nearly 20% of critical cases of COVID-19, including pneumonia,” said Jean-Laurent Casanova, M.D., Ph.D., of the Rockefeller Institute and the Howard Hughes Medical Institute. His research is funded by NIAID.

Casanova co-leads the [COVID Human Genetic Effort](#), a collaboration of more than 400 researchers and 40 sequencing hubs in 50 countries. He and his collaborators sequenced the genomes of thousands of patients with life-threatening COVID-19, publishing four

papers detailing genetic similarities between people hospitalized with critical COVID-19 pneumonia.

The group found that, in some cases, the immune system mistakenly flagged and attacked proteins called type I interferons (IFNs) that help regulate immune response. Normally, antibodies attach to foreign invaders, marking them for destruction by other immune cells. But the researchers discovered that some people made “auto-antibodies,” antibodies against their own type I IFNs. Nearly 20% of the people who died from COVID-19 created auto-antibodies.

“Their immune systems mistakenly depleted their IFNs, making them more susceptible to the virus,” Casanova said. These IFN-targeting auto-antibodies are found in uninfected individuals too, suggesting that they may contribute to the development of severe COVID-19. “Critical COVID-19 pneumonia probably results from insufficient type I IFNs in the first days of infection,” Casanova said.

Finding the Genes That Create Resistance to COVID-19

Neville Sanjana, Ph.D., of the New York Genome Center and New York University, was interested in another part of the question: What genes protect cells from SARS-CoV-2 infection? With support in part from NHGRI, Sanjana and his colleagues used the gene editing tool CRISPR to figure out which host genes were essential to viral infection, and to what degree.

The team used CRISPR to disable each individual gene that makes up the human genome in human lung cells. Lungs are known to be the primary site of infection for SARS-CoV-2, though other organs and tissues are also affected.

With the help of Benjamin tenOever, Ph.D., of New York University, the researchers then exposed the cells to SARS-CoV-2. Disabling most of the genes didn't make a difference, and the cells died when infected with SARS-CoV-2. But some genes seemed to help SARS-CoV-2 infect cells; when those genes were disabled, most of the cells survived the infection.

The researchers then ranked every gene by how easily SARS-CoV-2 was able to infect host cells after the gene was inactivated. The gene that codes for a protein on the surface of lung cells, known as the ACE-2 receptor, ranked eighth out of 20,000 genes, confirming for Sanjana and his colleagues that their CRISPR assay could pinpoint essential genes for infection. The ACE-2 receptor was already known to be required for SARS-CoV-2 infection, because SARS-CoV-2 needs to bind to it before entering the cell.

Their experiment confirmed that several genes that were already suspected are important for SARS-CoV-2 infection, but the team also found other genes had undiscovered roles. And those genes were active beyond the lung, possibly helping explain how the virus can spread beyond the lungs to other organs. "Some of the key human genes that the virus seems to depend on for successful infection are actually broadly expressed throughout the body. And, as we have seen, once it gets in the lung, the virus seems to spread to

other tissues in people with severe infections,” Sanjana said.

Sanjana and his colleagues did not want to stop there. They searched for existing drugs that might be used to inhibit the top-ranked genes and tested whether those drugs blocked SARS-CoV-2 infection. Several drugs appeared to do just that, including amlodipine, a medication used to treat high blood pressure. Moving forward, scientists can use this research in their search for promising COVID-19 treatments.

The Unpredictability of Emerging Variants

These research groups continue to work at an unprecedented pace, given the unpredictability of emerging SARS-CoV-2 variants. For example, Sanjana and colleagues are developing tests to understand the mechanisms that enable new variants to become more infectious. They showed how one of the first spike protein mutations, D614G, allowed SARS-CoV-2 variants to enter host cells more easily than the original viral strain could. Today, all currently circulating strains of SARS-CoV-2, including Delta and Omicron, carry the D614G mutation. The Sanjana lab is also developing new genome editing tools to explore the interactions between specific human genetic mutations and viral variants.

For many scientists, the pandemic has led to an “all-

hands-on-deck” response. The scientific community has shown that it can work together in new ways, step outside of usual specialties, produce rapid results, and influence the course of the pandemic. “As basic geneticists,” Sanjana said, “we rarely get the chance to do this kind of translational work where we can literally hand off drug candidates to drug developers around the world. It’s wonderful to see how people have picked up different aspects of the work and are running with it.”

Chapter 13.

Developmental Genetics

Human Reproduction and Development

All mammals, including humans, reproduce sexually.

The human reproductive system consists of a pair of gonads (ovaries or testes) where the gametes are produced, a system of ducts that house and conduct the gametes, and structures that facilitate copulation.

The human female has a pair of ovaries that contain egg cells (oocytes) surrounded by nourishing cells. An egg cell surrounded by these supportive cells is called a follicle. An egg cell is released from the follicle during ovulation into an oviduct (fallopian tube).

The uterus (the womb) houses the embryo (0 – 9 weeks) and the fetus (9th week – birth) during pregnancy.

The male reproductive system consists of a pair of testes that produce sperm; a system of ducts that moves sperm out of testes, and a set of glands that produce supportive fluids. Sperm mixed with the secretions of the glands forms semen that is discharged during ejaculation.

The formation of gametes (sperm and ova) requires meiosis.

Spermatogenesis is the formation of sperm cells.

The first division of spermatogenesis involves the production

of haploid secondary spermatocytes from diploid primary spermatocytes.

Haploid sperm cells form during the second division of spermatogenesis.

Oogenesis is the formation of an egg (ovum).

Oogenesis begins prior to birth. At birth, follicles contain diploid primary oocytes (arrested at the tetrad stage of Prophase I of Meiosis I).

At puberty, a single primary oocyte completes Meiosis I each month, producing a haploid secondary oocyte and a polar body (smaller cell, not essential in reproduction) and progresses to Metaphase II of Meiosis II.

The second division of oogenesis takes place when a sperm cell penetrates the secondary oocyte, and yields a haploid ovum and a second polar body.

Fertilization stimulates the ovum to complete the meiotic division.

Human development from conception to birth is divided into three trimesters (three months each). Childbirth is hormonally induced and occurs in three stages.

Stages of early human development: fertilization, cleavage, gastrulation.

During fertilization, the haploid nucleus of the ovum fuses with the haploid nucleus of the sperm cell, producing a zygote.

Fertilization: the union of egg and sperm:

- occurs in fallopian tubes (oviducts);
- out of 200 million sperm released per ejaculation, only one sperm penetrates the zona pellucida (clear zone around egg); upon this event, cortical granules are released that prevent any other sperm from penetrating the egg;
- after fertilization, the egg and sperm nuclei fuse, and a new diploid human zygote is formed.

After fertilization, the zygote undergoes a series of cell divisions (cleavage), producing a multicellular embryo. Cleavage partitions the embryo into developing regions. Cleavage produces a blastocyst, which contains a group of differentiated cells on the outer surface (the trophoblast cell layer) and still undifferentiated inner cell mass.

Day 1 post-conception: first cleavage – one cell becomes two.

Day 2: second cleavage – 4-cell stage.

Day 3: 6-12 cell stage – can test at this stage for genetic diseases by Preimplantation Genetic Diagnosis, PGD (if fertilization done in vitro (IVF)).

Day 4: 16-32 cell stage – solid ball of cells – morula.

Day 5: Solid morula develops into hollow, fluid-filled blastula.

Blastocyst implants in the uterine wall lining by about 7th day after fertilization. Implantation is aided by the enzymes released by the trophoblast cells. At the time of implantation, the upper layer of the inner cell mass forms a separate group of cells called an epiblast, the lower (outer) layer is called the hypoblast.

Gastrulation occurs by the inward movement of epiblast cells and forms the three tissue layers of the embryo: ectoderm, endoderm, and mesoderm. At later developmental stages, the ectoderm produces skin tissue and nervous system, the endoderm produces the lining of the digestive and respiratory tracts, and the mesoderm forms most other organs and tissues (heart, muscle, bone etc.).

Cells of the trophoblast give rise to chorion (the embryonic portion of the placenta); the hypoblast cells form the yolk sac (contains no yolk) where early blood cells form and later migrate to the embryo proper.

Day 6 –7, implantation: Blastocyst attaches to the endometrium and burrows in.

Days 10 – 14: Pregnancy becomes established:

- Fluid filled amniotic cavity starts to form;
- Yolk sac starts to form (will make blood cells, germ cells); -!
- Embryo starts to form from embryonic disc;
- Chorion (placenta) starts to form.

Developmental processes that take place after gastrulation:

Organ formation:

neural tube (future spinal cord and brain) and heart are formed by the 26th day of embryonic development;

pattern formation (the shaping of an animal's major body structures).

The embryo and placenta take shape during the first month of pregnancy.

Week 3 – Week 8 (Embryo) – Development of all organ systems

Days 15 – 21: Emergence of the vertebrate body plan:

- Primitive streak starts to form – this is the site of gastrulation (formation of the 3 tissue layers – ecto-, endo-, and mesoderm);
- Neural groove (future spinal cord and brain) begins to form;
- Somites (bands of tissue that will become muscles and bones) begin to form;
- Pharyngeal arches (future face, neck, mouth, nose) begin to form.

Day 22: Cardiac cells (early heart) begins to beat.

Week 9 – Week 40: (Fetus):

After ~12 weeks, the organ and pattern formation is largely “finished” – except brain and lung development.

Gene Regulation During Eukaryotic Cell

Differentiation And Development

Embryonic development depends on genetic as well as environmental influences that are temporally and spatially adjusted to one another. The factors that, for example, determine the interactions between the tissues, the migration and differentiation of the cells, the proliferation of the cell colonies, as well as the apoptosis (programmed death of cells) are numerous. Embryonic development is a process of growth and differentiation in which the embryo becomes increasingly complex and is more and more enhanced with structures and functions. The growth depends on the somatic multiplication of the cells through mitosis. One of the most fascinating aspects of embryonic development is the fact that out of a simple zygote (fertilized oocyte) an organism arises that consists of billions of cells.

A human body consists of about 250 cell types, each with a different morphology, biochemistry and role they play in the organism, while each cell type (except for reproductive cells) are genetically equivalent, i.e., they all have the same amount of DNA. The distinctions between cell types are the result of differences in gene expression patterns. Thus, somatic cells differ in structure and function not because they contain different genes, but because they express different portions of a common genome.

Some eukaryotic genes are only expressed during development. Most eukaryotes (and some bacteria) switch genes off and on at different stages in their development. Developmental regulation (also called the long-term regulation of gene expression) enables the biochemistry of an organism to be altered in response to the particular requirements of each stage of its life cycle.

Master control genes

The human body is composed of several trillion cells. All of them originate from the same single fertilized egg, yet almost all of them have become specialized to perform certain tasks. But if all of the body's cells (except for the lymphocytes) contain the same genes, how does one cell become a nerve cell rather than, say, a liver cell? The answer lies in the fact that not all the genes in a cell are activated at the same time. Some proteins in the nuclei of cells will, by binding to their DNA, trigger the expression of only certain genes of this DNA. And products of these particular genes will be proteins whose function will contribute to making the cell more specialized.

MyoD and muscle cell differentiation

Muscle cells develop from embryonic precursors that have the potential to become a number of alternative cell types, including cartilage cells, fat cells, or multinucleate muscle cells. As the muscle cells differentiate, they become myoblasts and begin to synthesize muscle-specific proteins. They fuse to form mature, elongated, multinucleate skeletal muscle cells.

Researchers developed a hypothesis that certain muscle-specific regulatory genes are active in myoblasts, leading to muscle cell determination. To test this, researchers isolated mRNA from cultured myoblasts and used reverse transcriptase to produce genomic DNA complementary to isolated RNA (called complementary DNA or cDNA) containing all the

genes that are expressed in cultured myoblasts. Transplanting these cloned genes into embryonic precursor cells led to the identification of several “master regulatory genes” that, when transcribed and translated, commit the cells to become skeletal muscle.

One of these master regulatory genes is called *myoD*. The *myoD* gene encodes MyoD protein, a transcription factor that binds to specific control elements and stimulates the transcription of various genes, including some that encode for other muscle-specific transcription factors. These secondary transcription factors activate the muscle protein genes.

MyoD also stimulates expression of the *myoD* gene itself, perpetuating its effect in maintaining the cell's differentiated state. MyoD protein is capable of changing fully differentiated non-muscle cells into muscle cells. However, not all cells will transform. Non-transforming cells may lack a combination of regulatory proteins, in addition to MyoD.

Certain eukaryotic genes are expressed in a sequence during development. A master control gene is the first gene activated in a hierarchy that leads to differentiation along a particular pathway. Master control genes encode the first transcription factor in a hierarchy. A master control gene product activates the next set of genes that encodes the next set of transcription factors, and the cascade of gene expression has been set in motion.

This cascade of gene expression starts at first mitotic division of a developing zygote and is controlled by cytoplasmic determinants in the oocyte. This type of control is called

autonomous regulation. Examples of such determinants are maternal effect genes (rather, the RNA copies of those genes).

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Maternal effect genes

Maternal factors govern early embryonic development before embryonic genome activation takes place. Maternal genetic determinants are distributed asymmetrically in the cytoplasm of the oocyte, which leads to unequal distribution among cells after cell division, subsequently affecting the activity of genes these maternal determinants are designed to regulate. The maternal genetic determinants in the oocyte influence the course of early development by producing a set of proteins (transcription factors) that initiate the transcription of the genes of the developing embryo. A subsequent cascade of gene activations sets up the developmental process that gives rise to the various parts of the organism (morphogenesis).

Most of the maternal effect genes are known from non-mammalian organisms. The presence and importance of maternal effect genes in early mammalian development has been only inferred until the recent identification of maternal effect genes in mice, such as *Mater* and *Hsf1*. A human homologue of *Mater* has been identified as well. To date, little is known about the mechanisms by which maternal products direct early development in mammals.

Homeobox genes

Some of the most essential embryonic genes are called homeo- box genes. Every organism has a unique body pattern because of the influence of homeobox genes. These specify how different areas of the body develop their individual structures (arms, legs etc.).

Homeotic genes are regulatory genes that determine where certain anatomical structures, such as appendages, will develop in an organism during morphogenesis. Homeotic genes encode homeotic proteins that function as transcription factors that switch on other genes. The homeobox is a sequence within a homeotic gene that contains 180 nucleotide-long sequences, which code for a DNA-binding domain (homeodomain).

In humans, the homeobox gene family contains an estimated 235 functional genes. Homeobox genes are present on every human chromosome, and they often appear in clusters. Mutations in homeobox genes are responsible for a variety of developmental disorders (craniofacial abnormalities etc.). Some homeobox genes act as tumor suppressors, which means they help prevent cells from growing and dividing too rapidly or in an uncontrolled way. For example, mutations in the HOXB13 gene can lead to prostate cancer.

Cell-to-cell signaling and transduction pathway during embryonic development

Adjacent cells in the developing embryo are able to send signals to each other (cell-to-cell signaling). This signaling allows the cells to determine their fates during development. Such cellular induction is another major mechanism that regulates embryonic development.

The induction signal can be a diffusible molecule that is secreted by the inductor and binds to the surface receptor on the reacting cell. The signal molecule can also travel along the extracellular matrix and bind to the reacting cell. Induction may also take place via direct cellular contact, during which the signal molecule leaves the inductor and enters the reactant cell via a gap junction.

Cells in the developing organism also communicate remotely through the signal- transduction pathways. A signal-transduction pathway is a series of molecular changes that convert a signal received by a target cell's surface into a specific response inside the cell.

Genes that perform key functions during development are conserved among the eukaryotic organisms. For example, the genes involved in the development of the *Drosophila*, with few exceptions, are the same as those involved in the development of higher organisms. Developmental biology studies the sequential activation and interaction of genes, in relation to developing morphology.

Cell differentiation can be reversed. The nuclei of most differentiated animal cells have all the genes required for making an entire organism. Animal cloning is a procedure that involves the replacement of the nucleus from an animal egg cell with the nucleus from a somatic cell of another animal of the same species, and an induction (chemical and physical) of cell division in the modified egg.

Cell differentiation begins at early stages of embryonic development. A mammalian embryo that underwent several stages of cell division after fertilization consists of a ball of cells called blastocyst. The inner cell mass cells of the blastocyst can be extracted and grown on a nutrient medium outside of the blastocyst. Such cells will have a potential to become any cell of the body (liver, muscle, heart etc.). These cells are called

embryonic stem cells. Stem cells are sometimes found in adult tissues and organs.

Chapter 14. Molecular Genetic Technology

Cloning And Stem Cell Research

Cloning

The word “cloning” refers to making an identical copy of something (usually several times over).

Ex.: skin cells clone themselves through mitotic division.

Cloning, as applied to eukaryotic organisms, can be divided into three categories.

Gene cloning (producing recombinant DNA) is an insertion of a genetic sequence of interest into a vector (plasmid) and propagation of the resulting construct until a sufficient amount of inserted (recombinant) DNA is produced.

Reproductive cloning refers to a generation of a multicellular organism from a DNA sample from an oocyte carrying the nucleus from a somatic cell of the individual being cloned. The goal of reproductive cloning is to produce a viable individual genetically identical to the donor of the somatic nucleus (for example, cloning an entire sheep from an udder cell of that sheep).

Therapeutic cloning is the generation of embryonic or induced pluripotent stem cells (iPSCs) containing the donor's DNA for subsequent growth of tissue for donor's transplantation needs.

Therapeutic cloning is concerned with the propagation of cells

of a particular individual for therapeutic purposes and not for re-productive purposes.

Therapeutic Cloning and Embryonic Stem Cells

The word “cloning” in this context refers to a process of initiation of embryonic cell division from a genetic material that replaces the genetic material of the ovum, thus by-passing the fertilization of an egg by a sperm. As such, cloning refers specifically to propagating of totipotent embryonic cells till the blastocyst stage of development. Consequently, there is no intent to actually clone a human being through this procedure. At the same time, since the use of embryonic stem cells has been marred in controversy, therapeutic cloning has recently taken advantage of the induced pluripotent stem cells (iPSCs) derived from adult multipotent stem cells.

The potency of stem cells

The potency of a cell specifies its differentiation potential, or potential to differentiate into different cell types. Totipotency is the ability of a single cell to divide and produce all the differentiated cells in an organism, including extraembryonic tissues. After reaching the 16-cell stage post-fertilization in a developing embryo, the totipotent cells of the morula differentiate into cells that will eventually become either the blastocyst's Inner cell mass or the outer trophoblasts. Approximately four days after fertilization and after several cycles of cell division, these totipotent cells begin to specialize. The inner cell mass, the source of embryonic stem cells, is pluripotent, not totipotent.

Pluripotency (from the Latin *plurimus*, meaning very many, and *potens*, meaning having power) refers to a stem cell that has the potential to differentiate into any of the three germ layers: endoderm (interior stomach lining, gastrointestinal tract, the lungs), mesoderm (muscle, bone, blood, urogenital), or ectoderm (epidermal tissues and nervous system). Pluripotent stem cells can give rise to any fetal or adult cell type. However, alone they cannot develop into a fetal or adult organism because they lack the potential to contribute to extraembryonic tissue, such as the placenta.

Multipotent stem cells are a specialized cell type that is produced from pluripotent stem cells. They are found in adult tissues. A multipotent stem cell can give rise to other types of cells but it is limited in its ability to differentiate. These other types of cells are also limited in numbers. Examples of multipotent stem cells include those in the brain that give rise to different neural cells and glia or haematopoietic cells, which can give rise to different blood cell types, but they can't create brain cells. Bone marrow also contains multipotent stem cells that give rise to all blood cell types but not other cells. Multipotent stem cells are essentially committed to produce specific cell types.

Induced Pluripotent Stem Cells (iPSCs)

iPSCs are derived from adult skin or blood stem cells or the fully differentiated somatic cells that have been reprogrammed back into an embryonic-like pluripotent state that enables the development of an unlimited source of any type of human cell needed for therapeutic purposes. For example, iPSC can be prodded into becoming beta islet cells to treat diabetes, blood cells to create new blood free of cancer cells for a leukemia patient, or neurons to treat neurological disorders.

Originally, iPSCs were expected to replace therapies based on

the use of embryonic stem cells in regenerative medicine. The idea was to take a person's skin, blood or other cells, reprogram them into iPS cells, and then use those to grow liver cells, neurons or whatever was needed to treat a disease. This personalized therapy would get around the risk of immune rejection as well as sidestepping the ethical concerns of using cells derived from embryos. However, with time the goals have shifted — in part because those therapies have proved challenging to develop. The only clinical trial using iPSC cells was halted in 2015 after just one person had received a treatment.

But iPS cells have made their mark in a different way. They have become an important tool for modelling and investigating human diseases, as well as for screening drugs. Improved ways of making the cells, along with gene-editing technologies, have turned iPSC cells into a lab workhorse — providing an unlimited supply of once-inaccessible human tissues for research.

This has been especially valuable in the fields of human development and neurological diseases. Nevertheless, a number of potentially therapeutic applications have been developed in the last few years, such as the production of neural stem cells from fibroblasts to fight glioblastoma tumors.

Could Skin Cells Be Used To Kill Cancer?

By Kathleen Lees

First Posted: Feb 24, 2016 06:42 PM EST

<https://www.scienceworldreport.com/articles/37684/20160224/could-skin-cells-be-used-to-kill-cancer.htm>

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Researchers have discovered a way to turn skin cells into cells that fight cancer. Researchers at the University of North Carolina discovered this technique that fights remnant brain tumor cells in mice. In humans with glioblastomas—an aggressive form of brain cancer in which—only 30 percent patients survive past the two-year mark following a diagnosis.

“We wanted to find out if these induced neural stem cells would home in on cancer cells and whether they could be used to deliver a therapeutic agent,” Dr.

Shawn Hingtgen, an assistant professor at the University of North Carolina, said in a news release.

“This is the first time this direct reprogramming technology has been used to treat cancer.” During the study, researchers reprogrammed

fibroblasts—otherwise known as skin cells— in order to make them neural stem cells and create a tumor-killing protein. The cells, which in skin produce collagen and connective tissue, hunt and kill cancer cells, according to UPI.

Based on the type of tumor, researchers were able to increase survival time in the mice from 160 to 220 percent. With future studies, researchers hope to focus on human stem cells and testing more effective anti-cancer drugs that can be loaded into tumor-seeking neural stem cells.

“Our work represents the newest evolution of the stem-

cell technology that won the Nobel Prize in 2012,” Hingtgen said, via Medical Xpress. “We wanted to find out if these induced neural stem cells would home in on cancer cells and whether they could be used to deliver a therapeutic agent. This is the first time this direct reprogramming technology has been used to treat cancer.” The findings are published in the journal *Nature Communications*.

Recombinant DNA Technology

Recombinant DNA (rDNA) molecules are DNA sequences that result from the use of laboratory methods (molecular cloning) to bring together genetic material from multiple sources, creating sequences that would not otherwise be found in biological organisms. Recombinant DNA is possible because DNA molecules from all organisms share the same chemical structure; they differ only in the sequence of nucleotides within that identical overall structure. Consequently, when DNA from a foreign source is linked to host sequences that can drive DNA replication and then introduced into a host organism, the foreign DNA is replicated along with the host DNA.

Cloning vectors

Formation of recombinant DNA requires a cloning vector, a DNA sequence capable of carrying DNA from another source that can replicate within a living cell. Vectors are generally derived from plasmids or viruses.

A plasmid is a small DNA molecule that is physically separate

from, and can replicate independently of, chromosomal DNA within a cell. Most commonly found as small circular, double-stranded DNA molecules in bacteria, plasmids are sometimes present in archaea and eukaryotic organisms. In nature, plasmids carry genes that may benefit survival of the organism (e.g., antibiotic resistance), and can frequently be transmitted from one bacterium to another, even of another species. Artificial plasmids are widely used as vectors in molecular cloning, serving to drive the replication of recombinant DNA sequences within host organisms.

Restriction Enzymes

DNA restriction endonucleases, or restriction enzymes, recognize short, specific palindromic sequences of DNA double-stranded nucleotides and make breaks in the sugar-phosphate backbone of the DNA in the region of the recognized sequence.

Example: EcoRI restriction enzyme:

5'GAATTC

3'CTTAAG

Restriction enzymes, also called restriction nucleases) surround the DNA molecule at the recognized sequence (GAATTC for EcoRI). They cut one strand of the DNA double helix at one point and the second strand at a different, complementary point (for EcoRI, between the G and the A base).

DNA micromanipulations: Recombinant plasmids and DNA cloning

The use of cloning is interrelated with recombinant DNA in classical biology, as the term “clone” refers to a cell or organism

derived from a parental organism, with modern biology referring to the term as a collection of cells derived from the same cell that remain identical. In the classical instance, the use of recombinant DNA provides the initial cell from which the host organism is then expected to recapitulate when it undergoes further cell division, with bacteria remaining a prime example due to the use of viral vectors in medicine that contain recombinant DNA inserted into a structure known as a plasmid.

Plasmids are extrachromosomal self-replicating circular forms of DNA present in most bacteria, such as *Escherichia coli* (*E. coli*), containing genes related to catabolism and metabolic activity, and allowing the carrier bacterium to survive and reproduce in adverse environments. Genes carried by plasmids include those of antibiotic resistance. Plasmids' DNA is usually several thousand base pairs in size, and can be modified by extracting them from the host bacterium, and inserting or removing genetic sequences with the use of restriction endonucleases. By accepting a DNA sequence of interest, a plasmid becomes a cloning vector, which can be re-inserted back into a bacterial cell, which would then be allowed to multiply, and, thus, to increase the copy number of the recombinant DNA plasmids they carry. The size of a genetic sequence a plasmid can accept is limited, and better integration rates are usually achieved with fragments comparable with the plasmid's own size. For larger genetic sequences, alternative vectors, such as phagemids or artificial bacterial or yeast chromosomes (BACs and YACs), are used.

Plasmids contain three common genetic features—a replicator, a selectable marker and a cloning site. The replicator or “ori” refers to the origin of replication of plasmid's DNA. The marker refers to a particular gene that usually contains resistance to an antibiotic, but may also be a gene that is attached alongside the desired one, such as that which confers

luminescence to allow identification of successfully recombined DNA. The cloning site is a sequence of nucleotides representing one or more positions where cleavage by restriction endonucleases occurs.

A Petri plate showing the growth of host cells after uptake of recombinant plasmids. The medium on the plates contains a compound called X-gal. DNA inserts in the plasmid vector and disrupts the gene responsible for the formation of blue colonies. Cells in blue colonies do not carry any cloned DNA inserts, whereas white colonies contain vectors carrying the DNA of interest.

Biotechnology and human disease: manufacturing insulin

Eukaryotic genes can also be artificially expressed in prokaryotic cells (*E. coli*) by a physical transfer into a prokaryotic genome. Prokaryotic cells expressing eukaryotic genes can produce large amounts of eukaryotic protein like insulin.

Biotechnology is the manipulation of biological systems to manufacture important foods, drugs, biopolymers and other materials. This is illustrated below by the production of the protein human insulin in bacterial culture.

1. A recombinant DNA plasmid is produced to include the piece of human DNA containing instructions for making proinsulin. Proinsulin is the precursor for insulin.
2. The plasmid is then inserted into *E. coli* bacteria,

which can now produce human proinsulin, along with bacterial own proteins.

3. The bacteria multiply and produce large quantities of human proinsulin in a fermentor.

4. Bacteria are then killed by heat sterilization, leaving behind the proinsulin.

5. Proinsulin is enzymatically cleaved to produce human insulin. Enzymatic cleavage is the slicing of a molecule by an enzyme.

6. The solution is centrifuged to remove cell matter and purified by liquid chromatography and crystallization.

Polymerase Chain Reaction (PCR)

Polymerase chain reaction (PCR) is a form of gene cloning that does not involve plasmids. PCR is a technique to amplify a single or a few copies of a piece of DNA across several orders of magnitude, generating thousands to millions of copies of a particular DNA sequence. The method relies on thermal cycling, consisting of cycles of repeated heating and cooling of the reaction for DNA denaturation and re-annealing and enzymatic replication of the DNA, using a thermostable DNA polymerase.

PCR reaction components include DNA template that contains

the DNA region (target) to be amplified; two primers that are complementary to the 3' ends of each of the strand of the DNA target; DNA polymerase with a temperature optimum at around 70 °C, and deoxynucleoside triphosphates (dNTPs), the building blocks from which the DNA polymerases synthesizes a new DNA strand. A PCR reaction also needs to be supplied with a buffer solution, providing a suitable chemical environment for optimum activity and stability of the DNA polymerase. Divalent cations, such as magnesium or manganese ions, are important polymerase co-factors and need to be supplied in the buffer as well.

Quantitative PCR (qPCR)

Quantitative PCR (qPCR) is a PCR technique that allows researchers to estimate the quantity of starting material in a DNA sample. Please watch the following Youtube video for additional details about the qPCR technique and its application in detecting the presence of COVID-19 virus in a biological sample.



One or more interactive elements has been excluded from this version of the text. You can view them online here: <https://pressbooks.gvsu.edu/humangenetics/?p=93#oembed-1>

DNA sequencing

Sanger dideoxynucleotide terminator-cycle sequencing

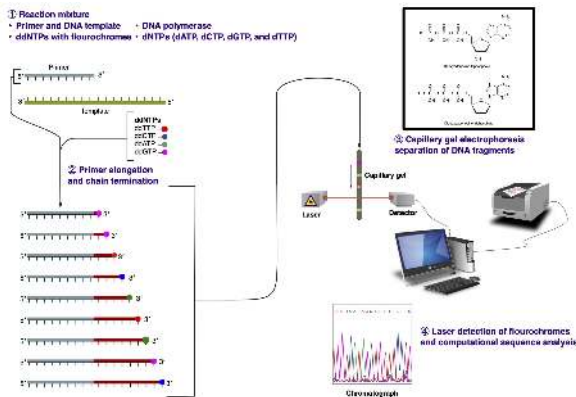
In 1977, Frederick Sanger devised the chain-termination dideoxynucleotide method for sequencing. The fluorescent dye-terminator cycle sequencing method, adapted for PCR amplification in 1980s, is based on this concept.

The PCR reaction includes the following ingredients: DNA template, primer, Taq DNA polymerase, unlabeled dNTP's, fluorescently labeled ddNTP's, buffer, and distilled water. The fluorescently labeled ddNTP's are called dideoxynucleotides, named so because they lack a 3' OH. Due to that chemical modification, no additional nucleotides can be added to the chain once a ddNTP has been added; therefore, the sequence terminates in a fluorescently labeled ddNTP. Each base has a differently colored dye that can be recognized by a laser.

During PCR, the temperature is increased to ~95 degrees Celsius to denature the double-stranded DNA. Then the temperature drops to ~50 degrees Celsius to allow the primers to anneal to the template. Finally, Taq polymerase adds nucleotides to the growing chain at ~60 degrees Celsius. This cycle repeats 25-30 times. During the last step, the Taq polymerase adds either the dNTP's or the ddNTP's to the chain randomly. Therefore, when PCR is complete, you are left with PCR fragments of all different lengths; for whenever a ddNTP was added, the elongation stopped.

You can then load the entire PCR reaction into one lane of a polyacrylamide gel in a special laser sequencer. The sequencer can read the different dyes as they pass the laser at the bottom of the gel. Because the different fragments migrate toward

the positively-charged pole at different rates, the shorter fragments migrate toward the bottom of the gel faster, so you read it bottom->top (5'->3'). The computer then compiles these data into the image of a gel. The colors are then read as bases to produce a chromatogram of the piece of DNA sequenced. The sequence can be taken right from the chromatogram with an extremely high level of accuracy.



Sanger chain-termination DNA sequencing technique.
 By Estevezj – Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=23264166>

The main limitation of the Sanger sequencing method is in the size of the DNA fragments that can be sequenced. Typically, the best reading resolution is achieved for sequences 100 to 300 nucleotides in length (excluding primer sequences). It is not uncommon to get a decent readout for sequences in the 500 nucleotide range, but the equality of the sequencing read starts to drop precipitously after that.

Next generation sequencing (NGS)

Next generation sequencing (NGS) is often referred to as massively parallel sequencing, which means that millions of small fragments of DNA can be sequenced at the same time, creating a massive pool of data. This pool of data can reach gigabytes in size, which is the equivalent of 1 billion (1,000,000,000) base pairs of DNA. In comparison, previous methods could sequence one DNA fragment at a time, perhaps generating 500 to 1000 base pairs of DNA in a single reaction.

‘Traditional’ genetic testing involved the sequence analysis of a number of genes in a step-wise manner to try to identify where a genetic alteration may be. An example of this could be the genetic testing involved in cases of inherited breast cancer where two genes, BRCA1 and BRCA2 are usually sequenced in the laboratory to look for genetic alterations in affected patients. This could require sequencing of more than 60 fragments or pieces of DNA to identify where a single DNA base change may be present. Using NGS these genes (and perhaps others that have potential clinical relevance) could be sequenced all at once in a single test. This clearly has the potential to provide information on lots of genes all at once.

Another example might be where there are lots of genes involved in the development of a tumor such as lung cancer. Looking at the molecular profile (or genetic make-up) of the tumor sample using NGS might lead to the identification of an alteration that could suggest the best drug treatment option for a patient or perhaps a clinical trial that the patient could be entered into in the development of new and effective cancer treatments. This is in the area of stratified medicine.

Extending this technology further, it is becoming possible to

undertake whole exome or genome analysis for patients. This is a particular area that would not have been possible without the development of NGS. In this type of analysis, it is possible for scientists to look at the entire genetic make-up of a patient. The genome is all of the genetic material in an individual and includes all of the genes and DNA that each cell in the human body contains.

This is an incredibly powerful development, and whilst offering the potential to identify exactly where a gene alteration might be present in a patient, it does also provide the potential to identify some unexpected findings as well. Consequently this testing is being used very carefully and with the full consent and understanding of the patient.

The introduction of NGS also has a significant part to play in the area of genomics. This is particularly relevant in stratified or personalized medicine where it is possible to identify the genetic alterations that can predict how a patient might respond to a specific drug and decide what dose might be most effective. To achieve this it is necessary to 'profile' or sequence a number of genes to identify the most relevant genetic alterations.

Perhaps the biggest challenge for the genetics laboratory when creating large amounts of sequence data by NGS is the vast quantity of computer data that is produced. This can be as much as 600 gigabytes in a single run. That is the equivalent of perhaps 90,000 songs on an MP3 player. Bearing in mind this is data from a single NGS run, then over the weeks, months and years NGS will produce large amounts of data that require management and storage in a safe and secure manner. In addition there is significant reviewing and scientific interpretation of the data produced to translate into useful information that can be passed back to the clinician and patient.

Shotgun DNA Sequencing

Shotgun DNA sequencing refers to the NGS technique where researchers first shred DNA into small fragments, then sequence the fragments, followed by arranging the fragments in one linear sequence to reassemble the entire genome. Please read more about Shotgun DNA Sequencing under the following link:

<https://www.genome.gov/genetics-glossary/Shotgun-Sequencing>

Exome Sequencing

Exome Sequencing targets the protein-coding portion of the genome for sequencing. Protein-coding regions comprise about 2.5% of the human genome (see Chapter 1), which in reality amounts to a little over 1% if we consider just the exon (protein-coding) portion of the protein-coding DNA (excluding introns and various control sequences). The collection of all exon sequences in the genome is called an exome. At the same time, the exome harbors 85% of all disease-causing mutations in human DNA. For clinical purposes, exome sequencing is a more targeted and more cost-effective alternative to the whole genome sequencing.

Gene editing

CRISPRs (clustered regularly interspaced short palindromic repeats) are segments of prokaryotic DNA containing short repetitions of base sequences. The CRISPR/Cas system is a prokaryotic immune system that confers resistance to foreign

genetic elements such as plasmids and phages. The CRISPR/Cas system has been used for gene editing (adding, disrupting or changing the sequence of specific genes) and gene regulation. CRISPR/Cas is a radically fast and easy way to precisely cut and replace DNA sections in a living organism.

CRISPR/Cas DNA editing system consists of two components: a “guide” RNA (gRNA) and a non-specific CRISPR-associated endonuclease (Cas9). The gRNA is a short synthetic RNA composed of a “scaffold” sequence necessary for Cas9-binding and a user-defined ~20 nucleotide “spacer” or “targeting” sequence which defines the genomic target to be modified. Thus, one can change the genomic target of Cas9 by simply changing the targeting sequence present in the gRNA.



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Gene therapy

Gene therapy involves the transfer of a therapeutic or working gene copy into specific cells of an individual. It may be used to:

- Replace a faulty gene
- Introduce a new gene whose function is to cure or to favorably modify the clinical course of a condition
- Inactivate or “knock out” a faulty gene that is not functioning properly.

The potential of gene therapy is very broad, with research involving a number of diseases such as severe combined immune-deficiencies (SCID), hemophilia, Parkinson's disease, many forms of cancer and HIV.

There are many challenges to successful gene therapy.

Firstly, the condition in question must be well understood and the underlying causative gene identified. A working copy of the gene involved must be available and the specific cells in the body requiring treatment must be identified, accessible and a means of efficiently delivering working copies of the gene to these cells must be available.

Of all these challenges, the one that is most difficult is the problem of gene delivery i.e. how to get the new or replacement genes into the desired tissues.

One of the most promising methods currently being developed is the use of harmless viruses that can be used to carry genes into cells. Scientists now have the knowledge and skills to remove the virus' own genes and to replace them with working human genes. These altered viruses can then be used to deliver genes into cells with great efficiency. When viruses are used in this way they are known as vectors.

Some of these vectors are capable of not only carrying the gene into the cell but also of inserting the gene into the genetic material of the cell. Once in the right location within the cell of an affected person, the transplanted gene is switched on. The transplanted gene can then issue the instructions necessary for the cell to make the protein that was previously missing or altered.

Another technique with a promising potential is the use of stem cells in delivering gene therapy. Stem cells are cells that have not yet differentiated into a specific tissue or organ cell.

In this technique, stem cells are manipulated in the

laboratory in order to make them accept new genes that can then change their behavior. For example, a gene might be inserted into a stem cell that could make it better able to survive chemotherapy. This would be of assistance to those patients who could benefit from further chemotherapy following stem cell transplantation.



One or more interactive elements has been excluded from this version of the text. You

can view them online here:

<https://pressbooks.gvsu.edu/humangenetics/?p=93#oembed-3>

Applications of Molecular DNA Technology in Forensic Science and Human Identity Research

Forensic DNA analysis utilizes genetic tools and scientific methodology to examine biological evidence in the process of criminal and civil litigation. Human identity analysis is used in broader applications, such as ancestry analysis.

The techniques used by forensic and identity DNA analysis have vastly evolved in the recent decades. Depending on the information needed, there are several different techniques that can be used to type a DNA fragment. Short tandem repeat (STR) fragment analysis, Sanger sequencing followed by capillary electrophoresis are some of the techniques that have been used for the genetic analysis of DNA samples. NGS is the newest and

most revolutionary technology that is becoming a commonplace in the genetic analysis of forensic and identity samples.

The Ubiquity of Human DNA in the Environment

Molecular genetic technology can be used to identify individuals using the DNA they leave behind. Humans “shed” their DNA anywhere they go. Human DNA can be found in the soil, water, and air. While the ability to extract and analyze DNA from environment using eDNA techniques can be a useful tool in tracking down people who break the law, it can also be used for less legally justifiable purposes.

New threat to privacy? Scientists sound alarm about DNA tool

By Juliette Collen, AFP

Mon, May 15, 2023

<https://news.yahoo.com/threat-privacy-scientists-sound-alarm-150315334.html>

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The traces of genetic material that humans constantly shed wherever they go could soon be used to track individual people, or even whole ethnic groups, scientists said on Monday, warning of a looming “ethical quagmire”.

A recently developed technique can glean a huge amount of information from tiny samples of genetic material called environmental DNA, or eDNA, that humans and animals leave behind everywhere — including in the air.

The tool could lead to a range of medical and scientific advances, and could even help track down criminals, according to the authors of a new study published in the journal *Nature Ecology & Evolution*.

But it also poses a vast range of concerns around consent, privacy and surveillance, they added.

Humans spread their DNA — which carries genetic information specific to each person — everywhere, by shedding skin or hair cells, coughing out droplets, or in wastewater flushed down toilets.

In recent years, scientists have been increasingly collecting the eDNA of wild animals, in the hopes of helping threatened species.

For the new research, scientists at the University of Florida's Whitney Laboratory for Marine Bioscience had been focused on collecting the eDNA of endangered sea turtles.

– 'Human genetic bycatch' –

But the international team of researchers inadvertently collected a massive amount of human eDNA, which they called "human genetic bycatch".

David Duffy, a wildlife disease genomic professor at the Whitney Laboratory who led the project, said they were "consistently surprised" by the amount and quality of the human eDNA they collected.

"In most cases the quality is almost equivalent to if you took a sample from a person," he said.

The scientists collected human eDNA from nearby oceans, rivers and towns, as well as from areas far from human settlements.

Struggling to find a sample not tainted by humans, they went to a section of a remote Florida island inaccessible to the public.

It was free of human DNA — at least until a member of the team walked barefoot along the beach. They were then able to detect eDNA from a single footprint in the sand.

In Duffy's native Ireland, the team found human DNA all along a river, with the exception of the remote mountain stream at its source.

Taking samples from the air of a veterinary hospital, the team captured eDNA that matched the staff, their animal patient and viruses common in animals.

– ‘Perpetual genetic surveillance’? –

One of the study's authors, Mark McCauley of the Whitney Laboratory, said that by sequencing the DNA samples, the team was able to identify if a person had a greater risk of diseases such as autism and diabetes.

“All of this very personal, ancestral and health-related data is freely available in the environment, and it's simply floating around us in the air right now,” McCauley told an online press conference.

“We specifically did not examine our sequences in a way that we would be able to pick out specific individuals because of the ethical issues,” he said.

But that would “definitely” be possible in the future, he added.

“The question is how long it takes until we're at that stage.”

The researchers emphasised the potential benefits of collecting human eDNA, such as tracking cancer mutations in wastewater, discovering long-hidden archaeological sites or revealing the true culprit of a crime using only the DNA they left in a room.

Natalie Ram, a law professor at the University of Maryland not involved in the research, said the findings “should raise serious concern about genetic privacy and the appropriate limits of policing”.

“Exploiting involuntarily shed genetic information for investigative aims risks putting all of us under perpetual genetic surveillance,” she wrote in a commentary on the study.

The authors of the study shared her concerns.

McCauley warned harvesting human eDNA without consent could be used to track individual people or even target “vulnerable populations or ethnic minorities”.

It is why the team decided to sound the alarm, they said in a statement, calling for policymakers and scientists to start working on regulation that could address the “ethical quagmire”.

The article by Liam Whitmore et al., “Inadvertent human genomic bycatch and intentional capture raise beneficial applications and ethical concerns with environmental DNA”, can be accessed following this link:

<https://www.nature.com/articles/s41559-023-02056-2>

Forensic DNA analysis (DNA

fingerprinting)

Forensic science employs DNA analysis to identify individuals by their DNA profiles. Most commonly used forensic DNA approaches include the nuclear genome profile based on variable DNA repeats (VNTRs), Y-chromosome and mtDNA analysis.

Although 99.9% of human DNA sequences are the same in every person, enough of the DNA is different that it is possible to distinguish one individual from another, unless they are monozygotic (“identical”) twins. DNA profiling uses repetitive sequences that are highly variable, called variable number tandem repeats (VNTRs), in particular short tandem repeats (STRs). STRs are highly polymorphic regions that have short repeated sequences of DNA (the most common is 4 bases repeated, but there are other lengths in use, including 3 and 5 bases). VNTR loci are very similar between closely related humans, but are so variable that unrelated individuals are extremely unlikely to have the same VNTRs.

The system of DNA profiling used today is based on PCR and uses STRs. Because unrelated people almost certainly have different numbers of repeat units, STRs can be used to discriminate between unrelated individuals. These STR loci (locations on a chromosome) are targeted with sequence-specific primers and amplified using PCR. The DNA fragments that result are then separated and detected using electrophoresis.

In the US, the STR-based DNA profiling system screens for 13 polymorphic loci to establish a unique individual DNA profile.

The Combined DNA Index System (CODIS), a DNA database funded by the FBI, stores DNA profiles created

by federal, state, and local crime laboratories in the United States, with the ability to search the database to assist in the identification of suspects in crimes.

Y-chromosome analysis

Recent innovations have included the creation of primers targeting polymorphic regions on the Y-chromosome (Y-STR), which allows resolution of a mixed DNA sample from a male and female or cases in which a differential extraction is not possible. Y-chromosomes are paternally inherited, so Y-STR analysis can help in the identification of paternally related males.

Mitochondrial analysis

For highly degraded samples, it is sometimes impossible to get a complete profile of the 13 CODIS STRs. In these situations, mitochondrial DNA (mtDNA) is sometimes typed due to there being many copies of mtDNA in a cell, while there may only be one to two copies of the nuclear DNA. Forensic scientists amplify the region that controls replication of mtDNA (HV1 and HV2), and then sequence each region and compare single-nucleotide differences to a reference. Because mtDNA is maternally inherited, directly linked maternal relatives can be used as match references, such as one's maternal grandmother's daughter's son. MtDNA is useful in determining clear identities, such as those of missing people when a maternally linked relative can be found. mtDNA testing was used in determining that Anna Anderson was not the Russian princess she had claimed to be, Anastasia

Romanov. MtDNA can be obtained from such material as hair shafts and old bones/teeth.

Chapter 15. Population Genetics

Population Genetics is the study of the distribution of genes in a population and of how the frequencies of genes and genotypes are maintained or changed.

A population is a localized group of individuals belonging to the same species. A species is a group of organisms capable of interbreeding and producing fertile offspring, and separated from other such groups with which interbreeding does not (normally) happen.

Human Genetic Variation

Although the human chromosomes and loci they contain are identical in all members of the human species, allele frequencies at many loci vary widely among population groups.

Within each population there is extensive variation, greater on the average than the mean differences between groups.

Hardy-Weinberg Law

The Hardy-Weinberg principle is a general rule for allelic behavior in a large population with random mating. It states that in an infinitely large, randomly mating population in which selection, migration, and mutation do not occur, the frequencies of alleles and genotypes do not change (that is, they are at equilibrium) from generation to generation.

In the simplest case of a single locus with two alleles: the dominant allele is denoted A and the recessive a and their frequencies are denoted by p and q; $\text{freq}(A)=p$; $\text{freq}(a)=q$; $p+q=1$. If the population is in equilibrium, then we will have $\text{freq}(AA) = p^2$ for the AA homozygotes in the population, $\text{freq}(aa) = q^2$ for the aa homozygotes, and $\text{freq}(Aa) = 2pq$ for the heterozygotes.

A better, but equivalent, probabilistic description for the HWP is that the alleles for the next generation for any given individual are chosen randomly and independent of each other. Consider two alleles, A and a, with frequencies p and q, respectively, in the population. The different ways to form new genotypes can be derived using a Punnett square, where the fraction in each is equal to the product of the row and column probabilities.

These frequencies are called Hardy–Weinberg frequencies (or Hardy–Weinberg proportions). This is achieved in one generation, and only requires the assumption of random mating with an infinite population size.

It is important to understand that the infinite population size is a theoretical category and the Hardy–Weinberg principle is a theoretical model. That is, the Hardy–Weinberg equilibrium is impossible in nature. Genetic equilibrium is an ideal state that provides a baseline against which to measure change.

The following factors affect the Hardy–Weinberg equilibrium:

- Small population size

- Non-random mating (assortative mating, inbreeding)

- Selection

- Mutation

- Migration, gene flow

- Small population size

Small population size can cause a random change in allele frequencies. This is due to a sampling effect, and is called genetic drift. Sampling effects are most important when population sizes are small or the allele is rare.

Random mating in human populations

In human populations, mating can be random with respect to some traits, and nonrandom for others. Mating seems to be random for such characteristics as blood type groups (not observable). Mating may be nonrandom (assortative) for traits such as skin color and height (observable).

Non-Random Mating: the Dutch' story

(Adapted from By Richard Ingham, AFP, 2015)

The Netherlands is the land of giants: on average, its women stand almost 1.71 meters (5 feet 7 inches) tall, and its men 1.84 meters. But how the Dutch became the world's tallest people has been somewhat of a mystery. After all, two centuries ago they were renowned for being among the shortest. What happened since then?

A popular explanation is nutrition — a calorie-stuffed diet rich in meat and dairy products. But that can't be the whole story, experts say. Other European countries, too, have enjoyed similar prosperity and a rise in living standards, yet their citizens have not shot skywards as much.

The average male height in the Netherlands has gained 20 cm (eight inches) in the last 150 years, according to military records. By comparison, the height of the average American man has risen a mere six cm over the same period.

Researchers led by Gert Stulp, a specialist in population health at the London School of Hygiene and Tropical Medicine, combed a Dutch database for clues. Called LifeLines, the record contains exhaustive detail about the lives and health of more than 94,500 people who lived in the northern the Netherlands from 1935 to 1967. In this three-decade snapshot, the people who had the most children were tall men, and women of average height, the team found. For example, the most fertile men were 7 cm above the average height. Statistically, they had 0.24 more children on average than the least fertile men, who were about 14 cm below the average height.

Compared to counterparts in other countries where they often tended to have fewer children, taller women also reproduced more in the Netherlands. Many postponed having children until after their studies, but once they forged a successful relationship, often had a large family. The study did not involve genetic testing, but concluded from the observations that natural selection must have played a part: with time, more and more Dutch started sporting tall genes.

“Natural selection in addition to good environmental conditions may help explain why the Dutch are so tall,” said the study published in the Royal Society journal *Proceedings B* (Stulp et al., 2015).

“Height is very heritable — taller parents tend to have somewhat taller children than shorter parents,” Stulp told AFP by email. “Because taller individuals would have more offspring in the next generation who would be taller, the average height in that generation would a

bit taller on average than the preceding generation, if all else is equal.”

There seems to be a cultural preference as well. Stulp pointed to figures showing that, in the United States, shorter women and men of average height have the most reproductive success. “There is much variation in what men and women want,” he said.

“When it comes to choosing a mate, height tends to have (only) a small effect, which is not very surprising given the many other, more important, traits people value in their mate.”

Inbreeding and the frequency of rare traits

Inbreeding describes matings between individuals who are related by descent from a common ancestor. Both assortative mating and inbreeding can bring about an increase in the frequency of both homozygous genotypes and a decrease in the corresponding heterozygous form.

Selection

Selection, in general, causes allele frequencies to change, often quite rapidly. While directional selection (a mode of natural selection in which a single phenotype is favored, causing the allele frequency to continuously shift in one direction) eventually leads to the loss of all alleles except the favored one, some forms of selection, such as balancing selection, lead to equilibrium without loss of alleles.

Mutation

Mutation will have a very subtle effect on allele frequencies. Mutation rates are of the order 10^{-4} to 10^{-8} , and the change in allele frequency will be, at most, the same order. Recurrent mutation will maintain alleles in the population, even if there is strong selection against them.

Migration genetically links two or more populations together. In general, allele frequencies will become more homogeneous among the populations. Some models for migration inherently include nonrandom mating (Wahlund effect, for example). For those models, the Hardy–Weinberg proportions will normally not be valid.

Gene Flow

Defined as gradual diffusion of genes from one population to another by migration and mating. The process involves large population and gradual change in gene frequencies.

Classic example: transfer of genes between ethnic groups.

Ethnic Groups and Human Races

From a biological prospective, race is a synonym to population and is defined as a group of organisms that can interbreed and differs from other groups in the frequency of certain hereditary traits.

Human populations have distinct phenotypic characteristics that were historically used to separate humans into several phenotypical races: Mongoloid (Asian), Negro (African), Caucasian (European), Polynesian, Australian aborigine, Hindu, Native American, etc.

The concept of race in humans is socio-political and biologically non-applicable. Modern research has shown that human populations cannot be divided into clearly defined, biologically distinct groups. In the last 200+ years, genetic flow among different formerly isolated human populations increased and frequencies of many genetic traits became indistinguishable between individuals from different populations. Phenotypic differences began to disappear as well.

Genetically, all modern humans are indistinguishable and they are all related to a common ancestral group that lived in Africa 100,000 years ago. We know about the latter through the research on mtDNA.

Chapter 16.

Evolutionary Genetics



Mosaic medallion on the floor of the Jordan Hall of Science at the University of Notre Dame.
<https://uain.press/blogs/teodosij-do-bzhanskij-jogo-nauka-stanovila-vs-e-jogo-zhitt-ya-1159271>

Evolution: From Darwin to Modern Synthesis

The union of traditional Darwinian evolution with subsequent discoveries in classical and molecular genetics is termed the modern evolutionary synthesis.

The seminal work on modern evolutionary synthesis was published in 1937 by Theodosius Dobzhansky (1900-1975), a Ukrainian-American geneticist and evolutionary biologist

whose work had a major influence on 20th-century thought and research on genetics and evolutionary theory.

In his “Genetics and the Origin of Species” Dobzhansky defined evolution as “a change in the frequency of an allele within a gene pool”. Dobzhansky’s work established that mutations are the source of genetic variation from which natural selection chooses the variants that confer better survival and reproduction.

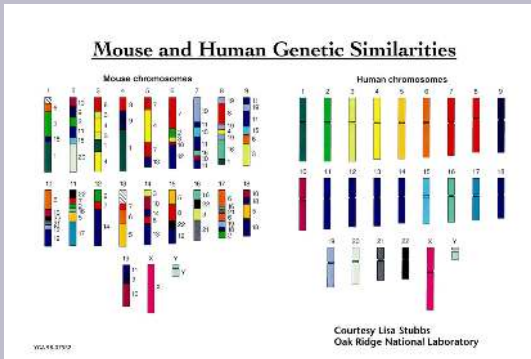
Genome rearrangements and evolution

As many as 99% of human genes are conserved across all mammals. The functionality of many genes is virtually the same among many organisms. It is highly unlikely that the same gene with the same function would spontaneously develop among all currently living species.

Humans and mice have similar genomes, but their genes are ordered differently. 245 chromosomal rearrangements separate human and mouse genome. Among these, there are reversals, fusions, fissions, and translocations.

Significant chromosomal rearranging occurred between the diverging point of humans and mice. For example, human chromosome 3 contains homologous sequences to at least five mouse chromosomes. The homology between chromosome regions of different species is called **synteny**.

Human and mice chromosome homology



This figure demonstrates the genetic similarity (synteny) of the superficially dissimilar mouse and human species. The similarity is such that human chromosomes can be cut (schematically at least) into about 150 pieces (only about 100 are large enough to appear here), then reassembled into a reasonable approximation of the mouse

genome.
The colors
and
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Role of Transposable Elements in Evolution

The proliferation and evolution of transposable elements have had multiple impacts on the genome. Retrotransposition can destabilize the genome, shaping genomic landscapes by insertional mutagenesis, deletions and gene rearrangements. Transposons can provide a substrate for illegitimate homologous recombination (ectopic recombination). One of the most important genetic “drivers” of human

evolution are the *Alu* family of retrotransposons. *Alu* elements are primate-specific TEs. There are more than a million *Alu* sequences, each about 300 nucleotides long, interspersed throughout the human genome. Scientists have used the occurrence and distribution of *Alu* sequences to confirm, for example, that flying lemurs are not part of the order Primata, because *Alu* elements have been detected in all primates tested but not in flying lemurs (Xing *et al.*, 2007). *Alu* elements have also been used in attempts to resolve what is known as the “trichotomy problem”: the evolutionary relationship among humans, chimpanzees, and gorillas. In one study, the distribution of more than 100 different *Alu* elements suggested that humans and chimpanzees are a sister clade, with gorillas being the outgroup (Salem *et al.*, 2003). This means that humans are more closely related to chimpanzees than to gorillas.

A substantial portion of the genomic differences between humans and chimpanzees was caused by *Alu* recombination (estimated time of divergence-6 million years ago). Although active at higher levels earlier in primate evolution, *Alu* elements continue their activity in modern human genomes, contributing to genetic diversity. Because there is no specific mechanism for removal of *Alu* insertions, *Alu* evolution is dominated by the accumulation of new *Alu* inserts. They are also a major factor contributing to non-allelic homologous recombination events causing copy number variation and disease.

The ubiquitous presence of *Alu* elements throughout the human genome has led to their presence in a large number of genes and their transcripts. Many individual *Alu* elements have wide-ranging influences on gene expression. The mobility of *Alu* elements is connected with major events in human evolution, such as the rapid increase in brain size in the human lineage after the human-chimpanzee lineage split (Britten, 2010).

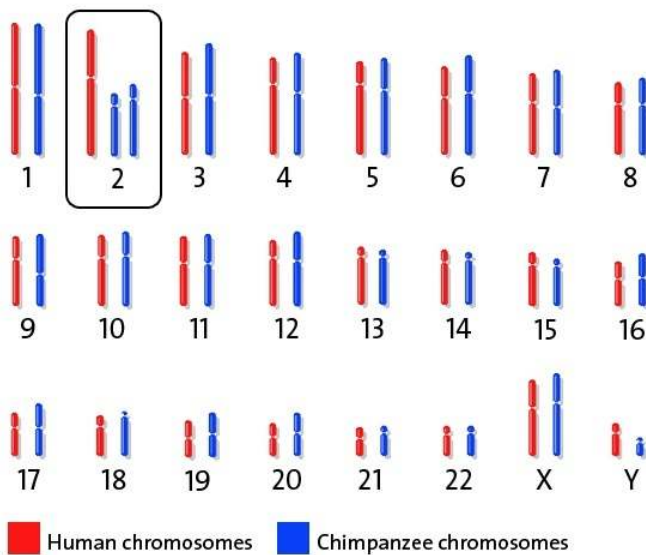
Transposons have also been notably involved in the emergence of new genes during mammalian evolution. The evolution of various physiological processes, such as pregnancy, in mammals happened not by the slow development of a completely new set of genes, but by change in the regulation of groups of genes expressed in other systems of the body.

Studies suggest that transposons spurred evolution of pregnancy in mammals (Lynch et al., 2011, 2015). About 13% of the 20,000 genes that regulate pregnancy in Eutherians have a Eutherian-specific transposable element (MER20) within 200 kb of their transcription start sites. MER20 binds transcription factors of the progesterone-induced signal transduction pathway, particularly MER20 drives the expression of prolactin, a protein-based hormone that plays an essential role in metabolism, regulation of the immune system and pancreatic development, as well as in pregnancy and nursing. The presence of MER20 in the promoter region of the prolactin gene induces the gene's tissue-specific expression in endometrium.

Some TEs contain DNA binding site motifs for transcription factors and have rewired the transcriptional regulatory networks in which these transcription factors function by introducing many new binding sites via their insertions throughout the genome. TE-derived sequences also appear in RNA transcripts where they can interact with RNA binding proteins (RBPs), which also often have preferred binding site motifs. In perhaps the most understood and interesting example, the antisense strand of Alu elements contains motifs that recruit the cell's splicing machinery and have thus introduced hundreds of novel exons into various protein coding genes. Sequence derived from TEs has also been implicated in both degradation and increasing the translation of RNA transcripts.

Chromosomal Differences Between Humans and the Great Apes

Humans share a branch of mammalian phylogenetic tree with the great apes of the Old World (chimpanzee, gorilla, and orangutan). Collectively, humans, the great apes, and their ancestors are called **hominids**. From the genetics standpoint, humans and the great apes are very similar in all but one major aspect. All great apes have 48 chromosomes ($n=24$), while humans have 46 ($n=23$). This karyotypic difference was caused by the fusion of two ancestral chromosomes in the ancestor of the **hominin** lineage to form human chromosome 2 and subsequent inactivation of one of the two original centromeres. The great apes retained the two ancestral chromosomes (2A and 2B).



As a result of this fusion, sequences that once resided near the ends of the ancestral chromosomes are now located in the middle of chromosome 2, near the borders of bands 2q13 and 2q14.1.

Human evolution continues: $2n=44$?

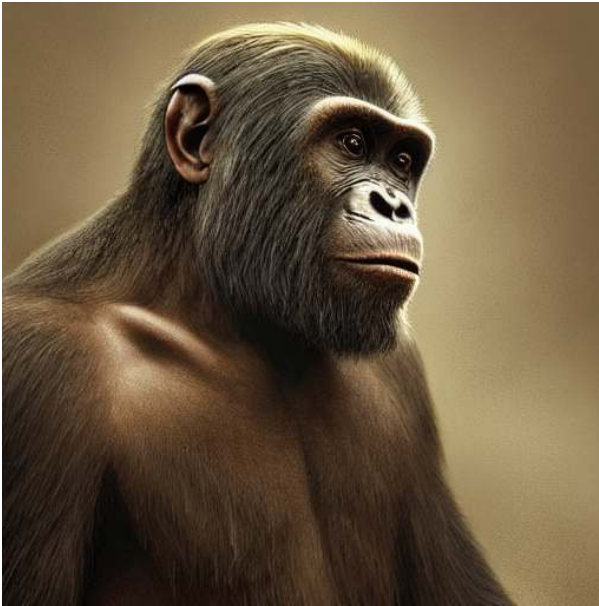
Reciprocal translocations are an exchange of material between non-homologous chromosomes (1/625 newborns). In Robertsonian translocations, long arms of acrocentric chromosomes fuse near the centromere region with loss of the short arms.

Robertsonian translocations, such as 14q/21q, can be inherited. A $2n=45$, 14q/21q individual would be fertile and would produce gametes with $n=23$ and $n=22$; 14q/21q.

Suppose a 45-chromosome human meets a mate with the same translocation and the same karyotype. If they were to have children, there would be a 50% chance for their offspring to have a $2n=44$ karyotype.

Such a scenario is not far fetched. There have been documented cases of fertile adult individuals with a 44-chromosome karyotype. A recent such case from China was described in 2013 (Wang et al., 2013). It describes a Robertsonian translocation family, one of them who came from a consanguineous marriage has the previously undescribed balanced human karyotype 44,XY,der(14;15)(q10;q10), der(14;15)(q10;q10) (image on the left). This man had fewer chromosomes but is actually missing very few genes. Instead, he has two chromosomes stuck to two other chromosomes. More specifically, both his chromosome 14's are stuck to his chromosome 15's. Thus, he is missing chromosome 15 pair and his chromosome 14 pair is a fusion chromosome. This is essentially what happened in the lineage leading to humans when we departed from the great apes due to a similar chromosomal fusion and became our own distinct species.

Human Origins



<https://deepai.org/>

The human family tree spans over 7 million years, when our ancestors split from the ancestors of modern chimpanzees. There are different scenarios proposed by human evolutionary biologists, anthropologists and paleontologists over the last 100 years regarding *Homo sapiens* evolution. There is no single scenario exists at the present time that satisfies everybody. It became even more contentious in the last two decades with the advent of ancient DNA (aDNA) analysis. One of the best supported by scientific evidence scenarios is presented here.

The earliest hominin fossils come from Bulgaria and Greece and belong to the genus *Graecopithecus* that lived during the late Miocene 7.2 million years ago (MYA) ([Fuss et al., 2017](#)). The emergence of the genus *Homo* is currently dated to about 3.3 MYA ([Püschel et al., 2021](#)) , overlapping with the latest surviving

members of the genus *Australopithecus*, proposed to have evolved in Africa some 4.4 MYA. *Australopithecines*, in turn, likely descended from an early African hominin genus called *Ardipithecus*, which origin is dated to around 6 MYA. Significant gaps still remain in our understanding of the early hominin evolution. It is even unclear whether any of the above-mentioned hominins are in the direct line of modern human descent.

The clear human ancestor, *Homo erectus*, emerged around 2.1 MYA. The *Erectus* fossils have been found in Africa, but most of the fossil evidence of the *Erectus* hominins comes from outside of Africa, from the Caucasus Mountains, China and Southeast Asia. The emergence of *H. erectus* followed what is, arguably, one of the most important genetic events in the ancestral human genome.

About 2.4 MYA a mutation in the Myosin Heavy Chain gene, MYH16, took place in the ancestor of modern humans. This frameshift mutation led to the inactivation of the the predominant myosin heavy chain, responsible for the powerful muscles supporting the mandibles of non-human hominids. The loss of this protein isoform led to significant size reductions in individual muscle fibers and entire masticatory muscles in the hominin lineage. These muscles connect at the sagittal crest, a bony ridge at the top of the skull of hominids. The pressure that masticatory muscles exert on the skull constricts the size of the cranial vault. The relaxing of this constriction due to the MYH16 mutation led to the shift of the bulk of the support of the chewing function to the products of the *MYH1*, *-2* and *-7* genes in hominins, the temporalis muscles that have a lower attachment point on the cranium (the temporal fossa). Such removal of the cranial pressure by the masticatory muscles may have led to the potentially unrestricted growth of the cranial capacity, resulting in a rapid increase in the brain size and complexity (encephalization) of the members of the *Homo* genus. This change is evident anatomically in the

remains of *H. erectus*, which display a relatively gracile masticatory apparatus starting 2 MYA ([Stedman et al., 2004](#)).

The MYH16 mutation appeared at a time of major environmental transformations. A major climatic shift was taking place at the onset of Quaternary Glaciation ca. 2.58 MYA, changing the landscape in east Africa from woodland to open grasslands. The ecological niche change combined with the MYH16 mutation conceivably led to an *in situ* speciation event, producing the *Homo* lineage.

Homo sapiens emerged as a species around 300 thousand years ago (KYA). Currently the oldest remains of early modern humans come from the site of Jebel Irhoud in Morocco and are dated to 286 ± 32 KYA ([Hublin et al., 2017](#)). The Anatomically modern human (AMH) remains with essentially the same cranial morphology as the humans living today come from Greece (Apidima Cave, 210 KYA), Israel (Mount Carmel, 177 KYA, Skhul and Qafzeh caves, 125 to 90 KYA), and China (Daioxian and Zhirendong, between 120 and 80 KYA).

Evidence from mitochondrial DNA (mtDNA) suggests an AMH dispersal out of Africa ca. 130 KYA reached southeast Asia, subsequently moving back to Africa ca. 70 KYA, and dispersing out of Africa again ca. 60-50 KYA ([Cabrera et al., 2018](#)). It is likely (though not well documented at the moment) that these “pendular” migrations in and out of Africa have been taking place continuously throughout the history of the *H. sapiens* species, and that the most “successful” dispersal episode led to a permanent colonization of the planet by *H. sapiens*, leaving a continuous genetic trail outside of Africa starting ca. 60-50 KYA.

Paleontological evidence suggests that there were several lineages of *Homo* co-existing in various parts of the world ca. 700-30 KYA. In Africa, *H. erectus* (2.1 MYA-120 KYA) coexisted with *H. sapiens idaltu*, an extinct *Sapiens* lineage (160 KYA), the Omo River hominins (possibly *H. sapiens*, 195 KYA), and Jebel Irhoud *H. sapiens* (286 KYA). In Asia, *H. erectus sensu*

lato co-existed with several distinct Homo lineages and Erectus subspecies, including *H. floresiensis* on the island of Flores in Indonesia (700-12 KYA), *H. luzonensis*, a dwarf lineage of *H. erectus* from Callao cave in the Philippines (67 KYA), *H. erectus soloensis* from the Solo river in Java, Indonesia (550-35 KYA), *Homo sp. altai* (Denisovans, 100-30 KYA, *H. neanderthalensis altai* (300-70 KYA), as well as Homo lineage hybrids. In Europe, *H. sapiens* overlapped with Neanderthals territorially and even culturally, until about 40 KYA.

There is ample genetic evidence suggesting that AMHs have interbred, on a number of occasions, with our closest cousins, Neanderthals (*Homo sapiens neanderthalensis*) and Denisovans (*Homo sp. Altai*). We know about the latter from just a phalanx and three molars, all found in the same cave, called Denisova, in the Altai Mountains (southwestern Siberia). Archaeogeneticists (scientists who work with ancient DNA) were able to extract DNA out of these remains and reconstruct the Denisovan genome.

Based on analyses of their nuclear genome, Denisovans are a sister group of Neanderthals, although the mtDNAs of Neanderthals and present-day humans share an mtDNA ancestor more recently with each other than with Denisovans.



<https://deepai.org/>

Based on the studies of the genomes of modern humans, Denisovans and Neanderthals, it is becoming apparent that

modern humans carry a small percentage of Neanderthal and Denisovan genomes and that, collectively, modern humans had retained almost 50% Neanderthal DNA. Interestingly, the percentage of Denisovan DNA in modern humans is higher in southeast Asia and Austronesia, while modern Europeans harbor more Neanderthal ancestry.

Nose shape gene inherited from Neanderthals

by University College London

MAY 8, 2023

Humans inherited genetic material from Neanderthals that affects the shape of our noses, finds a new study led by UCL researchers.

The new *Communications Biology* study finds that a particular gene, which leads to a taller nose (from top to bottom), may have been the product of natural selection as ancient humans adapted to colder climates after leaving Africa.

Co-corresponding author Dr. Kaustubh Adhikari (UCL Genetics, Evolution & Environment and The Open University) said, “In the last 15 years, since the Neanderthal genome has been sequenced, we have been able to learn that our own ancestors apparently interbred with Neanderthals, leaving us with little bits of their DNA.

“Here, we find that some DNA inherited from Neanderthals influences the shape of our faces. This

could have been helpful to our ancestors, as it has been passed down for thousands of generations.”

The study used data from more than 6,000 volunteers across Latin America, of mixed European, Native American and African ancestry, who are part of the UCL-led CANDELA study, which recruited from Brazil, Colombia, Chile, Mexico and Peru. The researchers compared genetic information from the participants to photographs of their faces—specifically looking at distances between points on their faces, such as the tip of the nose or the edge of the lips—to see how different facial traits were associated with the presence of different genetic markers.

The researchers newly identified 33 genome regions associated with face shape, 26 of which they were able to replicate in comparisons with data from other ethnicities using participants in east Asia, Europe, or Africa.

In one genome region in particular, called *ATF3*, the researchers found that many people in their study with Native American ancestry (as well as others with east Asian ancestry from another cohort) had genetic material in this gene that was inherited from the Neanderthals, contributing to increased nasal height. They also found that this gene region has signs of natural selection, suggesting that it conferred an advantage for those carrying the genetic material.

First author Dr. Qing Li (Fudan University) said, “It has long been speculated that the shape of our noses is determined by natural selection; as our noses can help us to regulate the temperature and humidity of the air

we breathe in, different shaped noses may be better suited to different climates that our ancestors lived in. The gene we have identified here may have been inherited from Neanderthals to help humans adapt to colder climates as our ancestors moved out of Africa.”

Co-corresponding author Professor Andres Ruiz-Linares (Fudan University, UCL Genetics, Evolution & Environment, and Aix-Marseille University) added, “Most genetic studies of human diversity have investigated the genes of Europeans; our study’s diverse sample of Latin American participants broadens the reach of genetic study findings, helping us to better understand the genetics of all humans.”

The finding is the second discovery of DNA from archaic humans, distinct from *Homo sapiens*, affecting our face shape. The same team discovered in a 2021 paper that a gene influencing lip shape was inherited from the ancient Denisovans.

The study involved researchers based in the UK, China, France, Argentina, Chile, Peru, Colombia, Mexico, Germany, and Brazil.

More information: Automatic landmarking identifies new loci associated with face morphology and implicates Neanderthal introgression in human nasal shape, *Communications Biology* (2023). [DOI: 10.1038/s42003-023-04838-7](https://doi.org/10.1038/s42003-023-04838-7)

Based on the mtDNA divergence the intermixing between Neanderthals and *H. sapiens* that can still be traced today in modern human genomes likely took place outside of Africa as early as 250 KYA, but no later than 110 KYA ([Kuhlwilmet al.,](#)

[2017](#)). However, there were multiple admixtures between the two hominins the descendants of which did not survive to the present day.

The [Oase 1](#) hominin from Romania, 39.5 ± 2.5 KYA, who derived 4.8-7.3% of genome from Neanderthals, had a Neanderthal ancestor 4-6 generations back, with largest Neanderthal-derived chromosomal segments located on chromosomes 4, 5, 6, 9, 12. The [Ust'-Ishim](#) individual from west Siberia, 45 KYA, had 2.3% of his genome coming from Neanderthals, with the latest Neanderthal ancestor 13-7 KYA back. His chromosome 21 was largely Neanderthal-derived.

However, current genetic evidence suggests that in the last five thousand years of Neanderthal and modern human overlap in Eurasia, there was little, if any, biological interaction between the two groups. Such as, the [Late Neanderthals](#) (47-39 KYA) from west Europe and the Caucasus show no genetic introgression from modern humans.

“Thus we have given to man a pedigree of prodigious length, but not, it may be said, of noble quality. The world, it has often been remarked, appears as if it had long been preparing for the advent of man: and this, in one sense is strictly true, for he owes his birth to a long line of progenitors. If any single link in this chain had never existed, man would not have been exactly what he now is. Unless we wilfully close our eyes, we may, with our present knowledge, approximately recognise our parentage; nor need we feel ashamed of it. The most humble organism is something much higher than the inorganic dust under our feet; and no one with an

unbiased mind can study any living creature, however humble, without being struck with enthusiasm at its marvellous structure and properties.”

Charles Darwin, “The Descent of Man, and Selection in Relation to Sex”, 1889, p. 165

Key Takeaways

- Recent data indicate that ca. 70KYA Eurasia was populated by archaic hominins such as Neanderthals, Denisovans, their hybrids, as well as descendants of H. Habilis and H. erectus in southeast Asia, plus transient Anatomically Modern Humans (AMHs).
- The bulk of the interbreeding between resident archaic hominins and transient AMHs took place prior to 150KYA.
- The latter transmitted admixed traits to the most recent African migrants of the most recent major dispersal throughout Eurasia ca. 60-50KYA.
- Biological contacts between archaic hominins and AMHs that left genetic traces in the latter were sporadic after 150KYA and ceased altogether after ca. 45 KYA.

- Reasons: separation of ecological niches and subsequent speciation.
- Archaic hominin contribution to modern genomes:
 - Neanderthals to Eurasians: 1-4%
 - Neanderthals to Africans: none directly; 0.3% from recent admixture with modern Eurasians.
- Collectively, 50% of Neanderthal DNA can be retrieved from modern humans.
 - 3–5% of the DNA of Melanesians and Aboriginal Australians and around 7%-8% in Papuans deriving from Denisovans.

Bibliography

Attanasio C. et al. (2013) Fine Tuning of Craniofacial Morphology by Distant-Acting Enhancers. *Science* 342: 441-448.

Bai M, Janicic N, Trivedi S, Quinn SJ, Cole DE, Brown EM, Hendy GN. (1997) Markedly reduced activity of mutant calcium-sensing receptor with an inserted Alu element from a kindred with familial hypocalciuric hypercalcemia and neonatal severe hyperparathyroidism. *J Clin Invest.* 99(8):1917-25. doi: 10.1172/JCI119359.

Britten RJ. (2010) Transposable element insertions have strongly affected human evolution. *Proc Natl Acad Sci USA*: 107(46):19945-8. doi: 10.1073/pnas.1014330107.

Cabrera VM, Marrero P, Abu-Amro KK, Larruga JM. (2018). Carriers of mitochondrial DNA macrohaplogroup L3 basal lineages migrated back to Africa from Asia around 70,000 years ago. *BMC evolutionary biology*, 18(1), 98. <https://doi.org/10.1186/s12862-018-1211-4>

Chen Y. et al. (2018) A single nucleotide incorporation step limits human telomerase repeat addition activity. *The EMBO Journal* 37: e97953, DOI 10.15252/embj.201797953.

Crick F. (1970) Central Dogma of Molecular Biology. *Nature* 227: 561-563.

Dobzhansky Th. 1937. *Genetics and the Origin of Species*. Columbia University Press, New York.

Dobzhansky, Th. (1973) Nothing in Biology Makes Sense Except in the Light of Evolution. *American Biology Teacher* 35 (3): 125-129.

Eiberg, H. et al. (2008) Blue eye color in humans may be caused by a perfectly associated founder mutation in a regulatory element located within the HERC2 gene inhibiting OCA2 expression. *Human Genetics*, 2008; 123 (2): 177, DOI: 10.1007/s00439-007-0460-x

Eisensmith, R. and S. Woo (1992) Molecular Basis of Phenylketonuria and Related Hyperphenylalaninemias: Mutations and Polymorphisms in the Human Phenylalanine Hydroxylase Gene. *Human Mutation* 1: 13-23.

Fuss J, Spassov N, Begun DR, Böhme M. (2017) Potential hominin affinities of *Graecopithecus* from the Late Miocene of Europe. *PLoS One* 12(5):e0177127. doi:10.1371/journal.pone.0177127

Greider, C.W. & Blackburn, E.H. (1985) Identification of a specific telomere terminal transferase activity in *Tetrahymena* extracts. *Cell* 43 (2 Pt 1): 405–413.

Harvati K. et al. (2019). Apidima Cave fossils provide earliest evidence of *Homo sapiens* in Eurasia. *Nature*, 571(7766), 500–504. <https://doi.org/10.1038/s41586-019-1376-z>

Hershenson, S. (1939) *Pryroda tak zvanykh henetychno-inertnykh chastyn khromosom*. Kyiv: Akademiya Nauk URSR. (Гершензон, С. (1939) Природа так званих генетично-інертних частин хромосом. Київ: Академія Наук УРСР.) In Ukrainian.

Gershenson S. (1928) A New Sex-Ratio Abnormality in *Drosophila obscura*. *Genetics*: 13(6):488-507. doi:10.1093/genetics/13.6.488

Hublin J-J. et al. (2017) New fossils from Jebel Irhoud, Morocco and the pan-African origin of *Homo sapiens*. *Nature* 546: 289–292. <https://doi.org/10.1038/nature22336>

Hutchinson GB, Andrew SE, McDonald H, Goldberg YP, Graham R, Rommens JM, Hayden MR. (1993) An Alu element retroposition in two families with Huntington disease defines a new active Alu subfamily. *Nucleic Acids Res*. 21(15):3379-83. doi: 10.1093/nar/21.15.3379.

Janicic N, Pausova Z, Cole DE, Hendy GN. (1995) Insertion of an Alu sequence in the Ca(2+)-sensing receptor gene in familial hypocalciuric hypercalcemia and neonatal severe hyperparathyroidism. *Am J Hum Genet*. 56(4):880-6.

de Koning APJ, Gu W, Castoe TA, Batzer MA, Pollock DD (2011)

Repetitive Elements May Comprise Over Two-Thirds of the Human Genome. *PLoS Genet* 7(12): e1002384. doi:10.1371/journal.pgen.1002384

Lyon MF. (1961) Gene action in the X chromosome of the mouse (*Mus musculus* L.). *Nature* 190:372–373.

Mueller JL. et al. (2013) Independent specialization of the human and mouse X chromosomes for the male germ line. *Nature Genetics* doi:10.1038/ng.2705.

Muratani K, Hada T, Yamamoto Y, Kaneko T, Shigeto Y, Ohue T, Furuyama J, Higashino K. (1991) Inactivation of the cholinesterase gene by Alu insertion: possible mechanism for human gene transposition. *Proc Natl Acad Sci USA*. 88(24):11315-9. doi: 10.1073/pnas.88.24.11315.

Nikitin AG, Shmookler Reis RJ. (1997). Role of transposable elements in age-related genomic instability. *Genetical research*, 69(3), 183–195. <https://doi.org/10.1017/s0016672397002772>

Ohno S. (1967) *Sex Chromosomes and sex-linked genes*. Berlin:Springer-Verlag.

Olalde I. et al. (2014) Derived immune and ancestral pigmentation alleles in a 7,000-year-old Mesolithic European. *Nature* 507, 225–228.

Olovnikov AM. (1973) A theory of marginotomy. The incomplete copying of template margin in enzymatic synthesis of polynucleotides and biological significance of the phenomenon. *J Theor Biol*.14;41(1):181-90.

Pidpala OV, Yatsishina AP, Lukash LL. (2008) Human Mobile Genetic Elements: Structure, Distribution and Functional Role. *Cytology and Genetics* 42(6): 420–43.

Pier GB. et al. (1998). *Salmonella typhi* uses CFTR to enter intestinal epithelial cells. *Nature*, 393(6680), 79–82. <https://doi.org/10.1038/30006>

Polderman TJ. et al.. (2015). Meta-analysis of the heritability of human traits based on fifty years of twin studies. *Nature genetics*, 47(7), 702–709. <https://doi.org/10.1038/ng.3285>

Püschel HP. et al. (2021). Divergence-time estimates for hominins provide insight into encephalization and body mass trends in human evolution. *Nature Ecology & Evolution* 5(6): 808–819. doi:[10.1038/s41559-021-01431-1](https://doi.org/10.1038/s41559-021-01431-1)

Ruby JG. et al. (2018). Estimates of the Heritability of Human Longevity Are Substantially Inflated due to Assortative Mating. *Genetics* 210(3):1109–1124. doi: 10.1534/genetics.118.301613.

Salem AH. et al. (2003) *Alu* elements and hominid phylogenetics. *Proceedings of the National Academy of Sciences* 100, 12787–12791.

Soudet J, Jolivet P, Teixeira M. (2014). Elucidation of the DNA end-replication problem in *Saccharomyces cerevisiae*. *Molecular cell*, 53(6), 954–964. <https://doi.org/10.1016/j.molcel.2014.02.030>

Stedman HH. et al. (2004). Myosin gene mutation correlates with anatomical changes in the human lineage. *Nature*, 428(6981), 415–418. <https://doi.org/10.1038/nature02358>

Stulp G, Barrett G, Tropf FC, Mills M. (2015) Does natural selection favour taller stature among the tallest people on earth? *The Royal Society Proceedings B*:DOI: 10.1098/rspb.2015.0211

Tong, Zhi-Bin, Bondy, Carolyn A., Zhou, Jian, Nelson, Lawrence M. (2002) A human homologue of mouse Mater, a maternal effect gene essential for early embryonic development. *Human reproduction* 17 (4): 903–911.

Turing AM. (1952) *Phil. Trans. R. Soc. Lond. B* 237: 37–72.

Valladas H. et al. (1998) Gif Laboratory dates for Middle Paleolithic Levant. In T. Akazawa, K. Aoki, and O. Bar-Yosef, eds. *Neandertals and modern humans in Western Asia*. New York: Plenum Press. pp. 69–75.

Wallace MR, Andersen LB, Saulino AM, Gregory PE, Glover TW, Collins FS. (1991) A de novo *Alu* insertion results in neurofibromatosis type 1. *Nature* 353(6347): 864–6. doi: 10.1038/353864a0.

Walser CB, Lipshitz HD. (2011) Transcript clearance during the maternal-to-zygotic transition. *Current Opinion in Genetics & Development* 21 (4): 431-443, <https://doi.org/10.1016/j.gde.2011.03.003>.

Woolf LI. (1986) The heterozygote advantage in phenylketonuria. *American Journal of Human Genetics* 38(5): 773-775.

Wu S, Powers S, Zhu W, Hannun YA. (2015) Substantial contribution of extrinsic risk factors to cancer development. *Nature* doi:10.1038/nature16166.

Xing J. et al. (2007) Mobile elements in primate and human evolution. *Yearbook of Physical Anthropology* 50: 2-19.

Xue Z. et al. (2013) Genetic programs in human and mouse early embryos revealed by single-cell RNA sequencing. *Nature* 500: 593-597, doi:10.1038/nature12364.

Yunis EJ, Svardal JM, Bridges RA. (1969) Genetics of the Bombay Phenotype. *Blood* 33(1): 124-132.

Zhu P. et al. (2018) Single-cell DNA methylome sequencing of human preimplantation embryos. *Nat. Genet.* 50, 12, <https://doi.org/10.1038/s41588-017-0007-6>.

Appendix

This is where you can add appendices or other back matter.